

IDENTIFICATION

PRODUCT CODE: MAINDEC-8E-D6AB-D

PRODUCT NAME: PDP8-E XY8-E PLOTTER
CONTROL AND DISPLAY
DIAGNOSTIC PROGRAM

DATE CREATED: MAY 18, 1971

MAINTAINER: DIAGNOSTIC PROGRAMMING
GROUP

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CAUTION NOTES

FAILURE TO COMPLY WITH THE CAUTION NOTES BELOW MAY
RESULT IN DAMAGE TO THE PLOTTING EQUIPMENT.

CAUTION

PRIOR TO RUNNING THE CONTROL AND DISPLAY TESTS, ENSURE THAT
ALL PLOTTER MANUAL COMMAND SWITCHES ARE IN THEIR NEUTRAL
POSITION.

CAUTION

IF THE "[M+]" COMMAND OPTION IS UTILIZED, ENSURE THAT NO
CONFLICTING COMMANDS ARE ENTERED INTO THE DIRECTION
REGISTER VIA THE SWITCH REGISTER.

CAUTION

THE PLOTTER MUST BE DEENERGIZED DURING THE CONTROL TEST.

N O T E

- - - -

THE DISPLAY TEST PORTION HAS BEEN CHECKED OUT
ONLY WITH A CALCOMP 500 SERIES PLOTTER WITH
8 VECTORS AND .01 INCH INCREMENT SIZE.

1. ABSTRACT

THE XY8-E PLOTTER OPTION CONTROL AND DISPLAY DIAGNOSTIC PROGRAM TESTS THE OVERALL OPERATION OF THE XY8-E CONTROL MODULE AND THE CALCOMP PLOTTER (SERIES 500 THROUGH 700), HOUSTON DP10 OR EDP10, OR EQUIVALENT, INSTALLED WITHIN A PDP8-E SYSTEM.

THIS PROGRAM IS SET UP TO USE DEVICE CODE 50. IF THE DEVICE CODE IN THE SYSTEM UNDER TEST IS OTHER THAN 50, LOAD ADDRESS 0225, CLEAR ALL SWITCHES, SET SR3-8 TO NEW DEVICE CODE, THEN DEPRESS CLEAR FOLLOWED BY CONTINUE. THE PROGRAM WILL HALT WITH THE MA=0227 AND THE DEVICE CODE IN THE AC.

ALL TIMING CHECKED IN THIS PROGRAM IS BASED UPON A 72.7 MILLISEC FLAG SETTING TIME FOR PEN UP AND PEN DOWN MOVEMENTS, AND A 7.5 MILLISEC FLAG SETTING TIME FOR ALL OTHER MOVEMENTS. IF THE SYSTEM IS CONFIGURED DIFFERENTLY, CHANGE THE CONTENTS OF THE FOLLOWING LOCATIONS AS SHOWN BELOW.

(A = PEN UP AND PEN DOWN FLAG SETTING TIME.)

(B = FLAG SETTING TIME FOR ALL OTHER MOVEMENTS.)

RELATIVE	ABSOLUTE	OLD	NEW
K70MIN	0115	0062	70% OF A
K70MAX	0116	0055	60% OF A
K5MIN	0117	0005	70% OF B
K5MAX	0120	0005	60% OF B
K5ALL	0121	0010	130% OF B

THE PROGRAM CONSISTS OF THE FOLLOWING MAJOR SEGMENTS.

1.1 XY8-E CONTROL TEST

THE XY8-E CONTROL TEST TESTS THE OPERATION OF THE CONTROL MODULE AND IS NOT DEPENDENT UPON A PLOTTER BEING INSTALLED. THE FOLLOWING TESTS ARE INCLUDED.

TEST0: TEST OF THE SIGNAL "INITIALIZE" TO CLEAR THE PLOTTER FLAG AND SET INTERRUPT ENABLE.

TEST1: FLAG TEST FOR PLPU AND PLPD IOT'S.

TEST2: FLAG TEST FOR PLLR WITH AC=31 AND 32.

TEST3: FLAG TEST FOR PLLR WITH AC=0 THROUGH 77 EXCEPT 31, 32 AND 33.

TEST4: FLAG TEST FOR CFLR WITH AC=31 AND 32.

TEST5: FLAG TEST FOR CFLR WITH AC=0 THROUGH 77 EXCEPT 31, 32 AND 33.

TEST6: INTERRUPT TEST.

1.2 XY8-E DISPLAY TEST

THE DISPLAY TEST IS CONTROLLED BY A SIMPLE TO USE CONVERSATIONAL MONITOR AND UPON COMMAND WILL DISPLAY SIX BASIC PATTERNS ON THE PLOTTER WHICH UTILIZE ALL FUNCTIONS EXCEPT ZIP MODE. ANY PORTION OF THESE PATTERNS MAY BE DISPLAYED SEPARATELY THROUGH THE USE OF COMMANDS ENTERED VIA THE TTY. A SPECIAL PATTERN (P07) IS AVAILABLE WHICH CONSISTS OF WORST CASE LINES WHICH ARE DISPLAYED ONLY UPON EXPLICIT USER COMMAND. IN ADDITION A COMMAND IS PROVIDED TO ALLOW THE USER TO CONTROL THE PLOTTER FROM THE SWITCH REGISTER. THE PATTERNS ARE VALID FOR ANY CALCOMP PLOTTER WITH THE FOLLOWING CHARACTERISTICS OR THE HOUSTON DP10/EDP10 (OR EQUIVALENT).
NOTE: PATTERNS 01-03 ARE TOO LARGE TO BE DISPLAYED ON THE DP10/EDP10.

PLOTTER SERIES:

5XX OR HOUSTON DP10/EDP10
6XX
7XX

PLOTTER TYPE AND SIZE:

11 INCH DRUM OR HOUSTON DP10/EDP10
28.55 INCH DRUM
31 BY 54 INCH FLATBED
54 BY 72 INCH FLATBED

(1.2 CONT'D)

VECTOR TYPES:*

8 VECTOR
24 VECTOR

*PATTERNS 4 AND 5 DIFFER FOR 8 AND 24 VECTOR PLOTTERS.

INCREMENT SIZES:

Ø.1 MM (8 VECTOR ONLY)
.ØØ125 INCH
.ØØ25 INCH
.ØØ5 INCH
.Ø1 INCH

2. REQUIREMENTS

2.1 EQUIPMENT

PDP8-E WITH XY8-E OPTION, CALCOMP OR HOUSTON PLOTTER WITH SPECIFICATIONS FITTING THOSE DELINEATED IN PARAGRAPH 1.2 ABOVE (FOR DISPLAY TEST ONLY), AND TELETYPE.

2.2 STORAGE

4K OF CORE REQUIRED (FIELD Ø).

3. LOADING PROCEDURE

LOAD THE PROGRAM USING THE BIN LOADER.

STARTING PROCEDURE

STANDARD TEST PROCEDURE

IF AN ERROR OCCURS DURING THE CONDUCT OF THIS PROGRAM REFER TO PARAGRAPHS 5 THROUGH 10 FOR RECOVERY INFORMATION AND PROGRAM OPTIONS.
THE FOLLOWING IS THE STANDARD TEST PROCEDURE.

- 4.1.1 DEENERGIZE THE PLOTTER;
- 4.1.2 LOAD ADDRESS 200.
- 4.1.3 CLEAR ALL SWITCHES.
- 4.1.4 DEPRESS CLEAR, THEN CONTINUE.
- 4.1.5 IF NO TEST Ø ERRORS OCCUR THE PROGRAM WILL INITIATE USER INTERROGATION.
- 4.1.6 ANSWER THE QUESTIONS USING THE METHOD DESCRIBED IN 5.1.2 BELOW.
- 4.1.7 AT THE COMPLETION OF INTERROGATION THE CONTROL TEST WILL BE RUN TEN TIMES, EACH PASS BEING INDICATED BY THE TTY BELL (EVERY 15 SECONDS).
- 4.1.8 AFTER COMPLETION OF THE CONTROL TEST, THE DISPLAY MONITOR ASSUMES CONTROL AND TYPES "*".
- 4.1.9 DISPLAY TEST PROCEDURE FOR CALCOMP PLOTTERS 5XX, 6XX, OR 7XX.

- A. ENERGIZE THE PLOTTER AND POSITION THE PEN AT LEAST 2 INCHES FROM ANY PHYSICAL STOP. THEN RETURN ALL SWITCHES TO NEUTRAL.
- B. TYPE "ALTMODE" FOLLOWED BY "D" FOLLOWED BY "RETURN".
- C. THE PROGRAM WILL NOW DRAW THE COMPLETE SET OF PATTERNS.
- D. WHEN ALL PATTERNS HAVE BEEN DISPLAYED, THE MONITOR WILL TYPE "*", THIS COMPLETES THE TEST AND IF NO ERRORS HAVE OCCURRED, NORMAL SYSTEM OPERATION MAY BE RESUMED.

4.1.10 DISPLAY TEST PROCEDURE FOR HOUSTON PLOTTERS DP10/EDP10.

- C = ALTMODE
- D = RETURN

PRIOR TO THE ACCOMPLISHMENT OF EACH STEP, A CLEAN SHEET OF PAPER SHOULD BE ON THE PLOT SURFACE, THE PEN POSITIONED TO THE CENTER OF THE PLOT AREA, ALL MANUAL CONTROL SWITCHES PLACED IN THEIR NEUTRAL POSITION, AND THE PLOTTER ENERGIZED. AN "*" IS TYPED AT THE COMPLETION OF EACH STEP.

- A. DRAW P04 BY TYPING "ID:P04-".
- B. DRAW P05 BY TYPING "ID:P05-".
- C. DRAW P06 BY TYPING "ID:P06-".
- D. THIS COMPLETES THE TEST, AND IF NO ERRORS HAVE OCCURRED, NORMAL SYSTEM OPERATION MAY BE RESUMED.

4.2 STARTING ADDRESSES (OPTIONAL)

STARTING ADDRESS	RESULT
200	RUN COMPLETE TEST.
201	RUN COMPLETE TEST WITHOUT INIT AND TEST 0.
202	INITIALIZE ONLY.
204	DISPLAY TEST ONLY.
205	TEST 0, THEN HALT.
210	TEST 1, THEN HALT.
212	TEST 2, THEN HALT.
214	TEST 3, THEN HALT.
216	TEST 4, THEN HALT.
220	TEST 5, THEN HALT.
222	TEST 6, THEN HALT.
225	CHANGE DEVICE CODE (REFER TO PARAGRAPH 1)

5. OPERATING PROCEDURE

5.1 CONTROL TEST

5.1.1 SWITCH REGISTER OPTIONS

SR BIT SET	YIELD
0	INHIBIT ERROR HALTS.
1	INHIBIT ERROR PRINTOUTS.
2	BELL ON ERROR.
3	LOOP 1.
4	LOOP 2.
5	LOOP 3.
9	INHIBIT RUNNING TEST0.
*10	TAKE ERROR CONTINUE EXIT.
11	LOOP ON CONTROL TEST (EXCEPT TEST0).

*SETTING SR10 MAY RESULT IN MISLEADING ERROR PRINTOUTS OCCURRING AFTER THE FIRST ERROR PRINTOUT. THIS OPTION SHOULD BE USED ONLY AFTER THE PROGRAM LISTING HAS BEEN CONSULTED TO DETERMINE THE CONSEQUENCES. (ERROR 6E IS NOT AFFECTED BY THIS OPTION.)

5.1.2 METHOD OF PLOTTER SPECIFICATION INPUT

WHEN REQUESTED BY THE INIT ROUTINE, RESPOND BY TYPING THE FOLLOWING NUMERIC CODES WHICH APPLY TO YOUR SITUATION TO PROVIDE PLOTTER SPECIFICATIONS TO THE PROGRAM.

"PLOTTER SERIES (1-3):"

1 = 500 SERIES OR HOUSTON DP10/EDP10
2 = 600 SERIES
3 = 700 SERIES

"PLOTTER TYPE (1-4):"

1 = 11 INCH DRUM OR HOUSTON DP10/EDP10
2 = 28.55 INCH DRUM
3 = 31 BY 54 INCH FLATBED
4 = 54 BY 72 INCH FLATBED

"VECTOR TYPE (1-2):"

1 = 8 VECTOR
2 = 24 VECTOR

"INCREMENT TYPE (1-5):"

1 = 0.1 MM
2 = .00125 INCH
3 = .0025 INCH
4 = .005 INCH
5 = .01 INCH

"CORRECT (Y OR N)?"

Y = YES.
N = NO.

5.1.3 NORMAL HALTS

PC	CAUSE
204	HALT AFTER INIT.
207	HALT AFTER TEST0.
212	HALT AFTER TEST1.
214	HALT AFTER TEST2.
216	HALT AFTER TEST3.
220	HALT AFTER TEST4.
222	HALT AFTER TEST5.
224	HALT AFTER TEST6.

5.2 DISPLAY TEST

CAUTION

PRIOR TO RUNNING THE CONTROL AND DISPLAY TESTS, ENSURE THAT ALL PLOTTER MANUAL COMMAND SWITCHES ARE IN THEIR NEUTRAL POSITION.

5.2.1 MONITOR COMMANDS

THE DISPLAY TEST IS CONTROLLED BY A MONITOR WHICH RESPONDS TO THE FOLLOWING COMMANDS FROM THE TTY. TO RUN COMPLETE DISPLAY TEST, POSITION PEN TO A POINT AT LEAST 2 INCHES FROM ANY PHYSICAL STOP AND TYPE "CD~".

NOTE: "~" INDICATES RETURN KEY.

"[" INDICATES ALTMODE.

"N" INDICATES ANY OCTAL NUMBER.

TYPING RUBOUT ABORTS THE COMPLETE COMMAND STRING.

COMMAND	RESULT
"CD~"	DRAW COMPLETE DISPLAY TEST
"CD;PNN~"	DRAW PATTERN NN ONCE - LIMITED TO ONE PATTERN
"CD;PNNLNN;PNNLNN~"	DRAW PATTERN NN LINE NN FOLLOWED BY PATTERN NN LINE NN - LIMITED TO 10 PATTERN/LINE COMBINATIONS.
"CL~"	LOOP ON COMPLETE DISPLAY TEST.
"CL;PNN~"	LOOP ON DRAWING PATTERN NN - LIMITED TO ONE PATTERN.
"CL;PNNLNN;PNNLNN~"	LOOP ON DRAWING PATTERN NN LINE NN FOLLOWED BY PATTERN NN LINE NN - LIMITED TO 10 PATTERN/LINE COMBINATIONS.
*"EC~"	CONTINUE EXECUTING PREVIOUS COMMAND.
*"ECL~"	CONTINUE EXECUTING PREVIOUS COMMAND AND UNCONDITIONALLY LOOP.
*"ECD~"	CONTINUE EXECUTING PREVIOUS COMMAND AND STOP LOOPING.
"ET~"	TERMINATE DISPLAY TEST MONITOR AND RETURN TO CONTROL TEST EXECUTIVE STARTING AT TEST1.
"EI~"	TERMINATE DISPLAY TEST MONITOR, GO TO INIT FOR NEW SPECIFICATION INPUT, AND RETURN TO DISPLAY MONITOR WHEN COMPLETED.

(5.2.1 CONT'D)

COMMAND

RESULT

CAUTION

IF THE "CM+" COMMAND OPTION IS UTILIZED, ENSURE THAT NO
CONFLICTING COMMANDS ARE ENTERED INTO THE DIRECTION REGISTER
VIA THE SWITCH REGISTER.

**"CM+" ENABLE USER TO COMMAND PLOTTER FROM
 THE SWITCH REGISTER USING THE FOLLOW-
 ING SETTINGS!

SR BIT

YIELD

0

M STRING HOLD. SET THIS BIT WHEN
CHANGING SR SETTING.

5

0 = PEN UP; 1 = PEN DOWN

6-11

DETERMINE STATE OF DIRECTION REGISTER
FLOPS. CONSULT DIRECTION CONSTANT
LISTS FOR EFFECT ON YOUR PARTICULAR
PLOTTER CONFIGURATION.

REFER TO TABLE 2 FOR A LIST OF THE DIRECTION CONSTANTS.

* "EC+"; "ECD+"; "ECL+" GENERATE A SYNTAX ERROR "?*" IF NO
"D" OR "L" TYPE COMMANDS ARE PENDING.

** "CM+" CAN BE EXITED ONLY BY STRIKING THE ALTMODE KEY.
CARE SHOULD BE EXERCISED TO ENSURE THAT CONFLICTING COM-
MANDS ARE NOT ENTERED VIA THE SR.

5.2.2 MONITOR TYPEOUTS

TYPEOUT

CAUSE

"*" (BELL)

NO COMMANDS PENDING; INPUT A COMMAND.

"?*" (BELL)

SYNTAX ERROR; CHECK YOUR PREVIOUS
INPUT FOR ERRORS AND INPUT AGAIN IN
CORRECT FORMAT.

5.2.3 MONITOR SYNTAX CHECKING

MONITOR CHECKS ALL INPUTS FOR CORRECT SYNTAX. SPACES ARE
ALLOWED AT ANY POINT IN THE FORMAT WITH THE EXCEPTION OF THE
ALTMODE INPUT. IF A SYNTAX ERROR IS DETECTED, MONITOR TYPES
OUT "?*" AND THE WHOLE COMMAND STRING MUST BE REINSERTED IN
CORRECT FORMAT.

6. ERRORS

6.1 CONTROL TEST ERRORS

6.1.1 ERROR HALTS

ONLY TWO ERROR HALTS BEAR EXPLANATION SINCE THEY HAVE NO ACCOMPANYING TYPEOUTS.

PC ERROR

1012 THE SIGNAL "INITIALIZE" DOES NOT CLEAR THE PLOTTER FLAG, OR PLSF SKIPS WHEN FLAG IS CLEAR, DEPRESS START TO RETEST.

1027 THE SIGNAL "INITIALIZE" DOES NOT SET INTERRUPT ENABLE, OR FLAG SET AND INTERRUPT ENABLE SET DO NOT RESULT IN AN INTERRUPT REQUEST, DEPRESS START TO RETEST.

6.1.2 ERROR HANGUP.

MA ERROR

1014/1015 PLLR WITH AC=00 DOES NOT SET PLOTTER FLAG OR PLSF DOES NOT SKIP WHEN FLAG IS SET, TO RECOVER, INHIBIT TEST0 AND START FROM 200.

6.1.3 ERROR PRINTOUTS.

DURING THE CONTROL TEST, IF AN ERROR OCCURS A GENERAL ERROR HEADING IS DISPLAYED AND SPECIFIC ERROR INFORMATION IS INSERTED UNDER THE APPLICABLE HEADING. THE HEADING AND RELATED DEFINITIONS ARE AS FOLLOWS:

"ERROR" ERROR CODE FILL IN.

"PC" ADDRESS AT WHICH THE ERROR WAS DETECTED.

"DR" THE DATA WHICH SHOULD HAVE BEEN LOADED INTO THE DIRECTION REGISTER AT THE TIME THE ERROR OCCURRED. (THIS IS DISPLAYED ONLY IF APPLICABLE TO THE ERROR.)

REFER TO TABLE 1 FOR CONTROL TEST ERROR PRINTOUT INTERPRETATION AND RECOMMENDED RECOVERY.

6.2 DISPLAY TEST ERRORS

ERRORS WHICH OCCUR DURING THE DISPLAY TEST ARE EVIDENCED ONLY BY NOTED DIFFERENCES BETWEEN THE PATTERNS DRAWN AND THOSE SHOWN IN FIGURE 1. ALL PORTIONS OF THE CONTROL MODULE EXCEPT FOR THE DIRECTION REGISTER SHOULD BE FUNCTIONING PROPERLY IF THE CONTROL TEST PASSED. IF THE PATTERN DEVIATIONS ARE SLIGHT, THE CAUSE IS PROBABLY DUE TO THE PLOTTER BEING OUT OF CALIBRATION. IF THE DEVIATIONS ARE GROSS, HOWEVER, CHANCES ARE THAT THE DIRECTION REGISTER IS NOT SETTING PROPERLY OR THAT A PORTION OF THE INTERFACE IS NOT FUNCTIONING AS IT SHOULD.

6.2.1 DISPLAY TEST ERROR RECOVERY

- A. ENSURE THAT THE PLOTTER SPECIFICATIONS SUPPLIED TO THE PROGRAM ARE CORRECT. IF NOT, COMMAND "CI+" AND REINITIALIZE, THEN REPEAT THE DISPLAY TEST.
- B. COMMAND "CT+" TO REPEAT THE CONTROL TEST TO ENSURE THAT THE MAJORITY OF THE CONTROL MODULE IS STILL FUNCTIONING PROPERLY.
- C. NOW ONE OF TWO METHODS CAN BE USED TO ISOLATE THE PROBLEM IF STEPS A AND B DO NOT.

- *C.1 USING THE "CD:PXXLXX+" OR "CL:PXXLXX+" FORMAT COMMANDS, SCOPE THE DR AND INTERFACE TO DETERMINE THE CAUSE OF THE FAILURE.

CAUTION

IF THE "CM+" COMMAND OPTION IS UTILIZED, ENSURE THAT NO CONFLICTING COMMANDS ARE ENTERED INTO THE DIRECTION REGISTER VIA THE SWITCH REGISTER.

- *C.2 USING THE "CM+" COMMAND, SET THE SWITCH REGISTER TO THE DESIRED DIRECTION CONSTANT AND SCOPE THE DR AND INTERFACE.

*NOTE: THE PLOTTER MAY BE TURNED OFF IF DESIRED, AND WILL NOT AFFECT THE OPERATION OF THE DR AND INTERFACE.

7. NORMAL PRINTOUTS

ALL NORMAL PRINTOUTS ARE SELF-EXPLANATORY.

8. RESTRICTIONS

CAUTION

PRIOR TO RUNNING THE CONTROL AND DISPLAY TESTS, ENSURE THAT ALL PLOTTER MANUAL COMMAND SWITCHES ARE IN THEIR NEUTRAL POSITION.

CAUTION

IF THE "CM-" COMMAND OPTION IS UTILIZED, ENSURE THAT NO CONFLICTING COMMANDS ARE ENTERED INTO THE DIRECTION REGISTER VIA THE SWITCH REGISTER.

9. MISCELLANEOUS.

9.1 EXECUTION TIME

9.1.1 CONTROL TEST

EACH PASS OF THE CONTROL TEST TAKES APPROXIMATELY 20 SECONDS.

9.1.2 DISPLAY TEST

THE EXECUTION TIME FOR THE DISPLAY TEST IS DEPENDENT ON THE SPECIFICATIONS OF THE PLOTTER BEING TESTED. IN GENERAL, EXECUTION TIME WILL VARY DIRECTLY WITH PLOTTER PHYSICAL SIZE AND INVERSELY WITH INCREMENT SIZE.

10. PROGRAM DESCRIPTION

10.1 CONTROL TEST

THE CONTROL TEST IS CONTROLLED BY AN EXECUTIVE ROUTINE (EXEC) WHICH TREATS EACH TEST AS A SUBROUTINE. EACH COMPLETE RUN OF THE EXECUTIVE INCLUDES 10 COMPLETE PASSES OF THE CONTROL TEST. THE END OF EACH PASS IS INDICATED BY THE TTY BELL. EACH CONTROL TEST PASS INCLUDES:

TEST 0 RUN ONCE ON STARTUP ONLY.

TESTS 1 THROUGH 6 10 TIMES EACH

AFTER A COMPLETE EXECUTIVE RUN, "CONTROL TEST COMPLETE" IS DISPLAYED. CONTROL IS THEN PASSED TO THE DISPLAY TEST MONITOR.

10.2 DISPLAY TEST

THE METHODS USED IN THE DISPLAY TEST ARE AMPLY DESCRIBED IN THE PROGRAM LISTING.

10.3 PATTERN LISTS

EACH PATTERN CONSISTS OF A LIST OF LINE IDENTIFIERS (LID) WHICH ARE INTERPRETED BY THE "LGEN" ROUTINE. THE STRUCTURE OF THESE LID'S AND THE GENERAL METHOD OF INTERPRETATION ARE DESCRIBED IN THE PROGRAM LISTING.

KEY TO TABLE 1 (CONTROL TEST ERROR RECOVERY TABLE)

ERROR: NX N INDICATES THE TEST WHICH FIRST CHECKS FOR THIS ERROR CONDITION (PRIMARY TEST). X INDICATES THE APPROXIMATE LOGICAL SEQUENCE WITHIN THAT TEST (IN MOST CASES).

PC: NNNN ADDRESS AT WHICH PROGRAM BRANCHED TO ERROR ROUTINE.

DR: NN CONTENTS OF AC (AND THEREFORE SHOULD BE CONTENTS OF DR) AFTER PLLR OR CFLR IOT'S.

TST: N TEST IN WHICH ERROR WAS DETECTED.

FAILURE: GENERAL ERROR DESCRIPTION.

RECOVERY: LOOP N (SCOPING PRIOR TO LOOPING MAY BE BENEFICIAL IN SOME CASES.) SET LOOP N SR BIT AND DEPRESS CONTINUE. SR BITS 0, 1 AND 2 MAY BE SET IN ANY CONFIGURATION DESIRED. THIS PROCEDURE WILL RESULT IN THE ERROR EXITING TO THE PRIMARY TEST REGARDLESS OF WHERE THE ERROR IS DETECTED. IF DESIRED, SR BIT 10 MAY BE SET WHICH RESULTS IN THE ERROR EXITING TO "PC"+1. IN THIS CASE THE SPECIFIED "LOOP N" DOES NOT NECESSARILY APPLY.

ERROR	PC	DR	TST	FAILURE	RECOVERY
1A	1035	--	1	PLCF DOES NOT CLEAR FLAG OR PLSF SKIPS WHEN FLAG CLEAR.	LOOP1 (CYCLE IN TEST1 ON PLPU-PLCF CHECK)
	1053				
	1075				
	1120		2		
	1136				
	1160				
	1212		3		
	1464		6		
	1505				
	1542				

TABLE 1 (CONT'D)

ERROR	PC	DR	TST	FAILURE	RECOVERY
1B	1043	--	1	PLPU IOT SETS FLAG PRIOR TO 50MS.	LOOP1 (CYCLE IN TEST 1 ON PLPU-PLCF CHECK.)
1C	1047	--	1	PLPU IOT DOES NOT SET FLAG IN 95MS.	LOOP1 (CYCLE IN TEST 1 ON PLPU-PLCF CHECK.)
1D	1065	--	1	PLPD IOT SETS FLAG PRIOR TO 50MS.	LOOP2 (CYCLE IN TEST 1 ON PLPD-PLCF CHECK.)
1E	1071	--	1	PLPD IOT DOES NO SET FLAG IN 95MS.	LOOP2 (CYCLE IN TEST 1 ON PLPD-PLCF CHECK.)
2A	1126	--	2	PLLR IOT WITH AC=31 SETS FLAG PRIOR TO 50MS.	LOOP1 (CYCLE IN TEST 2 ON PLLR31-PLCF CHECK.)
2B	1132	--	2	PLLR IOT WITH AC=31 DOES NOT SET FLAG IN 95MS.	LOOP1 (CYCLE IN TEST 2 ON PLLR31-PLCF CHECK.)
2C	1150	--	2	PLLR IOT WITH AC=32 SETS FLAG PRIOR TO 50MS.	LOOP2 (CYCLE IN TEST 2 ON PLLR32-PLCF CHECK.)
2D	1154	--	2	PLLR IOT WITH AC=32 DOES NOT SET FLAG IN 95MS.	LOOP2 (CYCLE IN TEST 2 ON PLLR32-PLCF CHECK.)
3A	1222	NN	3	PLLR IOT WITH AC=NN SETS FLAG PRIOR TO 5MS.	LOOP1 (CYCLE IN TEST 3 ON PRESENT PLLR NN CHECK.)
3B	1226	NN	3	PLLR IOT WITH AC=NN DOES NOT SET FLAG IN 10MS.	LOOP1 (CYCLE IN TEST 3 ON PRESENT PLLR NN CHECK.)

TABLE 1 (CONT'D)

ERROR	PC	DR	TST	FAILURE	RECOVERY
4A	1274	--	4	CFLR IOT WITH AC=31 SETS FLAG PRIOR TO 50MS.	LOOP1 (CYCLE IN TEST 4 ON CFLR31 CHECK.)
4B	1300	--	4	CFLR IOT WITH AC=31 DOES NOT SET FLAG IN 95MS.	LOOP1 (CYCLE IN TEST 4 ON CFLR31 CHECK.)
4C	1317	--	4	CFLR IOT WITH AC=32 SETS FLAG PRIOR TO 50MS.	LOOP2 (CYCLE IN TEST 4 ON CFLR32 CHECK.)
4D	1323	--	4	CFLR IOT WITH AC=32 DOES NOT SET FLAG IN 95MS.	LOOP2 (CYCLE IN TEST 4 ON CFLR32 CHECK.)
4E	1266	--	4	CFLR IOT DOES NOT INITIALLY CLEAR FLAG OR PLSF SKIPS WHEN FLAG IS CLEAR.	LOOP1 (CYCLE IN TEST 4 ON CFLR31 CHECK.)
	1311				
	1413		5		
5A	1420	NN	5	CFLR IOT WITH AC=NN SETS FLAG PRIOR TO 5MS.	LOOP1 (CYCLE IN TEST 5 ON PRESENT CFLR NN CHECK.)
5B	1424	NN	5	CFLR IOT WITH AC=NN DOES NOT SET FLAG IN 10MS.	LOOP1 (CYCLE IN TEST 5 ON PRESENT CFLR NN CHECK.)
6A	1501	--	6	INTERRUPT DOES NOT OCCUR AFTER PLSE IOT AND FLAG SET.	LOOP1 (CYCLE IN TEST 6 ON PLSE CHECK.)

TABLE 1 (CONT'D)

ERROR	PC	DR	TST	FAILURE	RECOVERY
6B	1514	--	6	INTERRUPT OCCURS AFTER PLSE IOT AND FLAG IS CLEAR.	LOOP1 (CYCLE IN TEST 6 ON PLSE CHECK.)
6C	1536	--	6	INTERRUPT OCCURS AFTER PLCE IOT AND FLAG IS SET,	LOOP2 (CYCLE IN TEST 6 ON PLCE CHECK.)
6D	1551	--	6	INTERRUPT OCCURS AFTER PLCE IOT AND FLAG IS CLEAR.	LOOP2 (CYCLE IN TEST 6 ON PLCE CHECK.)
6E	1471	--	6	CFLR IOT WITH AC=0 DOES NOT SET FLAG IN 10MS.	LOOP1 (CYCLE IN TEST 6 ON PLSE CHECK.)
	1526				

KEY TO TABLE 2

DIRECTION CONSTANTS

COORDINATE
TERMINOLOGY:

+Y CORRESPONDS TO PEN LEFT.
+X CORRESPONDS TO DRUM DOWN.

DIRECTION
TERMINOLOGY:

N CORRESPONDS TO PEN LEFT,
E CORRESPONDS TO DRUM DOWN,
S CORRESPONDS TO PEN RIGHT,
W CORRESPONDS TO DRUM UP,
L CORRESPONDS TO HALF VECTORS
FOR 24 VECTOR PLOTTERS.

NOTE THAT DIRECTION CONSTANTS 5X AND 7X APPLY TO 24 VECTOR PLOTTERS ONLY.

ALL BLANKS ARE INTENTIONAL AND IMPLY THAT THE PARTICULAR FUNCTION IS NON-EXISTENT FOR THAT TYPE PLOTTER.

THE 500 SERIES PLOTTERS USE THE PLPU(6503) AND PLPD(6505) IOT'S FOR PEN COMMANDS VICE DIRECTION CONSTANTS.

DIRECTION CONSTANTS TABLE 2

**NOTE: 500 SERIES FORMAT ALSO APPLIES TO HOUSTON DP10/EDP10

COORDINATE TERMINOLOGY	DIRECTION TERMINOLOGY	500 SERIES	600 SERIES	700 SERIES	REMARKS
+Y	N	20	10	10	
+Y +X	NE	30	11	11	
+X	E	10	12	12	
-Y +X	SE	50	13	13	
-Y	S	40	14	14	
-Y -X	SW	44	15	15	
-X	W	04	16	16	
+Y -X	NW	24	17	17	
+Y +X/2	NNE		72	72	---
+Y/2 +X	ENE		70	70	2
-Y/2 +X	ESE		74	74	4
-Y +X/2	SSE		76	76	
-Y -X/2	SSW		77	77	V
-Y/2 -X	WSW		75	75	E
+Y/2 -X	WNW		71	71	C
+Y -X/2	NNW		73	73	T
+Y/2	LN		50	50	O
+Y/2 +X/2	LNE		51	51	R
+X/2	LE		52	52	
-Y/2 +X/2	LSE		53	53	O
-Y/2	LS		54	54	N
-Y/2 -X/2	LSW		55	55	L
-X/2	LW		56	56	Y
+Y/2 -X/2	LNW		57	57	---

DIRECTION CONSTANTS

TABLE 2 (CONT'D)

COORDINATE TERMINOLOGY	DIRECTION TERMINOLOGY	500 SERIES	600 SERIES	700 SERIES	REMARKS
PEN UP	(01)	PLPU	31	31	
PEN DOWN			32	32	
SYNC	(02)	PLPD	37	37	NOT EXERCISED IN DISPLAY TEST
START ZIP				33	NOT EXERCISED IN DISPLAY TEST
BLOCK CODE			34	34	NOT EXERCISED IN DISPLAY TEST
PLOT CODE			35	35	NOT EXERCISED IN DISPLAY TEST
START INCR				36	NOT EXERCISED IN DISPLAY TEST

K TO FIGURE 1.

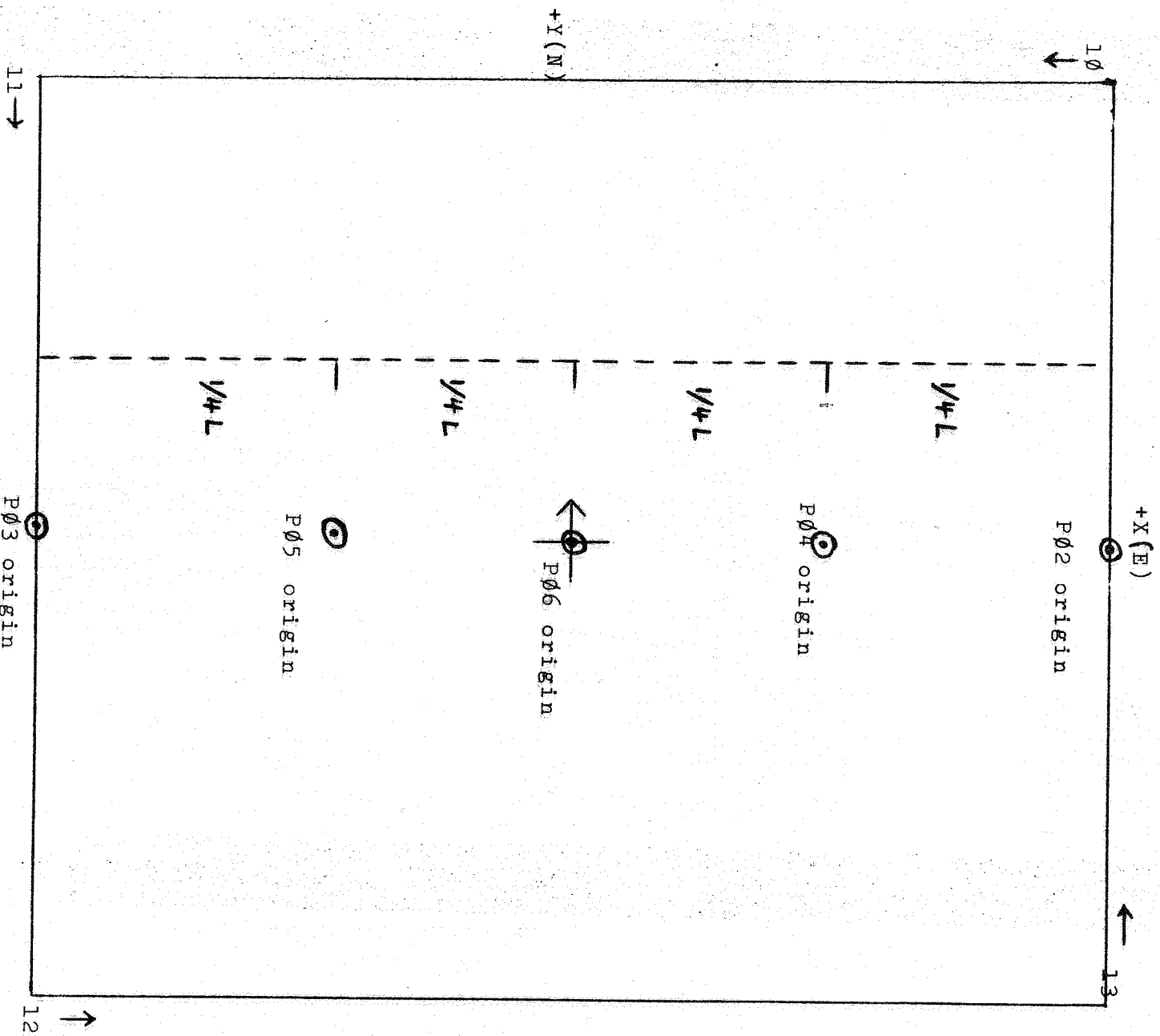
FIGURE 1 CONTAINS A REASONABLE FACSIMILE OF PATTERNS 01 THROUGH 06 (P01--P06). ONLY THE NECESSARY DIMENSIONS ARE LISTED.

ALL PATTERN NUMBERS AND LINE NUMBERS CONSIST OF TWO OCTAL DIGITS. LINES WHICH ARE TRACED WITH THE PEN UP ARE LISTED UNDER EACH PATTERN ALONG WITH THEIR PARTICULAR FUNCTION. ARROWS INDICATE THE DIRECTION IN WHICH THE LINES ARE DRAWN.

IF IT IS DESIRED TO UTILIZE P07, NOT SHOWN IN THIS FIGURE, CONSULT THE P07 LIST AT THE END OF THE PROGRAM LISTING FOR LINE FUNCTIONS.

P01 (8 and 24 vector)

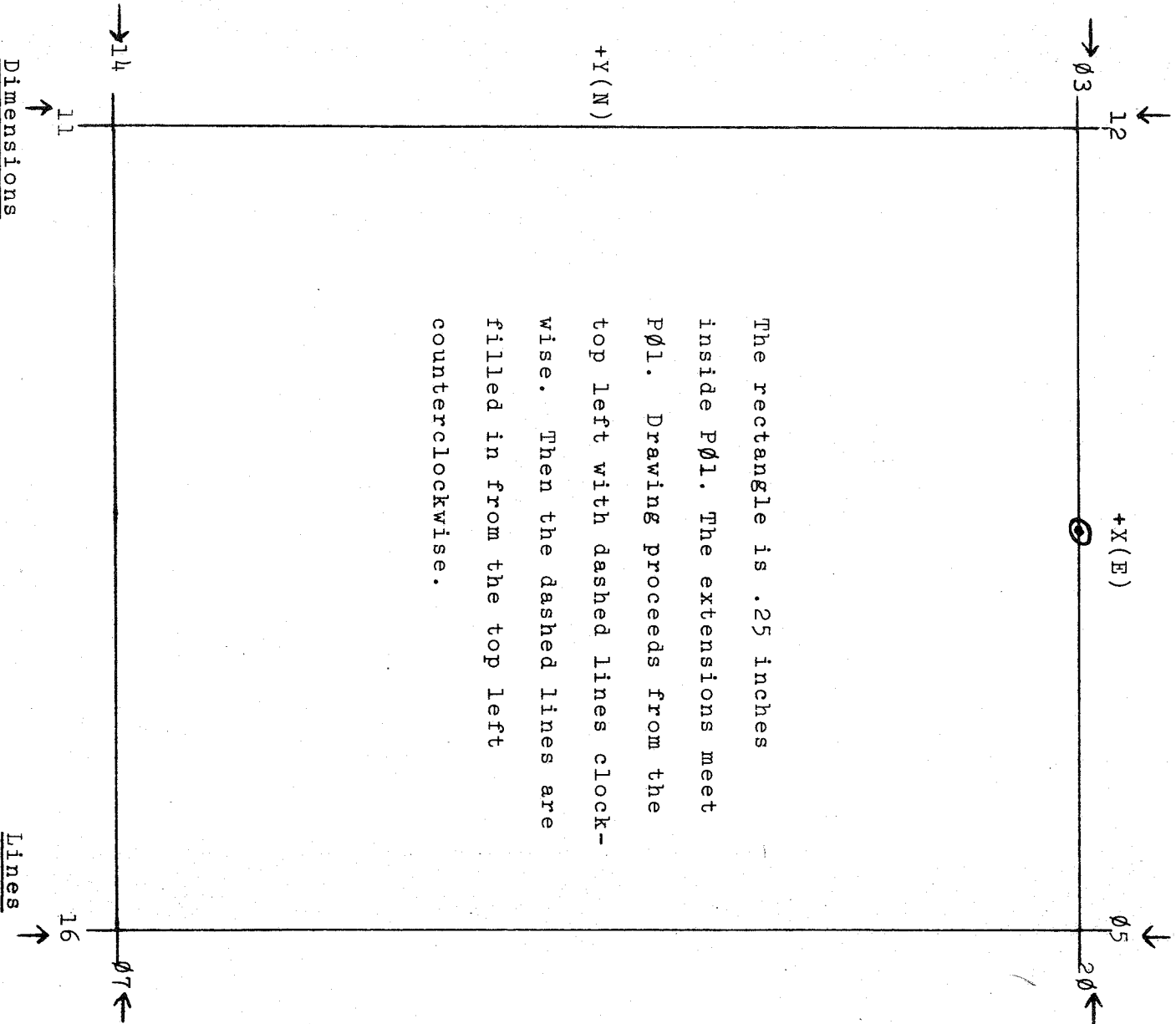
22



Plotter	Pattern Dimension	Lines
11 inch	10 x 30	01-07 orient to top left corner.
28.55 inch	27 x 32	14-15 orient to center of crosshairs
31 x 54 inch	30 x 53	16-31 draw crosshairs and arrowhead
54 x 71	53 x 71	32-33 pointing pen left.
		32-33 orient to P02 origin.

Figure 1-1

.25 Inch Dashed Lines Filled In



The rectangle is .25 inches inside P01. The extensions meet P01. Drawing proceeds from the top left with dashed lines clockwise. Then the dashed lines are filled in from the top left counterclockwise.

Dimensions

Lines

Including extensions the dimensions are the same as those of P01.

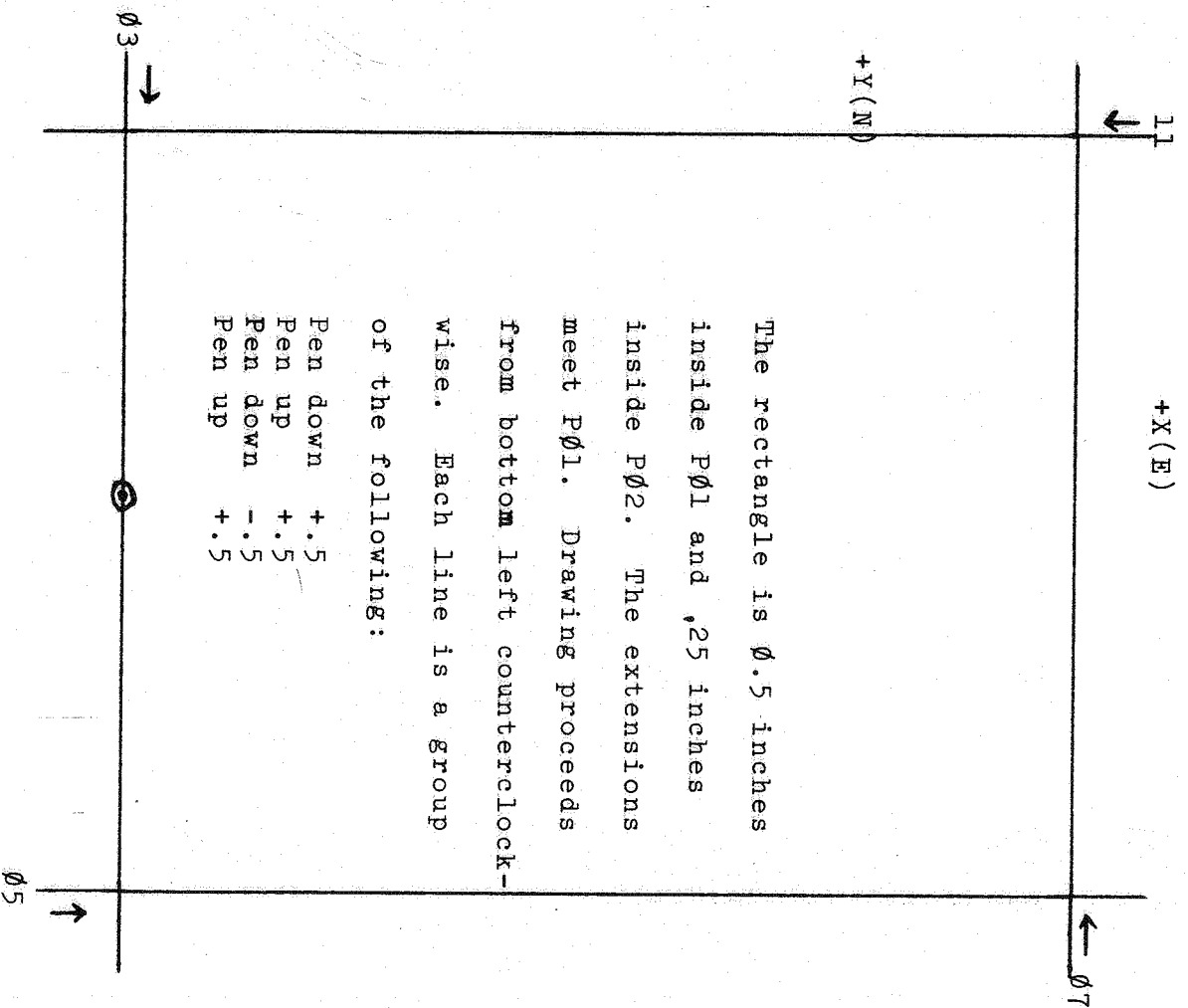
01-02 orient to start of 03.
04,06,10,13, 15, 17, 21all
orient to beginning of next line
22 orients to P02 origin.
23 orients to P03 origin.

Figure 1-2

P03 (8 and 24 Vector)

0.5 Inch Shuffle Lines

24



Dimensions

Including extensions the dimensions are the same as those of P01.

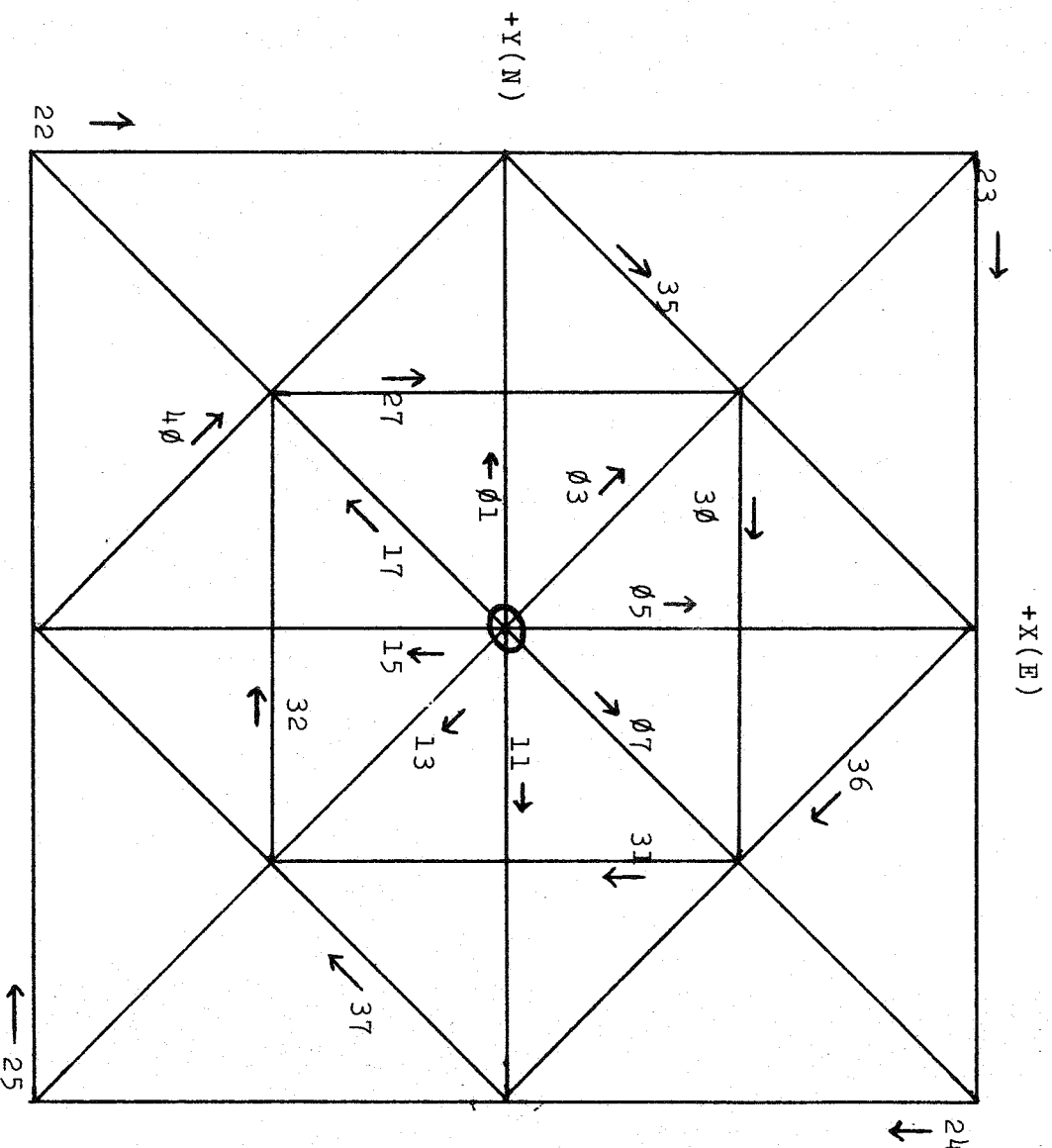
Lines

01-02 orient to start of 03.
04,06,10,12 orient to start of next line.
13 orients to P03 origin.
14 orients to P04 origin.

Figure 1-3

P04 (8 Vector only)

All Vector with Concentric and
Oblique Squares



Dimensions

The outer square is
8 inches and is not
dependent on plotter size.

Lines

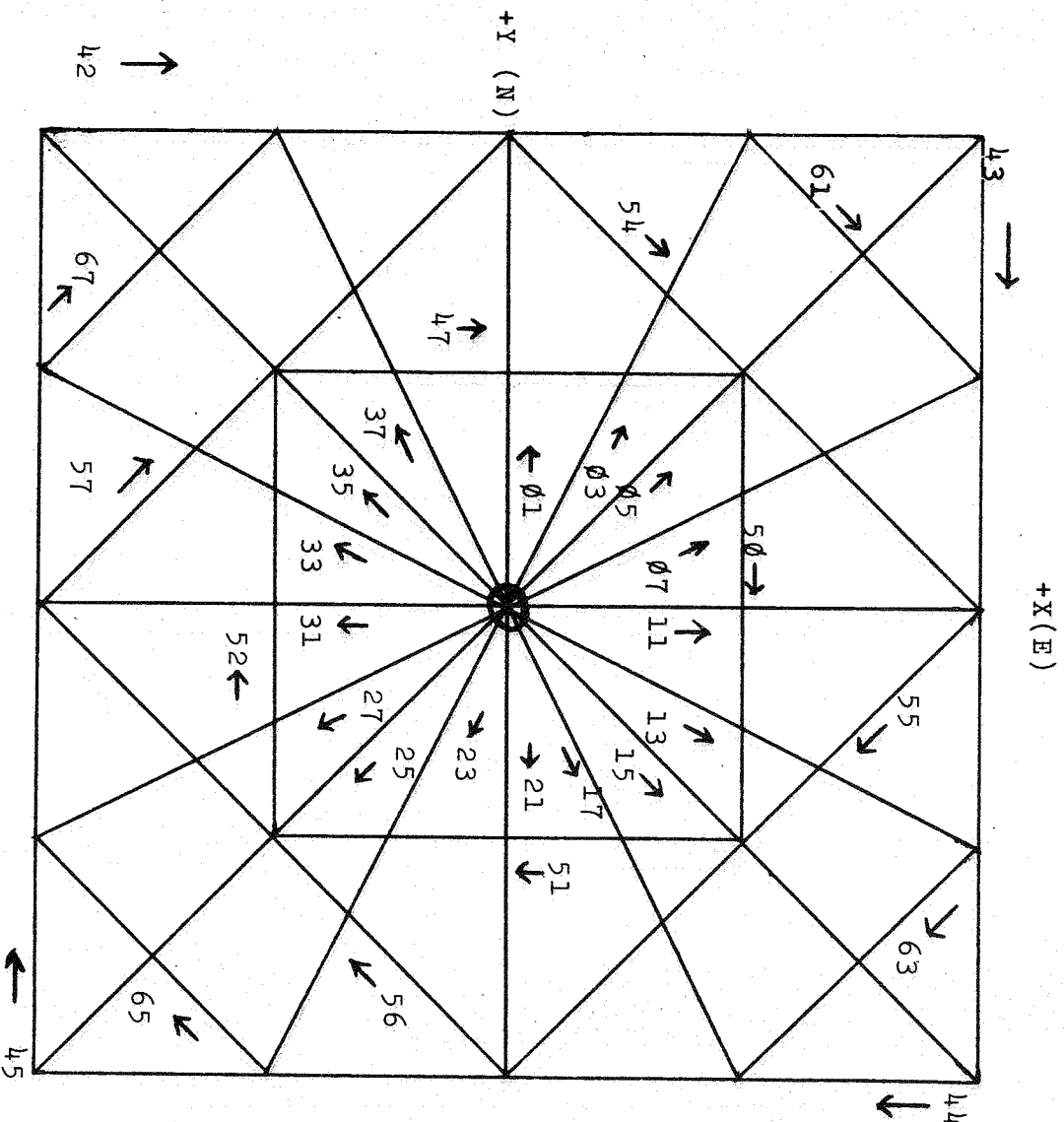
02,04,06,10,12,14, 16 and 20 are
returns to origin after each
preceding line.
21 orients to start of 22.
26 orients to start of 27.
33-34 orients to start of 35.
41 orients to P04 origin.
42 orients to P05 origin.

Figure 1-4-8

P04 (24 Vector only)

All Vectors with Concentric and Oblique Squares

26



Dimensions

The outer square is
8 inches and is not
dependent on plotter
size.

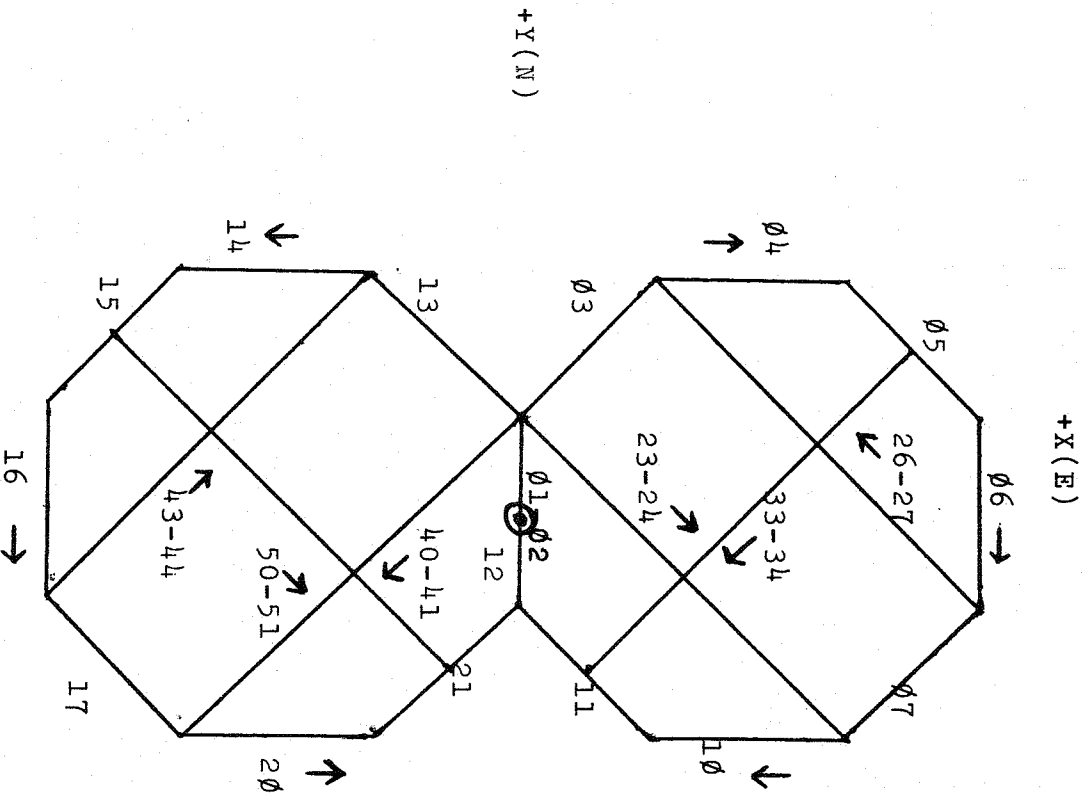
Lines

02,04,06,10,12,14,16,20,22,24,26,30
32,34,36, 40 all return to origin
after preceding line.
41 orients to start of 42.
46 orients to start of 47.
53 orients to start of 54.
60, 62,64,66 orient to 61, 63, 65,
67 respectively.
70-71 orient to P04 origin.
72 orients to P05 origin.

Figure 1-4-24

P05 (8 Vector only)

Joined Octagons with Diagonals



Dimensions

Diagonals are 6.0 inches.

Each octagon segment is 2 inches.

All dimensions are independent of plotter size

Lines

- 22 orients to 23.
- 25 orients to 26.
- 30-32 orient to 33.
- 35-37 orient to 40.
- 42 orients to 43.
- 45-47 orient to 50
- 52-55 orient to P05 origin.
- 56 orients to P06 origin

Figure 1-5-8

P05 (24 Vector only)

Joined 16 Sided Polygons with Diagonals.

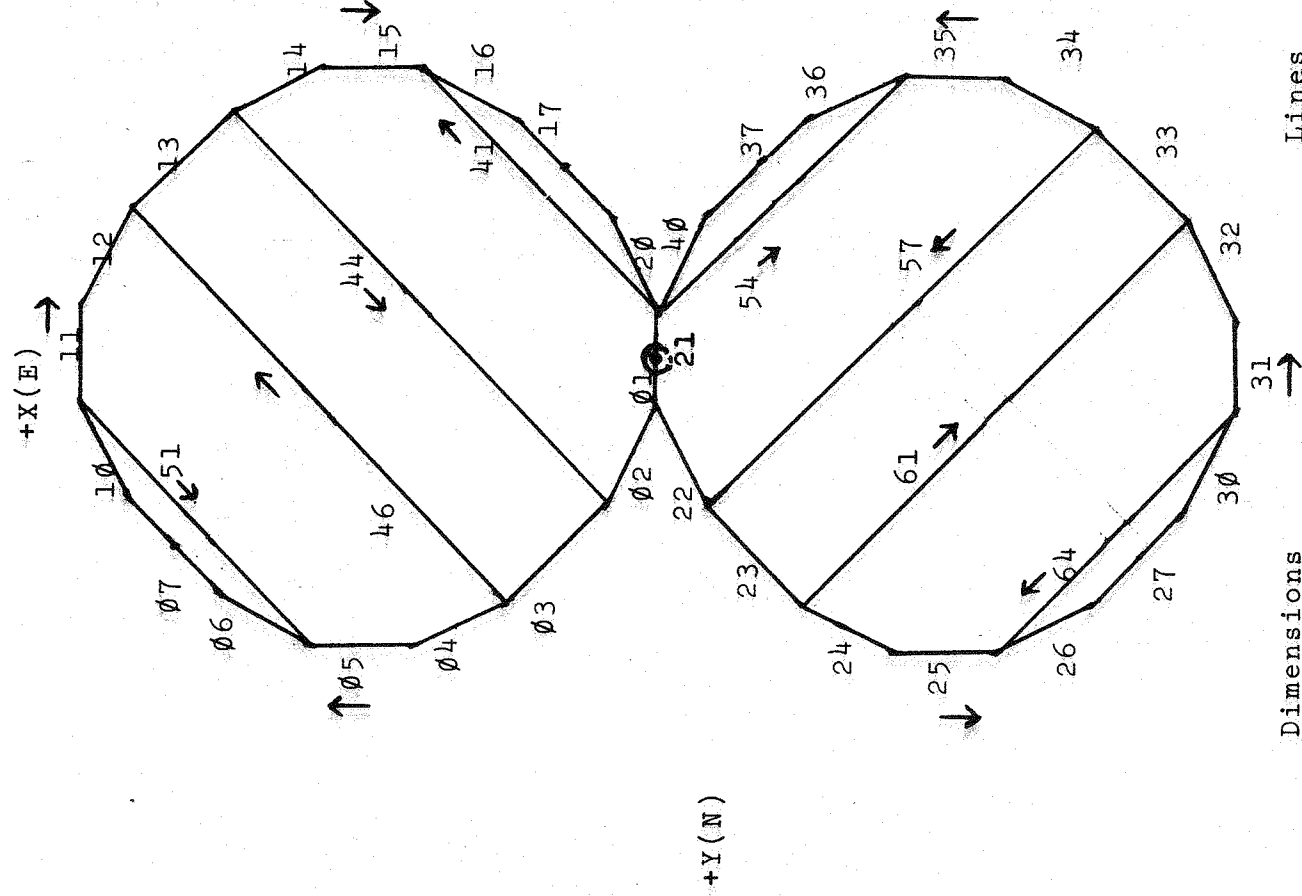
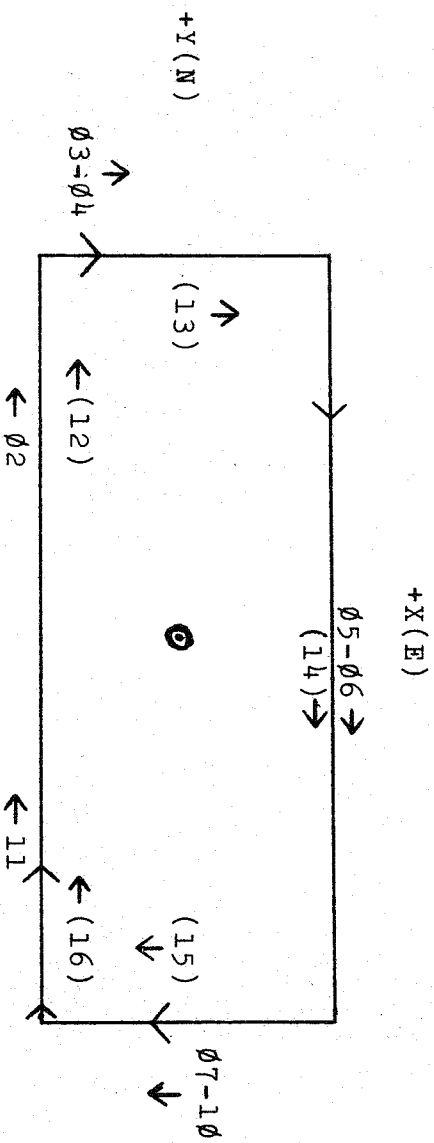


Figure 1-5-24

P06 (8 and 24 Vector).

Rapid Arrows on a Rectangle.



Dimensions

Rectangle = 6 x 2 inches.

Arrows are 0.1 inches apart and each arrow diagonal is 0.1 inches.

Dimensions are independent of plotter size.

Lines

Lines in parentheses refer to special LID's which generate arrows.

01 orients to 02.

17 orients to P06 origin.

20 orients to P03 origin.

Figure 1-6

PLC0D1.PA

/XY 8-E PLOTTER CONTROL AND DISPLAY DIAGNOSTIC PROGRAM.
/TAPE 1.

/TEST OF THE XY 8-E PLOTTER CONTROL AND THE SERIES 500
/THROUGH 700 CALCOMP PLOTTER CONNECTED.

/COPYRIGHT 1970-1971, DIGITAL EQUIPMENT CORPORATION, MAYNARD, MASS.

/THIS PROGRAM IS DIVIDED INTO TWO MAJOR PORTIONS:

/1. THE CONTROL TEST WHICH CHECKS THE XY 8-E PLOTTER CONTROL
/MODULE (WHETHER A PLOTTER IS CONNECTED OR NOT). THE CONTROL TEST
/CONSISTS OF:

/A. TEST0 - TEST OF ABILITY OF THE SIGNAL "INITIALIZE"
/TO CLEAR THE PLOTTER FLAG AND SET INTERRUPT ENABLE.
/B. TEST1 - FLAG TEST FOR PLPU AND PLPD IOT'S.
/C. TEST2 - FLAG TEST FOR PLLR WITH AC=31 OR 32.
/D. TEST3 - FLAG TEST FOR PLLR WITH AC NOT 31, 32, 33.
/E. TEST4 - FLAG TEST FOR CFLR WITH AC=31 OR 32.
/F. TEST5 - FLAG TEST FOR CFLR WITH AC NOT 31, 32, 33.
/G. TEST6 - INTERRUPT TEST.

/2. THE DISPLAY TEST WHICH CHECKS XY 8-E/PLOTTER INTERACTION
/AND PLOTTER ACCURACY BY DISPLAYING PRE-MEASURED PATTERNS.
/ALL PARTICULARS ARE DESCRIBED IN THE DOCUMENTATION AND
/IN THE LISTING.

/SWITCH REGISTER OPTIONS (EFFECTIVE EXCEPT DURING DISPLAY TEST).

/	/SRBIT SET	YIELD
/	0	INHIBIT ERROR HALTS (EXCEPT TEST 0 ERRORS).
/	1	INHIBIT ERROR PRINTOUTS.
/	2	BELL ON ERROR.
/	3	LOOP1.
/	4	LOOP2.
/	5	LOOP3.
/	6	NONE
/	7	NONE
/	8	NONE
/	9	INHIBIT RUNNING TEST 0.
/	10*	TAKE ERROR CONTINUE EXIT.
/	11	LOOP ON CONTROL TEST (EXCEPT TEST 0).

/*IF SR BIT 10 IS SET ALL ERRORS EXCEPT ERROR 6E
/EXIT TO TRACE +1. SETTING THIS BIT MAY RESULT IN MEANINGLESS ERROR
/PRINTOUTS. CONSULT DOCUMENTATION FOR USE OF THIS SWITCH FOR ERROR
/RECOVERY.

/STARTING ADDRESSES:

/	/ADDRESS	YIELD
/	200	RUN COMPLETE DIAGNOSTIC.

0346 2400
0347 3516
0350 4000

DISPLA
LGEN
LINCNT

/DISPLAY TEST
/MONITOR
/LINE GENERATOR (INTERPRETER)
/LINE COUNTING ROUTINE

0351 4400

SPLCE

/SUBROUTINES
/PLOT IOT SUBROUTINES. CHANGE THE DEVICE
/CODES IN THESE IOT SUBROUTINES IF YOUR
/SYSTEM DEVICE CODE IS OTHER THAN 50.
/PEN COMMAND SUBROUTINES
/GENERAL SUBROUTINES

0352 4054
0353 4140

CPUS
DSZ

0354 4600

MSG1

/MESSAGES

0355 5257
0356 5404
0357 5457
0360 5622

WLP
OKAYS
LGTHP
REPK

/PLOTTER RELATED DATA
/PLOTTER DIMENSION CONSTANTS
/DIRECTION REGISTER CONSTANTS
/LENGTH CONSTANTS
/REPETITION CONSTANTS

0361 5711
0362 6271
0363 6721

PAT18
PAT124
PAT07

/PATTERN LISTS
/8 VECTOR PATTERNS
/24 VECTOR PATTERNS
/SPECIAL CALL PATTERN LIST

/PLOTTER IOT DEFINITIONS:

4420 PLCE=JMS I SPLCEP
4421 PLSF=JMS I SPLSFP
4422 PLCF=JMS I SPLCFP
4423 PLPU=JMS I SPLPUP
4424 PLLR=JMS I SPLLRP
4425 PLPD=JMS I SPLPDP
4426 CFLR=JMS I SCFLRP
4427 PLSE=JMS I SPLSEP
6007 OAF=6007

/CLEAR INTERRUPT ENABLE
/SKIP IF PLOT FLAG = 1.
/CLEAR PLOT FLAG.
/PEN UP (500 SERIES ONLY).
/LOAD DR, SET FLAG.
/PEN DOWN (500 SERIES ONLY).
/CLEAR FLAG, LOAD DR, SET FLAG.
/SET INTERRUPT ENABLE.
/"PDP8/E "INITIALIZE".

0001 0001 *0001
0001 0001 JMP 1
0002 0002 2
0003 0003 3
0010 0010 *0010
0010 0000 AUTO0, 0
0011 0000 AUTO1, 0
0012 0000 AUTO2, 0
0013 0000 AUTO3, 0
0014 0000 AUTO4, 0
0015 0000 AUTO5, 0
0016 0000 AUTO6, 0

/SET UP FOR HIGH SPEED DUMP.
/CHANGED FOR INTERRUPT HANDLING BY PROGRAM.

/AUTO INDEXERS.

0017 0000 AUTO7, 0 /MSGN USES.

0020 #20 /POINTERS

/XY8E IOT SUBROUTINE POINTERS.

0020 4400 SPLCEP, SPLCE
 0021 4404 SPLSFP, SPLSF
 0022 4411 SPLCFP, SPLCF
 0023 4415 SPLPUP, SPLPU
 0024 4421 SPLLRP, SPLLR
 0025 4425 SPLPOD, SPLPD
 0026 4431 SCFLRP, SCFLR
 0027 4435 SPLSEP, SPLSE
 0030 4165 CRLFP, CRLF
 0031 4232 MSGNP, MSGN
 0032 4224 BELLP, BELL
 0033 4217 READNP, READN
 0034 4140 DSZP, DSZ
 0035 4156 TYPEP, TYPE
 0036 4274 DELNMP, DELNMS
 0037 1600 SETIPP, SETIP
 0040 1633 ERR1AP, ERR1A
 0041 2102 ERR4EP, ERR4E
 0042 2237 ERRORP, ERRORS
 0043 2353 QCONT, QCONT

/GENERAL PEN (BEAM) SUBROUTINE POINTERS. DEFINED BY SERIES PLOTTER
 /CONNECTED.

0044 0044 PENS, .
 0045 0000 CPUP, 0 /PEN UP OR BEAM OFF. (NO FLAG SENSE).
 0046 0000 CPDP, 0 /PEN DOWN OR BEAM ON. (NO FLAG SENSE.)
 0047 0000 DPUP, 0 /PEN UP OR BEAM OFF. (FLAG SENSE.)
 0050 0000 DPDP, 0 /PEN DOWN OR BEAM ON. (FLAG SENSE.)

0051 0000 SAVEAC, 0
 0052 0000 SAVEL, 0
 0053 0000 ALLSET, 0
 0054 0000 LOPFLG, 0
 0055 0000 PENFLG, 0
 0056 0000 FLAG24, 0
 0057 0000 FLGZIP, 0
 0060 0000 ACTFLG, 0
 0061 0000 MSGFLG, 0
 0062 0000 INTFLG, 0
 0063 0000 BUFF1, 0
 0064 0000 BUFF2, 0
 0065 5555 BUFF3, 5555
 /
 /

0066 0006 K6, 6

```

0067 0100 K100, 100
0070 0400 K400, 400
0071 0004 K4, 4
0072 0002 K2, 2
0073 0001 K1, 1
0074 0005 K5, 5
0075 0012 K12, 12
0076 0007 K7, 7
0077 0031 K31, 31
0100 0032 K32, 32
0101 0200 K200, 200
0102 0077 K77, 77
0103 0074 K74, 74
0104 0024 K24, 24
0105 7774 M4, -4
0106 7770 M10, -10
0107 7776 M2, -2
0110 7754 M24, -24
0111 7766 M12, -12
0112 7747 M31, -31
0113 7700 M100, -100
0114 5402 KJMP12, 5402
0115 0062 K70MIN, 62
0116 0055 K70MAX, 55
0117 0005 K5MIN, 5
0120 0005 K5MAX, 5
0121 0010 K5ALL, 10

```

```

/LOWER TOLERANCE FOR 70 MS.
/ADDITIVE FOR 70 MS UPPER TOLERANCE.
/LOWER TOLERANCE FOR 5MS.
/ADDITIVE FOR 5MS UPPER TOLERANCE.
/MAX TIME FOR 5MS ONE SHOT.

```

/ROUTINE TO CHANGE DEVICE CODES.

```

0122 0000 DEVCOD, 0
0123 7604 LAS
0124 0156 AND K770
0125 3010 DCA AUTO0
0126 1145 TAD DEVCOP
0127 3000 DCA 0
0130 1106 TAD M10
0131 3011 DCA AUTO1
0132 1400 TAD I 0
0133 3160 DCA DEVCOT
0134 1560 TAD I DEVCOT
0135 0137 AND K7007
0136 1010 TAD AUTO0
0137 3560 DCA I DEVCOT
0140 2000 ISZ 0
0141 2011 ISZ AUTO1
0142 0132 JMP .-10
0143 1010 TAD AUTO0
0144 5522 JMP I DEVCOD
0145 0146 DEVCOP, .+1
0146 4401 SPLCE+1
0147 4405 SPLSF+1
0150 4412 SPLCF+1
0151 4416 SPLPU+1
0152 4422 SPLLR+1

```

0153	4426	SPLPD+1
0154	4432	SCFLR+1
0155	4436	SPLSE+1
0156	0770	K770, 770
0157	7007	K7007, 7007
0160	0000	DEVCOT, 0
0161	7550	K818L, -230
0162	4320	M304P, M304

0200 *200
/JUMP TABLE.

0200	5230	JUMTAB, JMP	EXEC	/RUN COMPLETE DIAGNOSTIC.
0201	5236	JMP	EXEC2	/DITTO WITHOUT TST0 AND INIT.
0202	4777	JMS	INIT	/INIT AND HALT.
0203	7402	HLT		
0204	5776	JMP	DISPLA	/DISPLAY TEST MONITOR.
0205	4775	JMS	TST0	/TEST 0 AND HALT.
0206	7402	HLT		
0207	5205	JMP	.-2	/REPEAT TEST 0.
0210	4774	JT1, JMS	TST1	/TEST 1 AND HALT
0211	7402	HLT		
0212	4773	JMS	TST2	/TEST 2 AND HALT.
0213	7402	HLT		
0214	4772	JMS	TST3	/TEST 3 AND HALT.
0215	7402	HLT		
0216	4771	JMS	TST4	/TEST 4 AND HALT
0217	7402	HLT		
0220	4770	JMS	TST5	/TEST 5 AND HALT
0221	7402	HLT		
0222	4767	JMS	TST6	/TEST 6 AND HALT
0223	7402	HLT		
0224	5210	JMP	JT1	/GO TO TEST 1 AND HALT.
0225	4122	JMS	DEVCOD	/CHANGE DEVICE CODES.
0226	7402	HLT		
0227	5226	JMP	.-1	

/XY8E DIAGNOSTIC PROGRAM EXECUTIVE.
/1. RUNS COMPLETE CONTROL TEST (MAKES 10 COMPLETE PASSES.)
/THE CONTROL TEST CONSISTS OF THE FOLLOWING (1 PASS)
/TEST 0 ONLY ONCE (RUN ON FIRST PASS ONLY)
/TESTS 1 THRU 6 10 TIMES EACH.
/2. RUNS DISPLAY TEST UNTIL "CT=" COMMAND AT WHICH POINT
/STEP 1 IS REPEATED WITHOUT TEST 0, AND SO ON.
/3. THE TTY BELL SOUNDS ON EACH PASS OF THE CONTROL TEST.
/4. SR11 SET LOOPS ON WHOLE CONTROL TEST (WITHOUT TEST 0).
/5. "CONTROL TEST COMPLETE" DISPLAYED AT END OF STEP 1.

0230	4766	EXEC, JMS	PATCH	
0231	4775	JMS	TST0	/RUN TEST 0 ONCE.
0232	4430	JMS I	CRLF	
0233	1320	TAD	MSG1P	/PRINT TITLE MSG.

0234	4431	JMS	I	MSGNP	
0235	4777	JMS		INIT	/GO TO INIT.
0236	4430	EXEC2,	JMS	I	CRLFP
0237	1321	TAD		MSG14P	/PRINT CONTROL TEST MSG.
0240	4431	JMS	I	MSGNP	
0241	4430	JMS	I	CRLFP	
0242	3061	DCA		MSGFLG	/CLEAR ERRMSG FLAG.
0243	1111	TAD		M12	/SET EXEC FOR 10 PASSES.
0244	3322	DCA		EXECNT	
0245	1111	EXEC1,	TAD	M12	
0246	3323	DCA		TSTTAL	
0247	4774	JMS		TST1	/TEST 1 10 TIMES.
0250	2323	ISE		TSTTAL	
0251	5247	JMP		.-2	
0252	1111	TAD		M12	
0253	3323	DCA		TSTTAL	
0254	4773	JMS		TST2	/TEST 2 10 TIMES.
0255	2323	ISE		TSTTAL	
0256	5254	JMP		.-2	
0257	1111	TAD		M12	
0260	3323	DCA		TSTTAL	
0261	4772	JMS		TST3	/TEST 3 10 TIMES.
0262	2323	ISE		TSTTAL	
0263	5261	JMP		.-2	
0264	1111	TAD		M12	
0265	3323	DCA		TSTTAL	
0266	4771	JMS		TST4	/TEST 4 10 TIMES.
0267	2323	ISE		TSTTAL	
0270	5266	JMP		.-2	
0271	1111	TAD		M12	
0272	3323	DCA		TSTTAL	
0273	4770	JMS		TST5	/TEST 5 10 TIMES.
0274	2323	ISE		TSTTAL	
0275	5273	JMP		.-2	
0276	1111	TAD		M12	
0277	3323	DCA		TSTTAL	
0300	4767	JMS		TST6	/TEST 6 10 TIMES.
0301	2323	ISE		TSTTAL	
0302	5300	JMP		.-2	
0303	4432	JMS	I	BELLP	/RING BELL FOR END OF PASS.
0304	2322	ISE		EXECNT	/CONTROL TEST DONE 10 TIMES.
0305	5245	JMP		EXEC1	/NO. AGAIN.
0306	4430	JMS	I	CRLFP	
0307	1324	TAD		MSG17P	/YES. PRINT CONTROL TEST COMP. MSG.
0310	4431	JMS	I	MSGNP	
0311	4430	JMS	I	CRLFP	
0312	7604	LAS			/CONTINUOUS CONTROL TEST.
0313	0073	AND		K1	
0314	7640	SEA	CLA		
0315	5236	JMP		EXEC2	/YES. DO IT AGAIN.
0316	5776	JMP		DISPLA	/NO. GO TO DISPLAY TEST MONITOR.
0317	5236	EXEC3,	JMP	EXEC2	/RETURN FROM "CT".

0320 4600 MSG1P, MSG1
 0321 4761 MSG14P, MSG14
 0322 0000 EXECNT, 0
 0323 0000 TSTTAL, 0
 0324 5002 MSG17P, MSG17

/*****
 /LOCATIONS 330 THRU 362 USED FOR ASSEMBLY STORAGE OF PROGRAM DIRECTORY.
 /*****/

/PROGRAM INITIALIZATION ROUTINE.

/1. CARRIES ON A DIALOGUE WITH THE USER TO DETERMINE THE SPECIFICATIONS
 /OF THE PLOTTER TO BE TESTED. THE USER REPLIES WITH NUMERIC CODES AS
 /DEFINED BELOW.
 /2. WHEN ALL INPUTS HAVE BEEN RECEIVED THE INFORMATION IS ECHOED ON
 /THE TTY IN DECODED FORM FOR USER VERIFICATION. USER TYPES "Y" OR "N"
 / (FOR YES OR NO) IN ANSWER TO "CORRECT?" IF RESPONSE IS "N" THE DIALOGUE
 /IS REPEATED.
 /3. ALL LISTS ARE MODIFIED TO ENSURE PROPER FORMATTING DEPENDENT
 /ON PLOTTER SPECIFICATIONS.
 /IF AT ANY TIME THE INCREMENT SIZE IS MANUALLY CHANGED, IT WILL BE
 /NECESSARY TO GO THROUGH THE COMPLETE DIALOGUE AGAIN.

/THE FOLLOWING NUMERIC CODES ARE TO BE UTILIZED BY USER IN RESPONSE
 /TO THE DIALOGUE.

/PLOTTER SERIES CODES.

/"1" 500 SERIES
 /"2" 600 SERIES
 /"3" 700 SERIES

/PLOTTER TYPE CODES.

/"1" 11 INCH DRUM.
 /"2" 28.55 INCH DRUM (TREATED AS 28 INCHES WIDE)
 /"3" 31 X 54 INCH FLATBED
 /"4" 54 X 72 INCH FLATBED

/VECTOR TYPE CODES:

/"1" 8 VECTOR
 /"2" 24 VECTOR

/INCREMENT TYPE CODES:

/#1" 0.1 MM
 /#2" .00125 INCH
 /#3" .0025 INCH
 /#4" .005 INCH
 /#5" .01 INCH.

0366 0562
 0367 1456
 0370 1400
 0371 1260
 0372 1200
 0373 1113
 0374 1030
 0375 1000
 0376 2400
 0377 0400

0400 0400

0400 0000

0401 7300

0402 3057

0403 3056

0404 3053

0405 4430

0406 1336

0407 3010

0410 1347

0411 3011

0412 1105

0413 3332

0414 4430

0415 1410

0416 4431

0417 4432

0420 4433

0421 3411

0422 2332

0423 5214

0424 4430

0425 4430

0426 1333

0427 4431

0430 4430

0431 1347

0432 3011

0433 1354

0434 3012

0435 1105

0436 3332

0437 4430

0440 1410

0441 4431

0442 1411

*0400

INIT,

0

CLA CLL

DCA FLGZIP

DCA FLAG24

DCA ALLSET

JMS I CRLFP

TAD MSGPS

DCA AUTO0

TAD TYS

DCA AUTO1

TAD M4

DCA INITAL

INIL1,

JMS I CRLFP

TAD I AUTO0

JMS I MSGNP

JMS I BELLP

JMS I READNP

DCA I AUTO1

ISE INITAL

JMP INIL1

JMS I CRLFP

JMS I CRLFP

TAD MSG6P

JMS I MSGNP

JMS I CRLFP

TAD TYS

DCA AUTO1

TAD TYSP

DCA AUTO2

TAD M4

DCA INITAL

INIL2,

JMS I CRLFP

TAD I AUTO0

JMS I MSGNP

TAD I AUTO1

/SET INIT NOT DONE INDICATOR.

/INITIALIZE LOOP 1 FOR
 /INFORMATION REQUESTS.
 /REQUEST PLOTTER SERIES, PLOTTER
 /TYPE, VECTOR TYPE, INCREMENT
 /TYPE, AND STORE INFO.

/REQUEST VERIFICATION.

/SET UP LOOP 2 FOR VERIFICATION
 /MESSAGES.

0443	1412	TAD I	AUTO2	
0444	3361	DCA	INITEM	
0445	1761	TAD I	INITEM	
0446	4431	JMS I	MSGNP	
0447	2332	ISE	INITAL	
0450	5237	JMP	INIL2	
0451	4430	JMS I	CRLFP	
0452	4430	JMS I	CRLFP	
0453	1334	TAD	MSG13P	
0454	4431	JMS I	MSGNP	/REQUEST USER DECISION AS
0455	4432	JMS I	BELLP	/TO CORRECTNESS OF DATA.
0456	4433	JMS I	READNP	/LOOK FOR Y=YES; N=NO
0457	0073	AND	K1	
0460	7650	SNA CLA		/INFO CORRECT?
0461	5201	JMP	INIT+1	/NO. REQUEST INFO AGAIN.
0462	4430	JMS I	CRLFP	
0463	4434	SERIES, JMS I	DSEP	/IS SERIES 500?
0464	0550	PS		
0465	5273	JMP	.*6	
0466	1777'	TAD	P500	/YES. SET UP 500 SERIES
0467	3010	DCA	AUTO0	/SUBROUTINE POINTERS FOR
0470	1776'	TAD	DK500	/TRANSFER TO STACK.
0471	3012	DCA	AUTO2	
0472	5311	JMP	TRSER	/GO TO TRANSFER OF POINTERS.
0473	1775'	TAD	P678	/SERIES IS 600 OR 700.
0474	3010	DCA	AUTO0	/SET UP 600-700 SERIES SUBROUTINE
0475	1774'	TAD	DK678	/POINTERS FOR TRANSFER TO
0476	3012	DCA	AUTO2	/STACK.
0477	4434	JMS I	DSEP	/IS SERIES 600?
0500	0550	PS		
0501	7610	SKP CLA		/NO.
0502	7000	NOP		/YES.
0503	7000	NOP		
0504	4434	JMS I	DSEP	/IS SERIES 700?
0505	0550	PS		
0506	7610	SKP CLA		/NO. CLEAR ZIP FLAG.
0507	7240	CLA CMA		/YES. SET ZIP FLAG.
0510	3057	DCA	FLGZIP	
0511	1044	TRSER, TAD	PENS	/TRANSFER PEN SUBROUTINE POINTERS.
0512	3011	DCA	AUTO1	
0513	1105	TAD	M4	
0514	3332	DCA	INITAL	
0515	1410	TAD I	AUTO0	
0516	3411	DCA I	AUTO1	
0517	2332	ISE	INITAL	
0520	5315	JMP	.-3	
0521	1773'	TAD	OKAYS	/TRANSFER VARIABLE DIRECTION
0522	3013	DCA	AUTO3	/CONSTANTS.
0523	1106	TAD	M10	
0524	3332	DCA	INITAL	
0525	1412	TAD I	AUTO2	
0526	3413	DCA I	AUTO3	
0527	2332	ISE	INITAL	
0530	5325	JMP	.-3	

```

0531 5735      JMP I   CONFIP
0532 0000      INITAL, 0
0533 4673      MSG6P, MSG6
0534 4747      MSG13P, MSG13
0535 0600      CONFIP, CONFIG
0536 0536      MSG6PS, .
0537 4615      MSG2
0540 4631      MSG3
0541 4644      MSG4
0542 4657      MSG5
0543 4677      MSG7
0544 4710      MSG10
0545 4724      MSG11
0546 4736      MSG12
0547 0547      TYS, .
0550 0000      PS, 0
0551 0000      PT, 0
0552 0000      VT, 0
0553 0000      IT, 0
0554 0554      TYSP, .
0555 5300      PSP, PSM1P-1
0556 9303      PTP, PTH1P-1
0557 9307      VTP, VTM1P-1
0560 5311      ITP, ITM1P-1
0561 0000      INITEM, 0
0562 0000      PATCH, 0
0563 7320      CLA CLL CML
0564 6004      .
0565 7710      SPA CLA
0566 9371      JMP .+3
0567 1161      TAD K818L
0570 3562      DCA I M304P
0571 9762      JMP I PATCH

```

/POINTER TO NEXT SECTION.

/TEMP FOR PLOTTER SERIES CODE.
 /TEMP FOR PLOTTER TYPE CODE.
 /TEMP FOR VECTOR TYPE CODE.
 /TEMP FOR INCREMENT TYPE CODE.

/POINTER FOR PLOTTER SERIES MSGS.
 /POINTER FOR PLOTTER TYPE MSGS.
 /POINTER FOR VECTOR TYPE MSGS.
 /POINTER FOR INCREMENT TYPE MSGS.

```

0573 5404
0574 5446
0575 5377
0576 5435
0577 5372
0600 0600
0600 1777'    *0600
0601 3010      CONFIG, TAD      WLP
0602 4434      DCA      AUTO0
0603 0531      JMS I      DSEP
0604 7410      PT
0605 5211      SKP
0606 2010      JMP .+4
0607 2010      ISZ      AUTO0
0610 5202      ISZ      AUTO0
0611 1107      JMP      CONFIG+2
0612 3345      TAD      M2
0613 1776'      DCA      CONTA1
0614 3011      TAD      REPK
0615 1105      DCA      AUTO1
          CONL1, TAD      M4

```

/TRANSFER WIDTH AND LENGTH REPETITION
 /CONSTANTS.

```

0616 3346      DCA      CONCNT
0617 1410      TAD I    AUTO0
0620 3347      DCA      CONTEM
0621 1347      CONL2,  TAD      CONTEM
0622 3411      DCA I    AUTO1
0623 1347      TAD      CONTEM
0624 7104      CLL RAL
0625 3347      DCA      CONTEM
0626 2346      ISZ      CONCNT
0627 5221      JMP      CONL2
0630 2345      ISZ      CONTAL
0631 5215      JMP      CONL1
0632 4434      VECTOR, JMS I  DSZP
0633 0552      VT
0634 7610      SKP CLA
0635 7610      SKP CLA
0636 7240      CLA CMA
0637 3056      DCA      FLAG24
0640 4434      INCR,  JMS I  DSZP
0641 0553      IT
0642 5256      JMP      INCR1
0643 1775      TAD      METER
0644 3010      DCA      AUTO0
0645 1344      TAD      LGTP
0646 3011      DCA      AUTO1
0647 1350      TAD      M20
0650 3345      DCA      CONTAL
0651 1410      TAD I    AUTO0
0652 3411      DCA I    AUTO1
0653 2345      ISZ      CONTAL
0654 5251      JMP      .-3
0655 5307      JMP      PATTR
0656 1774      INCR1,  TAD      INSTP
0657 3010      DCA      AUTO0
0660 4434      JMS I    DSZP
0661 0553      IT
0662 5270      JMP      .+4
0663 1410      TAD I    AUTO0
0664 3302      DCA      INCR11
0665 1410      TAD I    AUTO0
0666 3303      DCA      INCR12
0667 5273      JMP      .+4
0670 2010      ISZ      AUTO0

```

/SET 24 VECTOR FLAG IF
/PLOTTER IS 24 VECTOR TYPE.

/SET UP LENGTH CONSTANTS
/ACCORDING TO INCREMENT SIZE
/USING .0025 INCHES AS BASIC UNLESS
/INCREMENT SIZE IS 0.1 MM.
/TRANSFER METRIC CONSTANTS.

/INCREMENT SIZE IS TYPE 2,3,4, OR 5.
/TRANSFER SCALED CONSTANTS

```

0671 2010      ISZ      AUTO0
0672 5260      JMP      INCR1+2
0673 1773      TAD      INCH
0674 3010      DCA      AUTO0
0675 1344      TAD      LGTP
0676 3011      DCA      AUTO1
0677 1352      TAD      M50
0700 3345      DCA      CONTAL
0701 1410      TAD I    AUTO0
0702 0000      INCR11, 0

```

/SCALE FACTOR INSTRUCTIONS

```

0703 0000 INCR12, 0 /FILL IN.
0704 3411 DCA I AUTO1
0705 2345 ISB CONTAL
0706 5301 JMP INCR11-1
0707 1056 PATTR, TAD FLAG24 /TRANSFER 24 VECTOR OR 8 VECTOR
0710 7090 SNA CLA /PATTERN POINTERS AND NEXT TO
0711 5316 JMP .+5 /LAST PATTERN LINE NUMBERS.
0712 1772 TAD PATS24
0713 3010 DCA AUTO0
0714 1771 TAD LAS24
0715 5321 JMP .+4
0716 1770 TAD PATS8
0717 3010 DCA AUTO0
0720 1767 TAD LAS8
0721 3012 DCA AUTO2
0722 1391 TAD M6
0723 5345 DCA CONTAL
0724 1766 TAD PATS
0725 3011 DCA AUTO1
0726 1765 TAD LASLIN
0727 3013 DCA AUTO3
0730 1410 TAD I AUTO0
0731 3411 DCA I AUTO1
0732 1412 TAD I AUTO2
0733 3413 DCA I AUTO3
0734 2345 ISB CONTAL /TRANSFER DONE?
0735 5330 JMP .-5 /NO.
0736 7240 CLA CMA /SET INIT DONE INDICATOR.
0737 3053 DCA ALLSET
0740 1764 TAD INIT /YES.
0741 3343 DCA .+2
0742 9743 JMP I .+1 /INITIALIZATION DONE. RETURN EXIT.
0743 0000 0
0744 5457 LGTP, LGTH-1
0745 0000 CONTAL, 0
0746 0000 CONCNT, 0
0747 0000 CONTEM, 0
0750 7760 M20, -20
0751 7772 M6, -6
0752 7730 M50, -50

```

/TAPE 2.

```

0764 0400
0765 5345
0766 5317
0767 5363
0770 5336
0771 5354
0772 5327
0773 5530
0774 5270
0775 5601
0776 5622

```

0777 5257
1000 *1000

/START CONTROL TESTS.

/TEST0: TEST OF THE ABILITY OF THE SIGNAL "INITIALIZE" TO CLEAR THE
/PLOTTER FLAG AND SET INTERRUPT ENABLE.
/THIS TEST IS RUN ONLY ONCE PRIOR TO ANY TYPEOUTS. IF AN
/ERROR OCCURS THE PROGRAM WILL HALT. IF THE FLAG WILL NOT
/SET OR PLSF WILL NOT SKIP, THE PROGRAM WILL HANG UP.
/ONLY AFTER THIS TEST PASSES WILL THE TITLE MESSAGE BE
/PRINTED.
/ALL ERROR RECOVERIES SHOULD BE MADE BY DEPRESSING START
/TO GENERATE ANOTHER "INITIALIZE" SIGNAL.
/THIS TEST MAY BE INHIBITED BY SETTING SR9.

1000	0000	TST0,	0	
1001	7200		CLA	/POP8-E USERS MAY INSERT CAF HERE.
1002	4437		JMS I SETIPP	/SET UP INTERRUPT LINK.
1003	7604		LAS	
1004	0071		AND K4	
1005	7640		SEA CLA	
1006	5600		JMP I TST0	/TEST 0 INHIBITED.
1007	4421		PLSF	
1010	5213		JMP .+3	
1011	7402	ERR0A,	HLT	/INITIALIZE DOES NOT CLEAR
1012	5201		JMP TST0+1	/FLAG OR PLSF SKIPS WHEN
				/FLAG IS CLEAR.
1013	4424		PLLR	/TRY TO SET FLAG.
1014	4421	HANGUP,	PLSF	/IF PROGRAM HANGS UP PLLR
1015	5214		JMP .-1	/((OR=0) DOES NOT SET FLAG.
1016	7200		CLA	/FLAG SETS. CHECK INTERRUPT.
1017	3062		DCA INTFLG	
1020	6001		ION	
1021	7000		NOP	
1022	6002		IOF	
1023	1062		TAD INTFLG	
1024	7640		SEA CLA	
1025	5600		JMP I TST0	/TST0 PASSES. EXIT.
1026	7402	ERR0B,	HLT	/INITIALIZE DOES NOT SET INT.
1027	5201		JMP TST0+1	/ENABLE OR (FLAG AND INT ENABLE)
				/DO NOT YIELD INT, REQUEST.

/TEST1: FLAG TEST FOR PLPU AND PLPD IOT'S.

/CHECKS THE FOLLOWING:

- /1. PLSF CLEARS FLAG.
- /2. PLSF SKIPS ONLY WHEN FLAG SET.
- /3. PLPU AND PLPD DO NOT SET FLAG PRIOR TO 70 MS.
- /4. PLPU AND PLPD SET FLAG AT 70 MS.

/ERRORS 1A THRU 1E ARE DETECTED.

/LOOP 1 CYCLES ON PLPU - PLSF CHECK.

/LOOP 2 CYCLES ON PLPD - PLSF CHECK.

/LOOP 3 CYCLES ON WHOLE TEST.

```

1030 0000 TST1, 0
/
/START LOOP 1 AND 3.
/
1031 7300 T1L1L3, CLA CLL /START LOOP 1 AND 3.
1032 4422 PLCF /CHECK FLAG CLEAR,
1033 4421 PLSF
1034 7410 SKP
1035 4440 JMS I ERR1AP /FLAG NOT CLEAR OR ILLEGAL SKIP
1036 4423 PLPU /TEST PLPU IOT,
1037 1115 TAD K70MIN /DELAY 60 MS.
1040 4436 JMS I DELNMP
1041 4421 PLSF /FLAG SET?
1042 7410 SKP
1043 4707 JMS I ERR1BP /YES, FLAG SET PRIOR TO 70 MS.
1044 1116 TAD K70MAX /DELAY 20 MORE MS.
1045 4436 JMS I DELNMP
1046 4421 PLSF /FLAG SET?
1047 4710 JMS I ERR1CP /NO, FLAG NOT SET BY 70 MS.
1050 4422 PLCF /CLEAR FLAG.
1051 4421 PLSF /FLAG CLEAR.
1052 7410 SKP
1053 4440 JMS I ERR1AP /NO, OR PLSF SKIPS WHEN FLAG CLEAR.
1054 7604 LAS /LOOP 1 REQUESTED?
1055 0070 AND K400
1056 7640 SZA CLA
1057 5231 JMP T1L1L3 /YES. LOOP1.

/END LOOP 1.
/START LOOP 2.
/
1060 4425 T1L2, PLPD /TEST PLPD IOT.
1061 1115 TAD K70MIN /WAIT 60 MS.
1062 4436 JMS I DELNMP
1063 4421 PLSF /FLAG SET?
1064 7410 SKP
1065 4711 JMS I ERR1DP /YES, FLAG SET PRIOR TO 70 MS.
1066 1116 TAD K70MAX /WAIT 20 MORE MS
1067 4436 JMS I DELNMP
1070 4421 PLSF /FLAG SET?
1071 4712 JMS I ERR1EP /NO, FLAG NOT SET BY 70 MS.
1072 4422 PLCF /CLEAR FLAG.
1073 4421 PLSF /FLAG CLEAR?
1074 7410 SKP
1075 4440 JMS I ERR1AP /NO, OR ILLEGAL SKIP.
1076 7604 LAS /LOOP2?
1077 0101 AND K200
1100 7640 SZA CLA
1101 5260 JMP T1L2 /YES. LOOP 2.

/END LOOP 2.
/
1102 7604 LAS /LOOP 3?
1103 0067 AND K100

```

```

1104 7640      SZA CLA
1105 5231      JMP      T1L1L3      /YES.  LOOP 3.
/
/END LOOP 3.
/

```

```

1106 5630      JMP I    TST1      /EXIT.

```

```

1107 1646      ERR1BP, ERR1B
1110 1661      ERR1CP, ERR1C
1111 1674      ERR1DP, ERR1D
1112 1707      ERR1EP, ERR1E

```

```

/TEST2: FLAG TEST FOR PLLR WITH DIRECTION REGISTER = 31 AND
/32. THIS IS EQUIVALENT FOR 600 - 700 SERIES TO PLPU AND
/PLPD.

```

```

/THE FOLLOWING CHECKS ARE MADE.

```

- /1. PLCF CLEARS FLAG.
- /2. PLSF SKIPS ONLY WHEN FLAG SET.
- /3. PLLR 31 AND 32 DO NOT SET FLAG PRIOR TO 70 MS.
- /4. PLLR 31 AND 32 SET FLAG AT 70 MS.

```

/ERRORS 1A, AND 2A THRU 2D ARE DETECTED.

```

```

/LOOP 1 CYCLES ON PLLR31 - PLCF CHECK.

```

```

/LOOP 2 CYCLES ON PLLR32 - PLCF CHECK.

```

```

/LOOP 3 CYCLES ON WHOLE TEST.

```

```

1113 0000      TST2,  0
/

```

```

/START LOOP 1 AND 3.
/

```

```

1114 7300      T2L1L3, CLA CLL      /START LOOP1 AND 3.
1115 4422      PLCF                /CLEAR FLAG.
1116 4421      PLSF                /FLAG CLEAR.
1117 7410      SKP
1120 4440      JMS I  ERR1AP        /NO.  FLAG SET OR ILLEGAL SKIP.
1121 4777      JMS      CPU8        /DO PLLR31.
1122 1115      TAD      K70MIN      /WAIT 60 MS.
1123 4436      JMS I  DELNMP
1124 4421      PLSF                /FLAG SET?
1125 7410      SKP
1126 4772      JMS I  ERR2AP        /YES.  FLAG SET PRIOR TO 70 MS.
1127 1116      TAD      K70MAX      /WAIT 20 MORE MS.
1130 4436      JMS I  DELNMP
1131 4421      PLSF                /FLAG SET?
1132 4773      JMS I  ERR2BP        /NO.  FLAG NOT SET AT 70 MS.
1133 4422      PLCF                /CLEAR FLAG.
1134 4421      PLSF                /FLAG CLEAR?
1135 7410      SKP
1136 4440      JMS I  ERR1AP        /NO.  FLAG SET OR ILLEGAL SKIP.
1137 7604      LAS
1140 0070      AND      K400
1141 7640      SZA CLA
1142 5314      JMP      T2L1L3      /YES.  LOOP 1.
/

```

```

/END LOOP 1.

```

```

/
/START LOOP 2.
/
1143 4776' T2L2, JMS CPD8 /DO PLLR32.
1144 1115 TAD K70MIN /WAIT 60 MS.
1145 4436 JMS I DELNMP
1146 4421 PLSF /FLAG SET?
1147 7410 SKP
1150 4774 JMS I ERR2CP /YES. FLAG SET PRIOR TO 70 MS.
1151 1116 TAD K70MAX /WAIT 20 MORE MS.
1152 4436 JMS I DELNMP
1153 4421 PLSF /FLAG SET.
1154 4775 JMS I ERR2DP /NO. FLAG NOT SET AT 70 MS.
1155 4422 PLCF /CLEAR FLAG.
1156 4421 PLSF /FLAG CLEAR?
1157 7410 SKP
1160 4440 JMS I ERR1AP /NO. FLAG SET OR ILLEGAL SKIP.
1161 7604 LAS /LOOP 2?
1162 0101 AND K200
1163 7640 SZA CLA
1164 5343 JMP T2L2 /YES. LOOP 2.

/END LOOP 2.
/
1165 7604 LAS /LOOP 3?
1166 0067 AND K100
1167 7640 SZA CLA
1170 5314 JMP T2L1L3 /YES. LOOP 3.

/END LOOP 3.
/
1171 5713 JMP I TST2 /EXIT.
1172 1722 ERR2AP, ERR2A
1173 1735 ERR2BP, ERR2B
1174 1750 ERR2CP, ERR2C
1175 1763 ERR2DP, ERR2D

1176 4110
1177 4102
1200 *1200

```

```

/TEST3: FLAG TEST FOR PLLR WITH DIRECTION REGISTER=N; WHERE
/N=0 THRU 77 EXCEPT FOR 31, 32 AND 33.
/THE FOLLOWING CHECKS ARE MADE.
/1. PLCF CLEARS FLAG.
/2. PLSF SKIPS ONLY WHEN FLAG SET.
/3. PLLR N DOES NOT SET FLAG PRIOR TO 5 MS.
/4. PLLR N DOES SET FLAG AT 5 MS.
/ERRORS 1A, AND 3A THRU 3B ARE DETECTED.
/LOOP 1 CYCLES ON PRESENT PLLR N.
/LOOP 2 CYCLES ON WHOLE TEST.

```

```

1200 0000 TST3, 0
/
/START LOOP 2.
/
1201 7300 T3L2, CLA CLL
1202 3253 DCA KLDRN /SET N = 0
1203 1112 TAD M31 /SET COUNTER TO SKIP DR = 31, 32 OR 33.
1204 3257 DCA T3M31
1205 1113 TAD M100 /SET MAIN COUNT = -64
1206 3254 DCA T3CNT
/
/START LOOP 1.
/
1207 4422 T3REP, PLGF /CLEAR FLAG.
1210 4421 PLSF /FLAG CLEAR?
1211 7410 SKP
1212 4440 JMS I ERR1AP /NO. FLAG SET OR ILLEGAL SKIP
1213 1253 TAD KLDRN /PUT CURRENT N IN AC.
1214 4424 PLLR /LOAD DIRECTION REGISTER.
1215 7200 CLA
1216 1117 TAD K5MIN /WAIT 4 MS.
1217 4436 JMS I DELNMP
1220 4421 PLSF /FLAG SET?
1221 7410 SKP
1222 4655 JMS I ERR3AP /YES. FLAG SET PRIOR TO 5 MS.
1223 1120 TAD K5MAX /WAIT 2 MORE MS.
1224 4436 JMS I DELNMP
1225 4421 PLSF /FLAG SET?
1226 4656 JMS I ERR3BP /NO. FLAG NOT SET AT 5 MS.
1227 7604 LAS /LOOP 1?
1230 0070 AND K400
1231 7640 SZA CLA
1232 5207 JMP T3REP /YES. LOOP 1.
/
/END LOOP 1.
/
1233 2253 ISZ KLDRN /UPDATE N.
1234 2257 ISZ T3M31 /N = 31?
1235 5244 JMP ,+7
1236 2253 ISZ KLDRN /YES. BYPASS N=31, 32 AND 33.
1237 2253 ISZ KLDRN
1240 2253 ISZ KLDRN
1241 2254 ISZ T3CNT
1242 2254 ISZ T3CNT
1243 2254 ISZ T3CNT
1244 2254 ISZ T3CNT
1245 5207 JMP T3REP /UPDATE MAIN COUNT
1246 7604 LAS /NOT DONE. CONTINUE.
1247 0101 AND K200 /LOOP 2?
1250 7640 SZA CLA
1251 5201 JMP T3L2 /YES. LOOP2.
/
/END LOOP 2.
/

```

```

1252 5600      JMP I   TST3      /EXIT.
1253 0000      KLDNR, 0
1254 0000      T3CNT, 0
1255 2000      ERR3AP, ERR3A
1256 2013      ERR3BP, ERR3B
1257 0000      T3M31, 0

```

```

/TEST4: FLAG TEST FOR CFLR WITH DIRECTION REGISTER * 31 AND 32.
/THIS TEST IS SIMILAR TO TEST 2 EXCEPT CFLR INITIALLY
/CLEARs FLAG THEN ACTs AS PLLR.
/THE FOLLOWING CHECKS ARE MADE:
/1. CFLR INITIALLY CLEARs FLAG.
/2. PLSF SKIPS ONLY WHEN FLAG SET.
/3. CFLR 31 AND 32 DO NOT SET FLAG PRIOR TO 70 MS.
/4. CFLR 31 AND 32 SET FLAG AT 70 MS.
/ERRORS 4A THRU 4E ARE DETECTED.
/LOOP 1 CYCLES ON CFLR 31 CHECK.
/LOOP 2 CYCLES ON CFLR 32 CHECK.
/LOOP 3 CYCLES ON WHOLE TEST.

```

```

1260 0000      TST4, 0
/
/START LOOP 1 AND 3.
/
1261 7300      T4L1L3, CLA CLL
1262 1077      TAD      K31      /PUT 31 IN AC
1263 4426      CFLR      /CLEAR FLAG, LOAD DR.
1264 4421      PLSF      /FLAG CLEAR?
1265 7410      SKP
1266 4441      JMS I   ERR4EP    /NO. FLAG SET OR ILLEGAL SKIP.
1267 7200      CLA
1270 1115      TAD      K70MIN   /WAIT 60 MS.
1271 4436      JMS I   DELNMP
1272 4421      PLSF      /FLAG SET?
1273 7410      SKP
1274 4735      JMS I   ERR4AP    /YES. FLAG SET PRIOR TO 70 MS.
1275 1116      TAD      K70MAX   /WAIT 20 MORE MS.
1276 4436      JMS I   DELNMP
1277 4421      PLSF      /FLAG SET?
1300 4736      JMS I   ERR4BP    /NO. FLAG NOT SET AT 70 MS.
1301 7604      LAS
1302 0070      AND      K400     /LOOP 1?
1303 7640      SZA CLA
1304 0261      JMP      T4L1L3   /YES. LOOP 1.
/
/END LOOP 1.
/
/START LOOP 2.
/
1305 1100      T4L2,  TAD      K32      /PUT 32 IN AC.
1306 4426      CFLR      /CLEAR FLAG, LOAD DR.
1307 4421      PLSF      /FLAG CLEAR?
1310 7410      SKP
1311 4441      JMS I   ERR4EP    /NO. FLAG SET OR ILLEGAL SKIP.

```

```

1312 7200      CLA
1313 1115      TAD      K70MIN      /WAIT 60 MS.
1314 4436      JMS I    DELNMP

1315 4421      PLSF      /FLAG SET?
1316 7410      SKP
1317 4737      JMS I    ERR4CP      /YES. FLAG SET PRIOR TO 70 MS.
1320 1116      TAD      K70MAX      /WAIT 20 MORE MS.
1321 4436      JMS I    DELNMP
1322 4421      PLSF      /FLAG SET?
1323 4740      JMS I    ERR4DP      /NO. FLAG NOT SET AT 70 MS.
1324 7604      LAS
1325 0101      AND      K200
1326 7640      SZA CLA
1327 5305      JMP      T4L2      /YES. LOOP 2.

```

/END LOOP 2.

```

1330 7604      LAS      /LOOP 3?
1331 0067      AND      K100
1332 7640      SZA CLA
1333 5261      JMP      T4L1L3    /YES. LOOP 3.

```

/END LOOP 3.

```

1334 5660      JMP I    TST4      /EXIT.
1335 2026      ERR4AP, ERR4A
1336 2041      ERR4BP, ERR4B
1337 2054      ERR4CP, ERR4C
1340 2067      ERR4DP, ERR4D

```

1400 *1400

/TEST5: FLAG TEST FOR CFLR WITH DIRECTION REGISTER = N; WHERE
 /N = 0 THRU 77 EXCEPT FOR 31, 32 AND 33. THE METHOD IS SIMILAR TO TEST 3.
 /THE FOLLOWING CHECKS ARE MADE:
 /1. CFLR INITIALLY CLEARS FLAG.
 /2. PLSF SKIPS ONLY WHEN FLAG SET.
 /3. CFLR N DOES NOT SET FLAG PRIOR TO 5 MS.
 /4. CFLR N SETS FLAG AT 5 MS.
 /ERRORS 4E, AND 5A THRU 5B ARE DETECTED.
 /LOOP 1 CYCLES ON PRESENT CFLR N.
 /LOOP 2 CYCLES ON WHOLE TEST.

```

1400 0000      TST5, 0
/
/START LOOP 2.
/

```

```

1401 7300      T5L2,    CLA CLL
1402 3251      DCA      KCFLRN    /SET N = 0
1403 1112      TAD      M31      /SET COUNTER TO SKIP DR = 31, 32, 33.
1404 3252      DCA      T5M31
1405 1113      TAD      M100      /SET MAIN COUNT = -64

```

```

1406 3253          DCA      T5CNT
/
/START LOOP 1.
/
1407 1251      T5REP, TAD      KCFLRN      /PUT N IN AC.
1410 4426          CFLR          /CLEAR FLAG, LOAD DR.
1411 4421          PLSF          /FLAG CLEAR?
1412 7610          SKP CLA
1413 4441          JMS I      ERR4EP      /NO. FLAG NOT CLEAR OR ILLEGAL SKIP.
1414 1117          TAD      K5MIN      /WAIT 4 MS.
1415 4436          JMS I      DELNMP
1416 4421          PLSF
1417 7410          SKP
1420 4654          JMS I      ERR5AP      /YES. FLAG SET PRIOR TO 5 MS.
1421 1120          TAD      K5MAX      /WAIT 2 MORE MS.
1422 4436          JMS I      DELNMP
1423 4421          PLSF
1424 4655          JMS I      ERR5BP
1425 7604          LAS
1426 0070          AND      K400
1427 7640          SZA CLA
1430 5207          JMP      T5REP      /YES. LOOP1
/
/END LOOP 1.
/
1431 2251          ISZ      KCFLRN      /UPDATE N.
1432 2252          ISZ      T5M31      /N = 31?
1433 5242          JMP      .+7
1434 2251          ISZ      KCFLRN      /YES. BYPASS N = 31, 32 AND 33.
1435 2251          ISZ      KCFLRN
1436 2251          ISZ      KCFLRN
1437 2253          ISZ      T5CNT
1440 2253          ISZ      T5CNT
1441 2253          ISZ      T5CNT
1442 2253          ISZ      T5CNT
1443 5207          JMP      T5REP
1444 7604          LAS
1445 0101          AND      K200
1446 7640          SZA CLA
1447 5201          JMP      T5L2      /YES. LOOP2.
/
/END LOOP 2.
/
1450 5600          JMP I      TST5
1451 0000      KCFLRN, 0
1452 0000      T5M31, 0
1453 0000      T5CNT, 0
1454 2115      ERR5AP, ERR5A
1455 2130      ERR5BP, ERR5B

```

/TEST6: INTERRUPT TEST: (USES CFLR 00 TO SET FLAG IN 5 MS.)
 /THE FOLLOWING CHECKS ARE MADE:
 /1. PLCF CLEARS FLAG.
 /2. PLSF SKIPS ONLY WHEN FLAG SET.
 /3. PLSE SETS INTERRUPT ENABLE.

/4. PLCE CLEARS INTERRUPT ENABLE,
 /5. INTERRUPT OCCURS ONLY AFTER PLSE AND FLAG SET.
 /ERRORS 1A, AND 6A THRU 6E ARE DETECTED.
 /LOOP 1 CYCLES ON PLSE CHECK.
 /LOOP 2 CYCLES ON PLCE CHECKS.
 /LOOP 3 CYCLES ON WHOLE TEST.

1456	0000	TST6, 0		
1457	4437	JMS I SETIPP		/SET INTERRUPT LINK.
/				
/START LOOP 1 AND 3.				
/				
1460	4420	T6L1L3, PLCE		/DISABLE INT
1461	4422	PLCF		/CLEAR FLAG.
1462	4421	PLSF		/FLAG CLEAR?
1463	7410	SKP		
1464	4440	JMS I ERR1AP		/NO. FLAG NOT CLEAR OR ILLEGAL SKIP.
1465	4426	CFLR		/LOAD DR.
1466	1121	TAD K5ALL		/WAIT 10 MS.
1467	4436	JMS I DELNMP		
1470	4421	PLSF		/FLAG SET?
1471	4763	JMS I ERR6EP		/NO. FLAG NOT SET AT 5 MS.
1472	3062	DCA INTFLG		/SET FLAG = 0
1473	4427	PLSE		/ENABLE INT
1474	6001	ION		/INT. ON
1475	7000	NOP		
1476	6002	IOF		/KILL INT
1477	1062	TAD INTFLG		/DID INT. OCCUR
1500	7700	SMA CLA		
1501	4764	JMS I ERR6AP		/NO INT AFTER PLSE AND FLAG SET
1502	4422	PLCF		/CLEAR FLAG.
1503	4421	PLSF		/FLAG CLEAR?
1504	7610	SKP CLA		
1505	4440	JMS I ERR1AP		/NO. FLAG NOT CLEAR OR ILLEGAL SKIP
1506	3062	DCA INTFLG		/SET FLAG TO 0
1507	6001	ION		/SET INT.
1510	7000	NOP		
1511	6002	IOF		/KILL INT
1512	1062	TAD INTFLG		/INT. OCCUR?
1513	7640	SZA CLA		
1514	4765	JMS I ERR6BP		/YES. INT AFTER PLSE AND FLAG CLEAR.
1515	7604	LAS		/LOOP 1?
1516	0070	AND K400		
1517	7640	SZA CLA		
1520	5260	JMP T6L1L3		
/				
/END LOOP 1.				
/				
/START LOOP 2.				
/				
1521	4427	T6L2, PLSE		/ENABLE INT
1522	4426	CFLR		/LOAD DR.

```

1523 1121      TAD      K5ALL      /WAIT 10 MS
1524 4436      JMS I    DELNMP
1525 4421      PLSF
1526 4763      JMS I    ERR6EP      /FLAG SET?
1527 4420      PLCE      /NO. FLAG NOT SET AT 5 MS
1530 3062      DCA      INTFLG      /DISABLE INT
1531 6001      ION
1532 7000      NOP
1533 6002      IOF
1534 1062      TAD      INTFLG      /SET FLAG = 0
1535 7640      SZA CLA
1536 4766      JMS I    ERR6CP      /SET INT
1537 4422      PLCF
1540 4421      PLSF
1541 7610      SKP CLA
1542 4440      JMS I    ERR1AP      /KILL INT
1543 3062      DCA      INTFLG      /INT. OCCUR?
1544 6001      ION
1545 7000      NOP
1546 6002      IOF
1547 1062      TAD      INTFLG      /YES. INT AFTER PLCE AND FLAG SET
1550 7640      SZA CLA
1551 4767      JMS I    ERR6DP      /CLEAR FLAG
1552 7604      LAS
1553 0101      AND      K200      /FLAG CLEAR?
1554 7640      SZA CLA
1555 9321      JMP      T6L2
1556 7604      LAS
1557 0067      AND      K100
1560 7640      SZA CLA
1561 9260      JMP      T6L1L3
1562 9656      JMP I    TST6
1563 2226      ERR6EP, ERR6E
1564 2143      ERR6AP, ERR6A
1565 2156      ERR6BP, ERR6B
1566 2200      ERR6CP, ERR6C
1567 2213      ERR6DP, ERR6D
1600          *1600
          /SET CONTROL TEST INTERRUPT LINK.
1600 0000      SETIP, 0
1601 7200      CLA
1602 1114      TAD      KJMPI2
1603 3001      DCA      1

```

1604	1207	TAD	CTINTP
1605	3002	DCA	2
1606	5600	JMP I	SETIP

/INTERRUPT SERVICE FOR CONTROL TESTS.

1607	1610	CTINTP,	.*1	
1610	3051	DCA	SAVEAC	
1611	7010	RAR		
1612	3052	DCA	SAVEL	
1613	6031	KSF		
1614	5217	JMP	.*3	
1615	6032	KCC		
1616	5226	JMP	FRT	
1617	6041	TSF		
1620	5223	JMP	.*3	
1621	6042	TCF		
1622	5226	JMP	FRT	
1623	7240	CLA CMA		
1624	3062	DCA	INTFLG	/SET INTFLG IF TTY NOT CAUSE
1625	5400	JMP I	0	
1626	1052	FRT,	TAD	SAVEL
1627	7004	RAL		
1630	1051	TAD	SAVEAC	
1631	6001	ION		
1632	5400	JMP I	0	

/TAPE 3.

/ERROR ROUTINES.

/

/

/ERROR1A: PLCF DOES NOT CLEAR FLAG OR PLSF SKIPS WHEN FLAG CLEAR.

1633	0000	ERR1A,	0
1634	7200	CLA	
1635	1245	TAD	K6101
1636	3063	DCA	BUFF1
1637	1233	TAD	ERR1A
1640	4442	JMS I	ERRORP
1641	7402	HLT	
1642	4443	JMS I	QCONTP
1643	5633	JMP I	ERR1A
1644	5777	JMP	TST1+1
1645	6101	K6101,	6101

/ERROR1B: PLPU IOT SETS FLAG PRIOR TO 70 MS.

1646	0000	ERR1B,	0
1647	7200	CLA	
1650	1260	TAD	K6102
1651	3063	DCA	BUFF1
1652	1246	TAD	ERR1B
1653	4442	JMS I	ERRORP
1654	7402	HLT	

```
1655 4443      JMS I   QCONTP
1656 5646      JMP I   ERR1B
1657 5777'     JMP     TST1+1
1660 6102      K6102, 6102
```

/ERROR1C: PLPU IOT DOES NOT SET FLAG IN 70 MS.

```
1661 0000      ERR1C, 0
1662 7200      CLA
1663 1273      TAD      K6103
1664 3063      DCA      BUFF1
1665 1261      TAD      ERR1C
1666 4442      JMS I   ERRORP
1667 7402      HLT
1670 4443      JMS I   QCONTP
1671 5661      JMP I   ERR1C
1672 5777'     JMP     TST1+1
1673 6103      K6103, 6103
```

/ERROR1D: PLPD IOT SETS FLAG PRIOR TO 70 MS.

```
1674 0000      ERR1D, 0
1675 7200      CLA
1676 1306      TAD      K6104
1677 3063      DCA      BUFF1
1700 1274      TAD      ERR1D
1701 4442      JMS I   ERRORP
1702 7402      HLT
1703 4443      JMS I   QCONTP
1704 5674      JMP I   ERR1D
1705 5777'     JMP     TST1+1
1706 6104      K6104, 6104
```

/ERROR1E: PLPD IOT DOES NOT SET FLAG IN 70 MS.

```
1707 0000      ERR1E, 0
1710 7200      CLA
1711 1321      TAD      K6105
1712 3063      DCA      BUFF1
1713 1307      TAD      ERR1E
1714 4442      JMS I   ERRORP
1715 7402      HLT
1716 4443      JMS I   QCONTP
1717 5707      JMP I   ERR1E
1720 5777'     JMP     TST1+1
1721 6105      K6105, 6105
```

/ERROR2A: PLLR IOT WITH AC=31 SETS FLAG PRIOR TO 70 MS.

```
1722 0000      ERR2A, 0
1723 7200      CLA
1724 1334      TAD      K6201
1725 3063      DCA      BUFF1
1726 1322      TAD      ERR2A
```

```
1727 4442      JMS I   ERRORP
1730 7402      HLT
1731 4443      JMS I   QCONTP
1732 5722      JMP I   ERR2A
1733 5776      JMP     TST2+1
1734 6201      K6201, 6201
```

/ERROR2B: PLLR IOT WITH AC=31 DOES NOT SET FLAG IN 70 MS.

```
1735 0000      ERR2B, 0
1736 7200      CLA
1737 1347      TAD     K6202
1740 3063      DCA     BUFF1
1741 1335      TAD     ERR2B
1742 4442      JMS I   ERRORP
1743 7402      HLT
1744 4443      JMS I   QCONTP
1745 5735      JMP I   ERR2B
1746 5776      JMP     TST2+1
1747 6202      K6202, 6202
```

/ERROR2C: PLLR IOT WITH AC=32 SETS FLAG PRIOR TO 70 MS.

```
1750 0000      ERR2C, 0
1751 7200      CLA
1752 1362      TAD     K6203
1753 3063      DCA     BUFF1
1754 1350      TAD     ERR2C
1755 4442      JMS I   ERRORP
1756 7402      HLT
1757 4443      JMS I   QCONTP
1760 5750      JMP I   ERR2C
1761 5776      JMP     TST2+1
1762 6203      K6203, 6203
```

/ERROR2D: PLLR IOT WITH AC=32 DOES NOT SET FLAG IN 70 MS.

```
1763 0000      ERR2D, 0
1764 7200      CLA
1765 1375      TAD     K6204
1766 3063      DCA     BUFF1
1767 1363      TAD     ERR2D
1770 4442      JMS I   ERRORP
1771 7402      HLT
1772 4443      JMS I   QCONTP
1773 5763      JMP I   ERR2D
1774 5776      JMP     TST2+1
1775 6204      K6204, 6204
```

```
1776 1114
1777 1031
      2000      *2000
```

/ERROR3A: PLLR IOT WHERE AC NOT 31, 32, OR 33 SETS FLAG PRIOR TO 5 MS.

```

2000 0000 ERR3A, 0
2001 7200 CLA
2002 1212 TAD K6301
2003 3063 DCA BUFF1
2004 1777 TAD KLDNRN
2005 3065 DCA BUFF3
2006 1200 TAD ERR3A
2007 4442 JMS I ERRORP
2010 7402 HLT
2011 5600 JMP I ERR3A
2012 6301 K6301, 6301

```

/ERROR3B: PLLR IOT WHERE AC NOT 31, 32, OR 33 DOES NOT SET FLAG IN 9 MS.

```

2013 0000 ERR3B, 0
2014 7200 CLA
2015 1225 TAD K6302
2016 3063 DCA BUFF1
2017 1777 TAD KLDNRN
2020 3065 DCA BUFF3
2021 1213 TAD ERR3B
2022 4442 JMS I ERRORP
2023 7402 HLT
2024 5613 JMP I ERR3B
2025 6302 K6302, 6302

```

/ERROR4A: CFLR IOT WITH AC=31 SETS FLAG PRIOR TO 70 MS.

```

2026 0000 ERR4A, 0
2027 7200 CLA
2030 1240 TAD K6401
2031 3063 DCA BUFF1
2032 1226 TAD ERR4A
2033 4442 JMS I ERRORP
2034 7402 HLT
2035 4443 JMS I QCONTP
2036 5626 JMP I ERR4A
2037 5776 JMP I TST4+1
2040 6401 K6401, 6401

```

/ERROR4B: CFLR IOT WITH AC=31 DOES NOT SET FLAG IN 70 MS.

```

2041 0000 ERR4B, 0
2042 7200 CLA
2043 1253 TAD K6402
2044 3063 DCA BUFF1
2045 1241 TAD ERR4B
2046 4442 JMS I ERRORP
2047 7402 HLT
2050 4443 JMS I QCONTP
2051 5641 JMP I ERR4B
2052 5776 JMP I TST4+1
2053 6402 K6402, 6402

```

/ERROR4C: CFLR IOT WITH AC=32 SETS FLAG PRIOR TO 70 MS.

2054	0000	ERR4C,	0
2055	7200		CLA
2056	1266		TAD K6403
2057	3063		DCA BUFF1
2060	1254		TAD ERR4C
2061	4442		JMS I ERRORP
2062	7402		HLT
2063	4443		JMS I QCONTP
2064	5654		JMP I ERR4C
2065	5776		JMP TST4+1
2066	6403	K6403,	6403

/ERROR4D: CFLR IOT WITH AC=32 DOES NOT SET FLAG IN 70 MS.

2067	0000	ERR4D,	0
2070	7200		CLA
2071	1301		TAD K6404
2072	3063		DCA BUFF1
2073	1267		TAD ERR4D
2074	4442		JMS I ERRORP
2075	7402		HLT
2076	4443		JMS I QCONTP
2077	5667		JMP I ERR4D
2100	5776		JMP TST4+1
2101	6404	K6404,	6404

/ERROR4E: CFLR IOT DOES NOT INITIALLY CLEAR FLAG OR
/PLSF SKIPS WHEN FLAG CLEAR.

2102	0000	ERR4E,	0
2103	7200		CLA
2104	1314		TAD K6405
2105	3063		DCA BUFF1
2106	1302		TAD ERR4E
2107	4442		JMS I ERRORP
2110	7402		HLT
2111	4443		JMS I QCONTP
2112	5702		JMP I ERR4E
2113	5776		JMP TST4+1
2114	6405	K6405,	6405

/ERROR5A: CFLR IOT WITH AC NOT 31, 32 OR 33 SET FLAG PRIOR TO 5 MS.

2115	0000	ERR5A,	0
2116	7200		CLA
2117	1327		TAD K6501
2120	3063		DCA BUFF1
2121	1775		TAD KCFLRN
2122	3065		DCA BUFF3
2123	1315		TAD ERR5A
2124	4442		JMS I ERRORP

```

2125 7402      HLT
2126 5715      JMP I   ERR5A
2127 6501      K6501, 6501

```

/ERROR5B: CFLR IOT WITH AC NOT 31, 32, OR 33 DOES NOT SET FLAG IN 5 MS.

```

2130 0000      ERR5B, 0
2131 7200      CLA
2132 1342      TAD      K6502
2133 3063      DCA      BUFF1
2134 1775      TAD      KCFLRN
2135 3065      DCA      BUFF3
2136 1330      TAD      ERR5B
2137 4442      JMS I   ERRORP
2140 7402      HLT
2141 5730      JMP I   ERR5B
2142 6502      K6502, 6502

```

/ERROR6A: INTERRUPT DOES NOT OCCUR AFTER PLSE IOT AND FLAG SET.

```

2143 0000      ERR6A, 0
2144 7200      CLA
2145 1355      TAD      K6601
2146 3063      DCA      BUFF1
2147 1343      TAD      ERR6A
2150 4442      JMS I   ERRORP
2151 7402      HLT
2152 4443      JMS I   QCONTP
2153 5743      JMP I   ERR6A
2154 5774      JMP      TST6+1
2155 6601      K6601, 6601

```

/ERROR6B: INTERRUPT OCCURS AFTER PLSE IOT AND FLAG CLEAR.

```

2156 0000      ERR6B, 0
2157 7200      CLA
2160 1370      TAD      K6602
2161 3063      DCA      BUFF1
2162 1356      TAD      ERR6B
2163 4442      JMS I   ERRORP
2164 7402      HLT
2165 4443      JMS I   QCONTP
2166 5756      JMP I   ERR6B
2167 5774      JMP      TST6+1
2170 6602      K6602, 6602

```

```

2174 1457
2175 1451
2176 1261
2177 1253
2200      *2200

```

/ERROR6C: INTERRUPT OCCURS AFTER PLCE IOT AND FLAG SET.

```

2200 0000 ERR6C, 0
2201 7200 CLA
2202 1212 TAD K6603
2203 3063 DCA BUFF1
2204 1200 TAD ERR6C
2205 4442 JMS I ERRORP
2206 7402 HLT
2207 4443 JMS I QCONTP
2210 5600 JMP I ERR6C
2211 5777 JMP TST6+1
2212 6603 K6603, 6603

```

/ERROR6D: INTERRUPT OCCURS AFTER PLCE IOT AND FLAG CLEAR.

```

2213 0000 ERR6D, 0
2214 7200 CLA
2215 1225 TAD K6604
2216 3063 DCA BUFF1
2217 1213 TAD ERR6D
2220 4442 JMS I ERRORP
2221 7402 HLT
2222 4443 JMS I QCONTP
2223 5613 JMP I ERR6D
2224 5777 JMP TST6+1
2225 6604 K6604, 6604

```

/ERROR6E: CFLR IOT WITH AC=0 DOES NOT SET FLAG IN 5 MS.

```

2226 0000 ERR6E, 0
2227 7200 CLA
2230 1236 TAD K6605
2231 3063 DCA BUFF1
2232 1226 TAD ERR6E
2233 4442 JMS I ERRORP
2234 7402 HLT
2235 5777 JMP TST6+1
2236 6605 K6605, 6605

```

/ERROR SERVICE ROUTINE. ENTERED WITH PC IN AC AND BUFFS 1 AND 3 STORED.

```

2237 0000 ERRORS, 0
2240 3064 DCA BUFF2 /STORE PC.
2241 7604 LAS /PRINTOUT INHIBITED?
2242 0262 AND K2000
2243 7640 SZA CLA
2244 7410 SKP
2245 4265 JMS ERRMSG /NO. PRINT UNIQUE ERROR MSG.
2246 1264 TAD K5555 /BLANK BUFF3.
2247 3065 DCA BUFF3
2250 7604 LAS
2251 0261 AND K1000
2252 7640 SZA CLA /BELL ON ERROR?

```

2253	4432	JMS I	BELLP	/YES. RING BELL.
2254	7604	LAS		/HALT INHIBITED?
2255	0263	AND	K4000	
2256	7640	SZA CLA		
2257	2237	ISZ	ERRORS	/YES. UPDATE RETURN ADDRESS.
2260	5637	JMP I	ERRORS	/EXIT.
2261	1000	K1000,	1000	
2262	2000	K2000,	2000	
2263	4000	K4000,	4000	
2264	5555	K5555,	5555	

/GENERAL ERROR MESSAGE ROUTINE.
 /ENTERED FROM ERRORS.
 /USING BUFFS 1 THRU 3, INSERTS APPLICABLE INFORMATION IN
 /MESSAGE 16.

2265	0000	ERRMSG, 0		
2266	7300	CLA CLL		
2267	1061	TAD	MSGFLG	/TIME FOR ERROR HEADING?
2270	7640	SZA CLA		
2271	5277	JMP	,+6	
2272	7040	CMA		
2273	3061	DCA	MSGFLG	
2274	1331	TAD	MSG15P	/YES.
2275	4431	JMS I	MSGNP	
2276	4430	JMS I	CRLFP	
2277	1063	TAD	BUFF1	/GET ERROR CODE.
2300	3776	DCA	M16P1	
2301	7040	CMA		
2302	1064	TAD	BUFF2	/GET PC.
2303	3330	DCA	ERRMST	
2304	1330	TAD	ERRMST	
2305	0113	AND	M100	
2306	7112	CLL RTR		
2307	7012	RTR		
2310	7012	RTR		
2311	4333	JMS	EDIT	
2312	3775	DCA	M16P2	
2313	1330	TAD	ERRMST	
2314	0102	AND	K77	
2315	4333	JMS	EDIT	
2316	3774	DCA	M16P3	
2317	1065	TAD	BUFF3	/GET DR IF APPLICABLE
2320	7510	SPA		
2321	7410	SKP		
2322	4333	JMS	EDIT	
2323	3773	DCA	M16P4	
2324	1332	TAD	MSG16P	/PRINT ERROR MSG.
2325	4431	JMS I	MSGNP	
2326	4430	JMS I	CRLFP	
2327	5665	JMP I	ERRMSG	/EXIT.
2330	0000	ERRMST, 0		
2331	4770	MSG15P, MSG15		

2332 5107 MSG16P, MSG16

/EDIT ROUTINE.

/ENTERED WITH 2 DIGIT OCTAL NUMBER IN AC RIGHT JUSTIFIED.
/EXITS WITH TWO NUMBERS IN AC IN PACKED ASCII.

```

2333 0000 EDIT, 0
2334 3347 DCA EDTEM1
2335 1347 TAD EDTEM1
2336 0352 AND K70
2337 7104 CLL RAL
2340 7006 RTL
2341 3350 DCA EDTEM2
2342 1347 TAD EDTEM1
2343 0076 AND K7
2344 1350 TAD EDTEM2
2345 1351 TAD K6060
2346 5733 JMP I EDIT
2347 0000 EDTEM1, 0
2350 0000 EDTEM2, 0
2351 6060 K6060, 6060
2352 0070 K70, 70
    
```

/ROUTINE TO CHECK FOR CONTINUE EXIT. EXITS TO ENTRY +1 IF CONTINUE
/REQUESTED. OTHERWISE EXITS TO ENTRY +2.

```

2353 0000 QCONT, 0
2354 7604 LAS /CONTINUE REQUEST?
2355 0072 AND K2
2356 7650 SNA CLA
2357 2353 ISZ QCONT /NO.
2360 5753 JMP I QCONT /YES.
    
```

/TAPE 4.

/DISPLAY TEST MONITOR. CONTROLLED VIA TTY WITH USER COMMANDS.
/1. MONITOR ACCEPTS TTY INPUTS AT ANY TIME DURING DISPLAY TEST.
/IF INCORRECT SYNTAX IS DETECTED, MONITOR REQUESTS NEW COMMAND STRING.
/THE FOLLOWING COMMANDS CONTROL MONITOR AND HAVE THE RESULT INDICATED.

/COMMAND (*=RETURN)([=ALTMODE) RESULT

```

/[D*- DRAW ALL PATTERNS ONCE.
/[DIP00*- DRAW PATTERN 00 ONCE. (LIMITED TO 1 PATTERN.)
/[DIP00L00:IP00L00*- DRAW PAT00 LINE00, ETC. (LIMITED TO 10 OPERANDS)
/[L*- LOOP ON ALL PATTERNS CONTINUOUSLY.
/[LIP00*- LOOP ON PATTERN 00 CONTINUOUSLY (1 PATTERN)
/[LIP00L00:IP00L00*- LOOP ON PAT00 LINE00, ETC. CONTINUOUSLY (10 OPERANDS)
/[C*- CONTINUE EXECUTING PREVIOUS COMMAND.
/[CL*- CONTINUE LAST COMMAND BUT SET LOOP MODE.
/[CD*- CONTINUE LAST COMMAND BUT CLEAR LOOP MODE.
/[T*- TERMINATE DISPLAY MONITOR; GO TO CONTROL TEST EXECUTIVE.
/[I*- TERMINATE DISPLAY TEST
/ MONITOR, GO TO INIT FOR
/ NEW SPECIFICATION INPUT,
    
```

```

/
/
/CM=
/
/      SR BIT      YIELD
/      0            M STRING HOLD. SET THIS BIT WHEN
/                   CHANGING SR SETTING.
/      5            0=PEN UP; 1=PEN DOWN.
/      6-11        DETERMINE STATE OF DIRECTION
/                   REGISTER FLOPS. CONSULT DIRECTION
/                   CONSTANT LISTS FOR EFFECT ON YOUR
/                   PARTICULAR PLOTTER CONFIGURATION.
/
/CM= CAN BE EXITED ONLY BY TYPING  ALTMODE KEY.
/
/THE COMPLETE COMMAND STRING CAN BE ABORTED BY TYPING "RUBOUT".
/

/NOTES:  PATTERN AND LINE NUMBERS ARE THOSE INDICATED IN
/DOCUMENTATION.
/SPACES MAY BE INPUT AT ANY POINT IN COMMAND STRING.
/EC WILL GENERATE SYNTAX ERROR IF PREVIOUS COMMAND HAS BEEN
/COMPLETED.
/IF IT IS ATTEMPTED TO RUN THE DISPLAY TEST BEFORE RUNNING
/INIT THE MONITOR WILL AUTOMATICALLY GENERATE A RETURN TO INIT
/COMMAND AFTER WHICH THE USER MUST INITIALIZE.

/SPECIAL MONITOR TYPE OUTS:
/"?"  SYNTAX ERROR
/""  REQUEST INPUT.  (RINGS TTY BELL.)

```

2373 5117
2374 5115
2375 5114
2376 5111
2377 1457
2400
2400 7300
2401 1053
2402 7700
2403 5777
2404 4445
2405 3055
2406 4420
2407 4430
2410 4430
2411 1114
2412 3001
2413 1230
2414 3002
2415 1231
2416 4431
2417 4430

```

*2400
DISPLA, CLA CLL
      TAD      ALLSET
      SMA CLA
      JMP      ILLDIS
      JMS I     CPUP
      DCA      PENFLG
      PLCE
      JMS I     CRLFP
      JMS I     CRLFP
      TAD      KJMPI2
      DCA      1
      TAD      ALTP
      DCA      0002
      TAD      MSG20P
      JMS I     MSGNP
      JMS I     CRLFP

```

```

/CHECK INIT DONE.
/GO TO INIT.
/PEN UP AND SET PLOTTER FLAG.
/INDICATE PEN UP.
/CLEAR PLOTTER INT. ENABLE.

/SET UP INTERRUPT LINK.

/PRINT TITLE MSG.

```

2420	6002	REQUES,	IOF		/KILL INTERRUPT.
2421	4430		JMS I	CRLFP	
2422	3060		DCA	ACTFLG	/CLEAR DISPLAY ACTIVE FLAG.
2423	7200	REQU,	CLA		
2424	1232		TAD	K252	/TYPE "*" BELL.
2425	4435		JMS I	TYPEP	
2426	4432		JMS I	BELLP	
2427	5246		JMP	ALT1	/GO TO ALT MODE STRING
2430	2433	ALTP,	ALT		
2431	5016	MSG20P,	MSG20		
2432	0252	K252,	252		
		/ALT MODE STRING. CHECKS FOR ALTMODE, D,L,C,I,M, OR T.			
2433	3051	ALT,	DCA	SAVEAC	/ENTER FROM INTERRUPT.
2434	7010		RAR		/SAVE AC AND LINK.
2435	3052		DCA	SAVEL	
2436	7040		CMA		/SET DISPLAY ACTIVE FLAG.
2437	3060		DCA	ACTFLG	
2440	6031		KSF		/KEYBOARD INTERRUPT?
2441	7410		SKP		
2442	5250		JMP	ALT1+2	/YES. GO ON TO ALTMODE STRING.
2443	4420		PLCE		
2444	6042		TCF		/CLEAR TIO FLAG AND PLOT INT ENABLE.
2445	5776		JMP	C	/NO. JUMP TO C STRING AND CONTINUE.
2446	6031	ALT1,	KSF		/ENTER FROM REQUES. WAIT FOR
2447	5246		JMP	.-1	/ALTMODE INPUT
2450	6036		KRB		
2451	1310		TAD	M375	
2452	7440		SEA		
2453	5775		JMP	QUES	/INPUT OTHER THAN ALTMODE.
2454	4430		JMS I	CRLFP	
2455	1307		TAD	K333	/INPUT IS ALTMODE.
2456	4435		JMS I	TYPEP	/TYPE "C".
2457	1774		TAD	PLCT	/SET COMMAND TEMPS TO 0.
2460	3305		DCA	ALTTEM	
2461	1110		TAD	M24	
2462	3306		DCA	ALTCNT	
2463	2305		ISE	ALTTEM	
2464	3705		DCA I	ALTTEM	
2465	2306		ISE	ALTCNT	
2466	5263		JMP	.-3	
2467	4773		JMS	LISN	/LOOK FOR NEXT INPUT.
2470	4772		JMS	QD	
2471	5771		JMP	D	/D STRING
2472	4770		JMS	QL	
2473	5767		JMP	L	/L STRING
2474	4766		JMS	QC	
2475	5776		JMP	C	/C STRING.
2476	4765		JMS	QT	
2477	5764		JMP	T	/T STRING.
2500	4763		JMS	QI	
2501	5762		JMP	IS	/I STRING.
2502	4761		JMS	QM	
2503	5760		JMP	MS	/M STRING.

2504 5775' JMP QUES /SYNTAX ERROR. "EX".

2505 0000 ALTTEM, 0
2506 0000 ALTCNT, 0
2507 0333 K333, 333
2510 7403 M375, -375

2560 2600
2561 3436
2562 3147
2563 3431
2564 3142
2565 3424
2566 3417
2567 2667
2570 3412
2571 2664
2572 3405
2573 4200
2574 5664
2575 3400
2576 3102
2577 3352
2600

*2600

/

/M STRING. ENABLES USER TO COMMAND PLOTTER FROM THE
/SWITCH REGISTER.

/M STRING CAN BE EXITED ONLY BY TYPING ALTMODE KEY.

2600 4777'	MS,	JMS	LISN	/LOOK FOR "CR".
2601 4776'		JMS	QCR	
2602 7610		SKP	CLA	
2603 5775'		JMP	QUES	/SYNTAX ERROR. "EMX"
2604 6001		ION		/SET INTERRUPT TO WAIT FOR EXIT.
2605 7040		CMA		
2606 3261		DCA	MSTEMP	/SET FOR COMPARE.
2607 4445		JMS I	CPUP	/INITIALIZE TO PEN UP.
2610 3055		DCA	PENFLG	
2611 7604	MS1,	LAS		/READ SR.
2612 7500		SMA		/MSTRING HOLD?
2613 5217		JMP	..+4	
2614 0262		AND	K177	/YES. SAVE SR 5-11.
2615 3261		DCA	MSTEMP	
2616 5211		JMP	MS1	
2617 0262		AND	K177	/NO. COMPARE PRESENT SR5-11
2620 7041		CIA		/WITH PREVIOUS SR5-11.
2621 1261		TAD	MSTEMP	
2622 7650		SNA	CLA	/ARE THEY THE SAME?
2623 5236		JMP	MGO	
2624 6002		IOF		/NO. INSTRUCT USER TO
2625 4430		JMS I	CRLFP	/SET SR0 TO A 1 WHEN CHANGING
2626 1263		TAD	MSG23P	/SR5-11.
2627 4431		JMS I	MSGNP	
2630 4430		JMS I	CRLFP	

2631	6001	ION			
2632	7604	LAS			/WAIT TILL USER SETS
2633	7700	SMA CLA			/SR0 TO A 1.
2634	5232	JMP	.-2		
2635	5211	JMP	MS1		
2636	1261	TAD	MSTEMP		/CONTINUE WITH M STRING.
2637	0067	AND	K100		/PEN DOWN?
2640	7640	SZA CLA			
2641	5247	JMP	+.6		
2642	1055	TAD	PENFLG		/NO. PEN UP.
2643	7650	SNA CLA			/PEN ALREADY UP?
2644	5253	JMP	MS2		/YES.
2645	4447	JMS I	DPUP		/NO. PUT PEN UP.
2646	5253	JMP	MS2		/GO TO NEXT PART.
2647	1055	TAD	PENFLG		/PEN DOWN.
2650	7640	SZA CLA			/PEN ALREADY DOWN?
2651	5253	JMP	MS2		/YES.
2652	4450	JMS I	DPDP		/NO. PUT PEN DOWN.
2653	1261	TAD	MSTEMP		/GET SR.
2654	0102	AND	K77		/MASK DR.
2655	4421	PLSF			/WAIT FOR PLOT FLAG.
2656	5255	JMP	.-1		
2657	4426	CFLR			/CLEAR FLAG. LOAD DR. SET FLAG.
2660	5211	JMP	MS1		/CHECK SR AGAIN.
2661	0000	MSTEMP, 0			/SR TEMP.
2662	0177	K177, 177			
2663	5063	MSG23P, MSG23			

/D STRING. CLEARS LOOP FLAG.

2664	7200	D,	CLA		
2665	3054		DCA	LOPFLG	/CLEAR LOOP FLAG
2666	5774		JMP	DL	/D-L STRING.

/L STRING. SETS LOOP FLAG.

2667	7240	L,	CLA CMA		
2670	3054		DCA	LOPFLG	/SET LOOP FLAG
2671	5774		JMP	DL	/D-L STRING.

/D-L STRING. DETERMINES OPERANDS FOR D AND L COMMANDS. IF LOOP
/FLAG IS SET (COMMAND L), LOOPS ON EXECUTION.

2774	3000				
2775	3400				
2776	3443				
2777	4200				
	3000	*3000			
3000	4445	DL,	JMS I	CPUP	
3001	3055		DCA	PENFLG	
3002	4777		JMS	LISN	

3003	4776'	JMS	QCR	
3004	5210	JMP	DLCR	/D CR OR L CR.
3005	4775'	JMS	QSEM	
3006	5231	JMP	DLSEM	/D; OR L;.
3007	5774'	JMP	QUES	/SYNTAX ERROR. DX OR LX.
3010	7201	DLCR,	CLA IAC	/D CR OR L CR.
3011	3773'		DCA STRTP	/SET START PAT NUM TO 1.
3012	7201		CLA IAC	
3013	3772'		DCA STRTL	/SET START LINE NUM TO 1.
3014	1086		TAD K6	
3015	3771'		DCA STPP	/SET STOP PAT NUM TO 6.
3016	1770'		TAD LASLIN	
3017	1086		TAD K6	
3020	3276		DCA DLTEM1	
3021	1676		TAD I DLTEM1	
3022	7001		IAC	
3023	3767'		DCA STPL	/SET STOP LINE NUM TO LAST LINE OF
3024	4766'		JMS LGEN	/PATTERN NUMBER 6. EXECUTE.
3025	1054		TAD LOPFLG	/LOOP FLAG SET?
3026	7640		SEA CLA	
3027	5210		JMP DLCR	/YES. DO IT AGAIN.
3030	5765'		JMP REQUES	/NO. REQUEST NEW COMMAND
3031	4764'	DLSEM,	JMS OP	/OP STRING. (D) OR (L);).
3032	1111		TAD M12	/SET COUNTER
3033	3277		DCA DLCNT	
3034	1763'		TAD PLCH	/SET PAT NUM HOLDER POINTER.
3035	3300		DCA DL1	
3036	1075		TAD K12	/SET LINE NUM HOLDER POINTER.
3037	1763'		TAD PLCH	
3040	3301		DCA DL2	
3041	2300	DLSEML,	ISE DL1	/UPDATE POINTERS
3042	2301		ISE DL2	
3043	1700		TAD I DL1	/GET PAT NUM.
3044	7450		SNA	
3045	5272		JMP DLSEMC	/PAT NUM=0. CHECK FOR EXIT.
3046	3773'		DCA STRTP	/PAT NUM OTHER THAN 0; PUT
3047	1773'		TAD STRTP	/IN START PAT NUM.
3050	3771'		DCA STPP	/PUT IN STOP PAT NUM.
3051	1701		TAD I DL2	/GET LINE NUM.
3052	7450		SNA	/LINE NUM=0?
3053	5260		JMP .+5	/YES.
3054	3772'		DCA STRTL	/NO. PUT IN START LINE NUM.
3055	1772'		TAD STRTL	/PUT IN STOP LINE NUM.
3056	3767'		DCA STPL	
3057	5267		JMP DLEX	/EXECUTE.
3060	1773'		TAD STRTP	/LINE NUM=0
3061	1770'		TAD LASLIN	/SET STOP LINE TO NEXT TO LAST
3062	3276		DCA DLTEM1	/LINE NUM.
3063	1676		TAD I DLTEM1	
3064	3767'		DCA STPL	
3065	7201		CLA IAC	/SET START LINE NUM=1.
3066	3772'		DCA STRTL	
3067	4766'	DLEX,	JMS LGEN	/EXECUTE.
3070	2277		ISE DLCNT	/UPDATE COUNTER. DONE?

```

3071 5241      JMP      DLSEML      /NO. GET NEXT PAT-LIN SET.
3072 1054      DLSEMC, TAD      LOPFLG      /YES. LOOP SET?
3073 7640      SZA CLA
3074 5232      JMP      DLSEM+1      /YES. DO IT ALL AGAIN
3075 5765      JMP      REQUES      /NO. ASK FOR NEXT COMMAND.
3076 0000      DLTEM1, 0
3077 0000      DLCONT, 0
3100 0000      DL1, 0
3101 0000      DL2, 0

```

/C STRING: CONTINUES PREVIOUS COMMAND IF NOT COMPLETED.
 /IF PREVIOUS COMMAND COMPLETED, RESULTS IN SYNTAX ERROR.
 /IF COMMAND IS "ECL" SET LOOP FLAG AND CONTINUE; IF "ECD"
 /CLEAR LOOP FLAG; IF "EC" LEAVE LOOP FLAG IN PRESENT STATUS.

```

3102 7200      G,      CLA
3103 1060      TAD      ACTFLG      /PATTERN INTERRUPTED?
3104 7650      SNA CLA
3105 5774      JMP      QUES      /NO. SYNTAX ERROR. NOTHING TO CONTINUE.
3106 4777      JMS      LISN      /LOOK FOR NEW CHARACTER.
3107 4776      JMS      QCR      /CR?
3110 5326      JMP      COUT      /YES. CONTINUE AS BEFORE.
3111 4762      JMS      QL      /"L"?
3112 5316      JMP      .+4      /YES.
3113 4761      JMS      QD      /"D"?
3114 5320      JMP      .+4      /YES.
3115 5774      JMP      QUES      /SYNTAX ERROR. "ECX"
3116 7240      CLA CMA      /"L". SET LOOP FLAG.
3117 7410      SKP
3120 7200      CLA      /"D". CLEAR LOOP FLAG.
3121 3054      DCA      LOPFLG
3122 4777      JMS      LISN      /LOOK FOR CR.
3123 4776      JMS      QCR      /CR?
3124 7410      SKP
3125 5774      JMP      QUES      /YES.
3126 4445      COUT, JMS I      CPUP      /NO. SYNTAX ERROR, "ECLX" OR "ECDX"
3127 1055      TAD      PENFLG      /YES. SET PLOT FLAG.
3130 7640      SZA CLA      /WAS PEN UP OR PEN DOWN?
3131 5334      JMP      .+3
3132 4447      JMS I      DPUP      /PEN UP.
3133 7410      SKP
3134 4450      JMS I      DPDP      /PEN DOWN.
3135 1052      TAD      SAVEL      /RESTORE LINK AND AC.
3136 7104      CLL RAL
3137 1051      TAD      SAVEAC
3140 6001      ION
3141 5400      JMP I      0000      /SET INTERRUPT.
                                   /EXIT.

```

/T STRING. KILLS MONITOR AND GOES TO CONTROL TEST.

```

3142 4777' T, JMS LISN /LOOK FOR "CR"
3143 4776' JMS QCR
3144 7610 SKP CLA
3145 5774' JMP QUES /SYNTAX ERROR. "[TX"
3146 5760' JMP EXECS

```

/I STRING. TERMINATES DISPLAY MONITOR, GOES TO INIT FOR NEW
/SPECIFICATION INPUT, AND RETURNS TO DISPLAY MONITOR.

```

3147 4777' IS, JMS LISN /LOOK FOR "CR"
3150 4776' JMS QCR
3151 7610 SKP CLA
3152 5774' JMP QUES /SYNTAX ERROR. "[IX"
3153 4757' JMS INIT /GO TO INIT.
3154 5756' JMP DISPLA /UPON RETURN GO TO MONITOR.

```

/OP STRING1 CHECKS PXXLXX SYNTAX AND SETS UP PATTERN AND LINE
/NUMBER TEMPORARY AND PERMANENT LISTS FOR EXECUTION
/OF D AND L COMMANDS.

```

3156 2400
3157 0400
3160 0317
3161 3405
3162 3412
3163 5637
3164 3200
3165 2420
3166 3516
3167 3537
3170 5345
3171 3540
3172 3535
3173 3536
3174 3400
3175 3450
3176 3443
3177 4200
3200 3200

```

*3200

```

3200 0000 OP, 0
3201 7300 CLA CLL /SET PAT-LINE TEMPORARIES POINTERS.
3202 1777' TAD PLCT
3203 3343 DCA OP1
3204 1343 TAD OP1
3205 1075 TAD K12
3206 3344 DCA OP2
3207 1351 TAD M13 /SET OPERAND LIMIT OF 10.
3210 3345 DCA OPCNT
3211 2343 OPUP, ISZ OP1 /UPDATE POINTERS.
3212 2344 ISZ OP2
3213 2345 ISZ OPCNT /LIMIT OF 10 OPERANDS EXCEEDED?
3214 7410 SKP

```

3215	5776'	JMP	QUES	/YES.
3216	7240	CLA CMA		/NO. SET PATTERN FLAG
3217	3346	DCA	OPFLG	
3220	3347	DCA	FINFLG	/CLEAR DONE FLAG.
3221	4247	JMS	OPQ	/GET INPUT AND CHECK SYNTAX.
3222	7200	CLA		
3223	3346	DCA	OPFLG	/SET NUMBER FLAG
3224	3347	DCA	FINFLG	/CLEAR DONE FLAG
3225	4247	JMS	OPQ	/GET INPUT AND CHECK SYNTAX.
3226	3743	DCA I	OP1	/STORE PAT NUM IN TEMP.
3227	7201	CLA IAC		/SET LINE FLAG.
3230	3346	DCA	OPFLG	
3231	7240	CLA CMA		/SET DONE FLAG.
3232	3347	DCA	FINFLG	
3233	4247	JMS	OPQ	/GET INPUT AND CHECK SYNTAX.
3234	7200	CLA		/SET NUMBER FLAG.
3235	3346	DCA	OPFLG	/
3236	3347	DCA	FINFLG	/CLEAR DONE FLAG.
3237	4247	JMS	OPQ	/GET INPUT AND CHECK SYNTAX.
3240	3744	DCA I	OP2	/PUT LINE NUM IN TEMP.
3241	7240	CLA CMA		
3242	3347	DCA	FINFLG	/SET DONE FLAG.
3243	7240	CLA CMA		
3244	3346	DCA	OPFLG	/REMOVE NUMBER FLAG.
3245	4775'	JMS	LISN	/GET NEXT INPUT.
3246	5255	JMP	OPQ1	/LOOK FOR CR OR ";"
3247	0000	OPQ,	0	
3250	4775'	JMS	LISN	
3251	4774'	JMS	QP	
3252	5302	JMP	P	/"P"
3253	4773'	JMS	QL	
3254	5307	JMP	YL	/"L"
3255	4772'	OPQ1,	JMS QSEM	
3256	5314	JMP	SEM	/";"
3257	4771'	JMS	QCR	
3260	5321	JMP	CR	/CR.
3261	4770'	JMS	QNUM	/IS INPUT NUMBER? (OCTAL)
3262	7410	SKP		
3263	5776'	JMP	QUES	/NO. SYNTAX ERROR.
3264	7104	CLL RAL		/YES. SAVE NUMBER
3265	7006	RTL		
3266	3350	DCA	OPTEM	
3267	4775'	JMS	LISN	/GET ANOTHER INPUT.
3270	4770'	JMS	QNUM	/IS INPUT NUMBER? (OCTAL)
3271	7410	SKP		
3272	5776'	JMP	QUES	/NO. SYNTAX ERROR.
3273	1350	TAD	OPTEM	/YES. PUT NUMBER IN PROPER FORM.
3274	3350	DCA	OPTEM	
3275	1346	TAD	OPFLG	/DID WE WANT NUMBERS?
3276	7640	SZA CLA		
3277	5776'	JMP	QUES	/NO. SYNTAX ERROR
3300	1350	TAD	OPTEM	/YES. EXIT.
3301	5647	JMP I	OPQ	
3302	7200	CLA		/"P"

3303	1346	TAD	OPFLG	/TIME FOR "P"?
3304	7500	SMA		
3305	5776	JMP	QUES	/NO. SYNTAX ERROR.
3306	5647	JMP I	OPQ	
3307	7200	CLA		/"L"
3310	1346	TAD	OPFLG	/TIME FOR "L"?
3311	7500	SPA SNA		
3312	5776	JMP	QUES	/NO. SYNTAX ERROR.
3313	5647	JMP I	OPQ	
3314	7200	CLA		/"I"
3315	1347	TAD	FINFLG	/TIME FOR "I"?
3316	7650	SNA CLA		
3317	5776	JMP	QUES	/NO. SYNTAX ERROR.
3320	5211	JMP	OPUP	
3321	7200	CLA		/CR?
3322	1347	TAD	FINFLG	/TIME FOR CR?
3323	7650	SNA CLA		
3324	5776	JMP	QUES	/NO. SYNTAX ERROR
3325	1777	TAD	PLCT	/YES. MOVE TEMPS TO HOLDS.
3326	3343	DCA	OP1	
3327	1767	TAD	PLCH	
3330	3344	DCA	OP2	
3331	1110	TAD	M24	
3332	3345	DCA	OPCNT	
3333	2343	ISE	OP1	
3334	2344	ISE	OP2	
3335	1743	TAD I	OP1	
3336	3744	DCA I	OP2	
3337	2345	ISE	OPCNT	
3340	5333	JMP	.-5	
3341	7300	CLA CLL		
3342	5600	JMP I	OP	/EXIT.
3343	0000	OP1,	0	
3344	0000	OP2,	0	
3345	0000	OPCNT,	0	
3346	0000	OPFLG,	0	
3347	0000	FINFLG,	0	
3350	0000	OPTEM,	0	
3351	7765	M13,	-13	
3352	4430	ILLDIS, JMS I	CRLFP	/ROUTINE TO TELL USER THAT
3353	1362	TAD	MSG21P	/AN ATTEMPT HAS BEEN MADE TO RUN THE
3354	4431	JMS I	MSGNP	/DISPLAY TEST WITHOUT INIT.
3355	4430	JMS I	CRLFP	/THEN GOES TO INIT.
3356	1363	TAD	MSG22P	
3357	4431	JMS I	MSGNP	
3360	4766	JMS	INIT	
3361	5765	JMP	DISPLA	
3362	5031	MSG21P, MSG21		
3363	5061	MSG22P, MSG22		

/QUESTION: SYNTAX ERROR.

3365 2400
3366 0400

3367 5637
 3370 3473
 3371 3443
 3372 3450
 3373 3412
 3374 3455
 3375 4200
 3376 3400
 3377 5664
 3400

*3400

3400 4430
 3401 1204
 3402 4435
 3403 5777
 3404 0277

QUES, JMS I CRLF
 TAD K277 /TYPE "?"
 JMS I TYPEP
 JMP REQU /EXIT TO REQUEST.
 K277, 277

/CHARACTER DETECTION SUBROUTINES.

3405 0000

QD, 0 /"D"?

3406 4262

JMS Q

3407 7474

-304

3410 2205

ISE QD

3411 5605

JMP I QD

3412 0000

QL, 0 /"L"?

3413 4262

JMS Q

3414 7464

-314

3415 2212

ISE QL

3416 5612

JMP I QL

3417 0000

QC, 0 /"C"?

3420 4262

JMS Q

3421 7475

-303

3422 2217

ISE QC

3423 5617

JMP I QC

3424 0000

QT, 0 /"T"?

3425 4262

JMS Q

3426 7454

-324

3427 2224

ISE QT

3430 5624

JMP I QT

3431 0000

QI, 0 /I?

3432 4262

JMS Q

3433 7467

-311

3434 2231

ISE QI

3435 5631

JMP I QI

3436 0000

QM, 0 /M?

3437 4262

JMS Q

3440 7463

-315

3441 2236

ISE QM

3442 5636

JMP I QM

3443 0000

QCR, 0 /CR?

3444 4262

JMS Q

3445 7563

-215

3446 2243

ISE QCR

3447 5643

JMP I QCR

3450 0000

QSEM, 0 /"I"?

3451 4262

JMS Q

3452 7505

-273

```

3453 2250      ISZ      QSEM
3454 5650      JMP I    QSEM
3455 0000      QP,      0          /"P"?
3456 4262      JMS      Q
3457 7460      -320
3460 2255      ISZ      QP
3461 5655      JMP I    QP
3462 0000      Q,      0          /QX SERVICE.
3463 3313      DCA      QHOLD
3464 1662      TAD I    Q
3465 2262      ISZ      Q
3466 1313      TAD      QHOLD
3467 7650      SNA CLA
3470 2262      ISZ      Q
3471 1313      TAD      QHOLD
3472 5662      JMP I    Q

3473 0000      QNUM,    0          /"NUMBER"?
3474 3314      DCA      QNUMT
3475 1314      TAD      QNUMT
3476 1315      TAD      M260
3477 7510      SPA
3500 5307      JMP      QNUM1
3501 1106      TAD      M10
3502 7700      SMA CLA
3503 5307      JMP      QNUM1
3504 1314      TAD      QNUMT
3505 0076      AND      K7
3506 9673      JMP I    QNUM
3507 2273      QNUM1,   ISZ      QNUM
3510 7200      CLA
3511 1314      TAD      QNUMT
3512 9673      JMP I    QNUM
3513 0000      QHOLD,    0
3514 0000      QNUMT,    0
3515 7520      M260,    -260

```

```

/TAPE 5.
/LINE GENERATOR INTERPRETIVE ROUTINE.
/1. ENTERED WITH PATTERN NUMBERS AND LINE NUMBERS SUPPLIED
/TO "STRTP", STRTL", "STPP", "STPL", BY ENTERING ROUTINE.
/2. USING PAT AND LINE NUMS COMPUTES START AND STOP POINTERS.
/3. INTERPRETS THE LIST OF LIDS FROM START THRU STOP AS FOLLOWS:
/3.1 IF LID IS SPECIAL TYPE (3XXY), TAKES NEXT Y LIDS
/AS A SEQUENCE AND INTERPRETS THEM (XX) TIMES.
/3.2 IF LID IS STANDARD, INTERPRETS TO FIND OUT FOLLOWING
/INFORMATION, THEN DRAWS SPECIFIED LINE.
/3.2.1 PEN UP OR DOWN.
/3.2.2 LENGTH OF VECTOR.
/3.2.3 DIRECTION OF VECTOR.
/4. EXITS WITH AC AND LINK CLEAR.
/

```

```

3516 0000
3517 4420

```

```

LGEN, 0
PLCE

```

/KILL PLOTTER INTERRUPT

3520	6001	ION		/SET INTERRUPT
3521	7200	CLA		
3522	1335	TAD	STRTL	
3523	3776	DCA	LINUM	
3524	1336	TAD	STRTP	
3525	4775	JMS	LINCNT	/GET START POINTER.
3526	3774	DCA	START	/SAVE.
3527	1337	TAD	STPL	
3530	3776	DCA	LINUM	
3531	1340	TAD	STPP	
3532	4775	JMS	LINCNT	/GET STOP POINTER.
3533	3773	DCA	STOP	/SAVE.
3534	3772	JMP	LGENL1	
3535	0000	STRTL,	0	
3536	0000	STRTP,	0	
3537	0000	STPL,	0	
3540	0000	STPP,	0	
/				
/				
/				
/				
3572	3600			
3573	3731			
3574	3730			
3575	4000			
3576	4050			
3577	2423			
	3600	*3600		
3600	7300	LGENL1,	CLA CLL	
3601	1730	TAD I	START	/IS PRESENT LID STANDARD OR SPECIAL.
3602	7006	RTL		
3603	7420	SNL		
3604	5243	JMP	STD	/STANDARD.
3605	7500	SMA		
3606	5243	JMP	STD	/STANDARD.
3607	7012	SPEC,	RTR	/SPECIAL (3XXY).
3610	0076	AND	K7	/EXTRACT Y, NEGATE AND SAVE.
3611	7041	CIA		
3612	3332	DCA	SPTALH	
3613	1730	TAD I	START	/EXTRACT XX, COMPUTE REPETITION
3614	7010	RAR		/CONSTANT POINTER, GET REPETITION
3615	7012	RTR		
3616	0102	AND	K77	
3617	1777	TAD	REP K	/CONSTANT, NEGATE, AND PUT IN
3620	3333	DCA	SPT1	/COUNTER.
3621	1733	TAD I	SPT1	
3622	7041	CIA		
3623	3334	DCA	SPREP	
3624	1330	SPECL1,	TAD	/SET UP SPECIAL LID STARTING POINTER.
3625	3335	DCA	SP1P	
3626	1332	TAD	SPTALH	/SET UP LID COUNTER.
3627	3336	DCA	SPTAL	

3630	2335	SPECL2,	ISE	SP1P	/SET UP TO GET NEXT LID.
3631	1735	TAD	I	SP1P	/GET NEXT LID (WHICH MUST BE
3632	4266	JMS		XLATE	/STANDARD) AND EXECUTE.
3633	2336	ISE		SPTAL	/ARE ALL LIDS IN SPECIAL LIST DONE?
3634	5230	JMP		SPECL2	/NO. DO NEXT LID.
3635	2334	ISE		SPREP	/HAS SEQUENCE OF LIDS BEEN REPEATED
3636	5224	JMP		SPECL1	/SPECIFIED NUMBER OF TIMES? NO.
3637	7240	CLA	CMA		
3640	1332	TAD		SPTALH	
3641	3332	DCA		SPTALH	
3642	5247	JMP		LGENCK	/YES. GO CHECK IF LGEN DONE.
3643	7012	STD,	RTR		/LID IS STANDARD.
3644	4266	JMS		XLATE	/EXECUTE LID.
3645	7040	CMA			/SET MODIFIER TO -1.
3646	3332	DCA		SPTALH	
3647	1330	LGENCK,	TAD	START	/LGEN DONE? (LID LIST BEEN
3650	7041	CIA			/(PROCESSED FROM START THRU STOP?)
3651	1331	TAD		STOP	
3652	7100	CLL			
3653	7050	SNA	CLA		
3654	5262	JMP		LGENRT	/YES. EXIT.
3655	1332	TAD		SPTALH	/NO. UPDATE START WITH MODIFIER
3656	7041	CIA			/AND GET NEXT LID.
3657	1330	TAD		START	
3660	3330	DCA		START	
3661	5200	JMP		LGENL1	
3662	7200	LGENRT,	CLA		
3663	1776	TAD		LGEN	
3664	3266	DCA		.+2	
3665	5666	JMP	I	.+1	
3666	0000	XLATE,	0		/ENTER WITH LID IN AC.
3667	3337	DCA		LID	/SAVE LID.
3670	1337	TAD		LID	/GET LID.
3671	7710	SPA	CLA		/PEN UP OR DOWN REQUESTED?
3672	5277	JMP		.+5	
3673	1055	TAD		PENFLG	/PEN UP. DOES PEN FLAG INDICATE
3674	7640	SEA	CLA		/PEN ALREADY UP?
3675	4447	JMS	I	DPUP	/NO. PEN UP.
3676	5302	JMP		.+4	
3677	1055	TAD		PENFLG	/PEN DOWN. DOES PEN FLAG INDICATE
3700	7650	SNA	CLA		/PEN ALREADY DOWN.
3701	4450	JMS	I	DPDP	/NO. PEN DOWN.
3702	1337	TAD		LID	/GET LID AND COMPUTE LENGTH CONSTANT
3703	0344	AND		K3740	/POINTER AND SAVE.
3704	7112	CLL	RTR		
3705	7012	RTR			
3706	7010	RAR			
3707	1775	TAD		LGTHP	
3710	3340	DCA		XLGP	
3711	1740	TAD	I	XLGP	/GET LENGTH CONSTANT AND SAVE.
3712	7041	CIA			
3713	3341	DCA		LENGTH	
3714	1337	TAD		LID	/GET LID AND COMPUTE DIRECTION

START,	0
STOP,	0
SPTALH,	0
SPT1,	0
SPREP,	0
SP1P,	0
SPTAL,	0
LID,	0
XLGP,	0
LENGTH,	0
DKP,	0
K37,	37
K3740,	3740

1

*4000
LINCNT, 0

```

      TAD      PATS      /COMPUTE PATTERN POINTER
      DCA      PATPT     /AND SAVE.
      TAD I      PATPT
      DCA      PATPT
      TAD      PATPT     /SET LID POINTER TO PATTERN
      DCA      LIDPT     /POINTER.
      TAD      LINUM     /SET LINE TALLY EQUAL TO
      CIA      /-LINE NUMBER.
      DCA      LINTAL
      DCA      ADDCNT     /SET ADDRESS COUNTER TO 0.
LINL1, TAD I      LIDPT  /GET LID.
      CLL RTL     /IS LID SPECIAL OR STD?
      SNL

```

```

4016 5230      JMP      LINCK      /STD.
4017 7500      SMA
4020 5230      JMP      LINCK      /STD.
4021 7012      RTR              /LID IS SPECIAL
4022 0076      AND      K7        /GET AND SAVE Y.
4023 3253      DCA      LINTEN
4024 1253      TAD      LINTEN
4025 1247      TAD      LIDPT      /UPDATE LID POINTER TO BYPASS
4026 3247      DCA      LIDPT      /LIDS IN SPECIAL GROUP.
4027 5232      JMP      ,+3
4030 7200      LINCK, CLA
4031 3253      DCA      LINTEN
4032 2251      ISZ      LINTAL
4033 7610      SKP     CLA
4034 5243      JMP      LINFIN
4035 2247      ISZ      LIDPT
4036 1253      TAD      LINTEN
4037 1252      TAD      ADDCNT
4040 3252      DCA      ADDCNT
4041 2252      ISZ      ADDCNT
4042 5213      JMP      LINL1
4043 1252      LINFIN, TAD      ADDCNT
4044 1246      TAD      PATPT
4045 5600      JMP      I      LINCNT
4046 0000      PATPT, 0
4047 0000      LIDPT, 0
4050 0000      LINUM, 0
4051 0000      LINTAL, 0
4052 0000      ADDCNT, 0
4053 0000      LINTEN, 0

```

/PEN SUBROUTINES: EXIT WITH AC CLEAR AND LINK UNCHANGED.

/CONTROL TEST TYPE, 500 SERIES

```

4054 0000      CPU5, 0
4055 7200      CLA
4056 4423      PLPU
4057 5654      JMP      I      CPU5      /PEN UP.
4060 0000      CPD5, 0
4061 7200      CLA
4062 4425      PLPD
4063 5660      JMP      I      CPD5      /PEN DOWN.

```

/DISPLAY TEST TYPE, 500 SERIES.

```

4064 0000      DPUS, 0
4065 7200      CLA
4066 4421      PLSF
4067 5266      JMP      , -1
4070 4423      PLPU
4071 3055      DCA      PENFLG
4072 5664      JMP      I      DPUS      /PEN UP.
4073 0000      DPD5, 0
4074 7240      CLA CMA
4075 4421      PLSF

```

4076	5275	JMP	.-1	
4077	4425	PLPD		/PEN DOWN.
4100	3055	DCA	PENFLG	/SET PEN FLAG.
4101	5673	JMP I	DPD5	

/

/CONTROL TEST TYPE, 600-700 SERIES.

4102	0000	CPU8,	0	
4103	7200		CLA	
4104	1077		TAD	K31 /PEN UP.
4105	4426		CFLR	
4106	7200		CLA	
4107	5702		JMP I	CPU8
4110	0000	CPD8,	0	
4111	7200		CLA	
4112	1100		TAD	K32 /PEN DOWN
4113	4426		CFLR	
4114	7200		CLA	
4115	5710		JMP I	CPD8

/DISPLAY TEST TYPE, 600-700 SERIES.

4116	0000	DPD8,	0	
4117	7200		CLA	
4120	1077		TAD	K31
4121	4421		PLSF	
4122	5321		JMP	.-1
4123	4426		CFLR	/PEN UP.
4124	7200		CLA	
4125	3055		DCA	PENFLG /CLEAR PEN FLAG.
4126	5716		JMP I	DPD8
4127	0000	DPD8,	0	
4130	7200		CLA	
4131	1100		TAD	K32
4132	4421		PLSF	
4133	5332		JMP	.-1
4134	4426		CFLR	/PEN DOWN.
4135	7240		CLA CMA	
4136	3055		DCA	PENFLG /SET PEN FLAG.
4137	5727		JMP I	DPD8

/DECREASE AND SKIP IF ZERO SUBROUTINE. (AC AND LINK ARE PRESERVED.)

/ENTERED WITH:

/	JMS I	DSZP	/LINK DSZP, DSZ
/	ABCD		/COUNTER POINTER.
/	NEXT INST.		

/DECREASES CONTENTS OF ABCD BY 1 AND SKIPS NEXT INST IF 0.

4140	0000	DSZ,	0	
4141	3354		DCA	DSZT1 /SAVE AC.
4142	1740		TAD I	DSZ /GET COUNTER POINTER.
4143	3355		DCA	DSZT2
4144	2340		ISZ	DSZ /UPDATE RETURN ADDRESS.
4145	7240		CLA CMA	/DECREASE COUNTER BY 1.
4146	1755		TAD I	DSZT2
4147	7490		SNA	/COUNTER=0?

```

0150 2340      ISZ      DSZ      /YES, SET SKIP.
0151 3755      DCA I    DSZT2    /RESTORE COUNTER.
0152 1354      TAD      DSZT1    /RESTORE AC.
0153 5740      JMP I    DSZ      /EXIT.
0154 0000      DSZT1, 0
0155 0000      DSZT2, 0

```

/TYPE SUBROUTINE
/ENTERED WITH ASCII IN AC. EXITS WITH TIO FLAG AND AC CLEAR.

```

0156 0000      TYPE, 0
0157 6046      TLS
0158 7200      CLA
0159 6041      TSF
0160 5361      JMP      .-1
0161 6042      TCF
0162 5756      JMP I    TYPE

```

/CARRIAGE RETURN LINE FEED.

```

0165 0000      CRLF, 0
0166 7200      CLA
0167 1375      TAD      K215      /CARRIAGE RETURN.
0168 4435      JMS I    TYPEP
0169 1374      TAD      K212      /LINE FEED.
0170 4435      JMS I    TYPEP
0171 5765      JMP I    CRLF

```

```

0174 0212      K212, 212
0175 0215      K215, 215

```

/READ CHARACTER FROM TTY, AND EXIT WITH ASCII IN AC.
/SPACES ARE ECHOED ON TTY BUT LISN WILL NOT EXIT.

```

0177 5317
0200 4200      *4200
0201 0000      LISN, 0
0202 6031      KSF
0203 5201      JMP      .-1      /WAIT FOR KEYBOARD FLAG.
0204 6036      KRB
0205 3215      DCA      LISNT    /READ BUFFER AND SAVE.
0206 1215      TAD      LISNT
0207 4435      JMS I    TYPEP    /ECHO BUFFER ON TTY.
0208 1215      TAD      LISNT    /DISREGARD "SPACE" IN SYNTAX.
0209 1216      TAD      M240
0210 7650      SNA CLA
0211 5201      JMP      LISN+1    /GET NEXT INPUT IF LAST WAS SPACE.
0212 1215      TAD      LISNT
0213 5600      JMP I    LISN     /EXIT WITH CHARACTER CODE IN AC.
0214 0000      LISNT, 0
0215 7540      M240, -240

```

/READ NUMBER SUBROUTINE. READS ONE NUMBER FROM TTY. EXITS WITH
/OCTAL NUMBER IN AC RIGHT JUSTIFIED.

```
4217 0000 READN, 0
4220 4200      JMS  LISN
4221 0223      AND  K17
4222 5617      JMP I  READN
4223 0017      K17, 17
      /RING TTY BELL SUBROUTINE.
```

```
4224 0000 BELL, 0
4225 7200      CLA
4226 1231      TAD  K207
4227 4435      JMS I  TYPEP
4230 5624      JMP I  BELL
4231 0207      K207, 207
```

/TYPE PACKED MESSAGE SUBROUTINE. ENTERED WITH POINTER IN AC.

```
4232 0000 MSGN, 0
4233 3270      DCA  MSGTEM
4234 7240      CLA CMA
4235 1270      TAD  MSGTEM
4236 3017      DCA  AUTO7
4237 1107      MSGNL1, TAD  M2
4240 3272      DCA  MSGCNT
4241 1417      TAD I  AUTO7
4242 7004      RAL
4243 3270      DCA  MSGTEM
4244 7620      MSGNL2, SNL CLA
4245 5250      JMP  .+3
4246 1101      TAD  K200
4247 5251      JMP  .+2
4250 1273      TAD  K300
4251 3271      DCA  MSGK
4252 1270      TAD  MSGTEM
4253 7006      RTL
4254 7006      RTL
4255 7006      RTL
```

```
4256 3270      DCA  MSGTEM
4257 1270      TAD  MSGTEM
4260 0102      AND  K77
4261 7450      SNA
4262 5632      JMP I  MSGN
4263 1271      TAD  MSGK
4264 4435      JMS I  TYPEP
4265 2272      ISZ  MSGCNT
4266 5244      JMP  MSGNL2
4267 5237      JMP  MSGNL1
4270 0000 MSGTEM, 0
4271 0000 MSGK, 0
4272 0000 MSGCNT, 0
4273 0300 K300, 300
```

/DELAY N MILLISEC ROUTINE. ENTERED WITH N IN AC.

```

4274 0000 DELNMS, 0
4275 7041 CIA
4276 3316 DCA DELC1
4277 1320 DELLOP, TAB M304
4300 3317 DCA DELC2
4301 7000 NOP
4302 2317 ISE DELC2
4303 5301 JMP .-2
4304 7000 NOP
4305 2316 ISE DELC1
4306 7410 SKP
4307 5674 JMP I DELNMS
4310 3317 DCA DELC2
4311 3317 DCA DELC2
4312 3317 DCA DELC2
4313 3317 DCA DELC2
4314 7000 NOP
4315 5277 JMP DELLOP
4316 0000 DELC1, 0
4317 0000 DELC2, 0
4320 7474 M304, -304

```

4400 *4400

/XY8E PLOTTER IOT SUBROUTINES.

```

4400 0000 SPLCE, 0 /PLCE.
4401 6500 6500
4402 5600 JMP I SPLCE
4403 7402 HLT /ILLEGAL SKIP.

4404 0000 SPLSF, 0 /PLSF.
4405 6501 6501
4406 5604 JMP I SPLSF
4407 2204 ISE SPLSF
4410 5604 JMP I SPLSF

4411 0000 SPLCF, 0 /PLCF
4412 6502 6502
4413 5611 JMP I SPLCF
4414 7402 HLT /ILLEGAL SKIP.

4415 0000 SPLPU, 0 /PLPU.
4416 6503 6503
4417 5615 JMP I SPLPU
4420 7402 HLT /ILLEGAL SKIP.

4421 0000 SPLLR, 0 /PLLR.
4422 6504 6504
4423 5621 JMP I SPLLR
4424 7402 HLT /ILLEGAL SKIP.

4425 0000 SPLPD, 0 /PLPD

```

4426	6505	6505		
4427	5625	JMP I	SPLPD	
4430	7402	HLT		/ILLEGAL SKIP.
4431	0000	SCFLR,	0	/CFLR.
4432	6506	6506		
4433	5631	JMP I	SCFLR	
4434	7402	HLT		/ILLEGAL SKIP.
4435	0000	SPLSE,	0	/PLSE
4436	6507	6507		
4437	5635	JMP I	SPLSE	
4440	7402	HLT		/ILLEGAL SKIP.

/MESSAGES.

4600	3031	*4600		
4601	4070	MSG1,	TEXT	"XY 0-E DIAGNOSTIC PROGRAM"
4602	5505			
4603	4004			
4604	1101			
4605	0716			
4606	1723			
4607	2411			
4610	0340			
4611	2022			
4612	1707			
4613	2201			
4614	1500			
4615	2014	MSG2,	TEXT	"PLOTTER SERIES (1-3): *"
4616	1724			
4617	2405			
4620	2240			
4621	2305			
4622	2211			
4623	0523			
4624	4050			
4625	6155			
4626	6351			
4627	7240			
4630	5200			
4631	2014	MSG3,	TEXT	"PLOTTER TYPE (1-4): *"
4632	1724			
4633	2405			
4634	2240			
4635	2431			
4636	2005			
4637	4050			
4640	6155			
4641	6451			
4642	7240			
4643	5200			
4644	2605	MSG4,	TEXT	"VECTOR TYPE (1-2): *"
4645	0324			
4646	1722			

4647 4024
4650 3120
4651 0540
4652 5061
4653 5562
4654 5172
4655 4052
4656 0000
4657 1116
4660 0522
4661 0515
4662 0516
4663 2440
4664 2431
4665 2005
4666 4050
4667 6155
4670 6551
4671 7240
4672 5200
4673 2605
4674 2211
4675 0631
4676 7200
4677 2014
4700 1724
4701 2405
4702 2240
4703 2305
4704 2211
4705 0523
4706 7240
4707 0000
4710 2014
4711 1724
4712 2405
4713 2240
4714 2311
4715 3205
4716 4001
4717 1604
4720 4024
4721 3120
4722 0572
4723 4000
4724 1625
4725 1502
4726 0522
4727 4017
4730 0640
4731 2605
4732 0524
4733 1722
4734 2372
4735 4000

MSG5, TEXT "INCREMENT TYPE (1-5): *"

MSG6, TEXT "VERIFY:"

MSG7, TEXT "PLOTTER SERIES: "

MSG10, TEXT "PLOTTER SIZE AND TYPE: "

MSG11, TEXT "NUMBER OF VECTORS: "

4736	1116	MSG12, TEXT	"INCREMENT SIZE: "
4737	0522		
4740	0515		
4741	0516		
4742	2440		
4743	2311		
4744	3205		
4745	7240		
4746	0000		
4747	0317	MSG13, TEXT	"CORRECT? (Y OR N) *"
4750	2222		
4751	0503		
4752	2477		
4753	4050		
4754	3140		
4755	1722		
4756	4016		
4757	5140		
4760	5200		
4761	0317	MSG14, TEXT	"CONTROL TEST."
4762	1624		
4763	2217		
4764	1440		
4765	2405		
4766	2324		
4767	5600		
4770	4040	MSG15, TEXT	" ERROR PC DR"
4771	0522		
4772	2217		
4773	2240		
4774	4040		
4775	4020		
4776	0340		
4777	4040		
5000	0422		
5001	0000		
5002	0317	MSG17, TEXT	"CONTROL TEST COMPLETE."
5003	1624		
5004	2217		
5005	1440		
5006	2405		
5007	2324		
5010	4003		
5011	1715		
5012	2014		
5013	0524		
5014	0556		
5015	0000		
5016	0411	MSG20, TEXT	"DISPLAY TEST MONITOR."
5017	2320		
5020	1401		
5021	3140		
5022	2405		
5023	2324		
5024	4015		

5025 1716
 5026 1124
 5027 1722
 5030 5600
 5031 2225
 5032 1640
 5033 1116
 5034 1124
 5035 1101
 5036 1411
 5037 3201
 5040 2411
 5041 1716
 5042 4002
 5043 0506
 5044 1722
 5045 0540
 5046 2225
 5047 1616
 5050 1116
 5051 0740
 5052 0411
 5053 2320
 5054 1401
 5055 3140
 5056 2405
 5057 2324
 5060 5600
 5061 3311
 5062 0000
 5063 2305
 5064 2440
 5065 2322
 5066 4040
 5067 2417
 5070 4001
 5071 4061
 5072 4002
 5073 0506
 5074 1722
 5075 0540
 5076 0310
 5077 0116
 5100 0711
 5101 1607
 5102 4023
 5103 2265
 5104 5561
 5105 4156
 5106 0000
 5107 4040
 5110 4040
 5111 0000
 5112 4040
 5113 4040

MSG21, TEXT "RUN INITIALIZATION BEFORE RUNNING DISPLAY TEST."

MSG22, 3311

/C,1

0000

MSG23, TEXT

"SET SR0 TO A 1 BEFORE CHANGING SR5-11."

MSG16, 4040

4040

M16P1, 0

4040

4040

/ERROR CODE FILLIN.

5114	0000	M16P2, 0	/PC FILL IN
5115	0000	M16P3, 0	
5116	4040	4040	
5117	0000	M16P4, 0	/DIRECTION REG. FILL IN.
5120	0000	0000	/END MSG CODE.

/VERIFICATION MESSAGE INSERTS.

5121	6560	PSM1, TEXT	"500 OR HOUSTON DP10/EDP10"
5122	6040		
5123	1722		
5124	4010		
5125	1725		
5126	2524		
5127	1716		
5130	4004		
5131	2061		
5132	6057		
5133	0504		
5134	2061		
5135	6000		
5136	6660	PSM2, TEXT	"600"
5137	6000		
5140	6760	PSM3, TEXT	"700"
5141	6000		
5142	6161	PTM1, TEXT	"11 INCH DRUM OR HOUSTON DP10/EDP10"
5143	4011		
5144	1603		
5145	1040		
5146	0422		
5147	2515		
5150	4017		
5151	2240		
5152	1017		
5153	2523		
5154	2417		
5155	1640		
5156	0420		
5157	6160		
5160	5705		
5161	0420		
5162	6160		
5163	0000		
5164	6270	PTM2, TEXT	"28.55 INCH DRUM"
5165	5665		
5166	6540		
5167	1116		
5170	0310		
5171	4004		
5172	2225		
5173	1500		
5174	6361	PTM3, TEXT	"31 BY 54 INCH FLATBED"
5175	4002		
5176	3140		
5177	6564		
5200	4011		

5201	1603		
5202	1040		
5203	0614		
5204	0124		
5205	0205		
5206	0400		
5207	6564	PTM4,	TEXT "54 BY 72 INCH FLATBED"
5210	4002		
5211	3140		
5212	6762		
5213	4011		
5214	1603		
5215	1040		
5216	0614		
5217	0124		
5220	0205		
5221	0400		
5222	7000	VTM1,	TEXT "8"
5223	6264	VTM2,	TEXT "24"
5224	0000		
5225	6056	ITM1,	TEXT "0.1 MM"
5226	6140		
5227	1515		
5230	0000		
5231	5660	ITM2,	TEXT ".00125 INCH"
5232	6061		
5233	6265		
5234	4011		
5235	1603		
5236	1000		
5237	5660	ITM3,	TEXT ".0025 INCH"
5240	6062		
5241	6240		
5242	1116		
5243	0310		
5244	0000		
5245	5660	ITM4,	TEXT ".005 INCH"
5246	6065		
5247	4011		
5250	1603		
5251	1000		
5252	5660	ITM5,	TEXT ".01 INCH"
5253	6140		
5254	1116		
5255	0310		
5256	0000		

5257	5257	/TAPE 6.
		WLP,
		DECIMAL
5260	0012	10
5261	0036	30
5262	0033	27
5263	0040	32

/PLOTTER DIMENSION CONSTANTS

/11 INCH DRUM WIDTH AND
/SIMULATED LENGTH.

/28.55 INCH DRUM WIDTH AND
/SIMULATED LENGTH.

5264	0036	30	/31 X 54 INCH FLATBED.
5265	0065	53	
5266	0065	53	/54 X 72 INCH FLATBED.
5267	0107	71	
		OCTAL	
5270	5270	INSTP.	/SCALE FACTOR INSTS FOR INIT.
5271	7104	CLL RAL	/,00125 INCH SCALE FACTOR (2.0)
5272	7000	NOP	
5273	7000	NOP	/,0025 INCH SCALE FACTOR (1.0)
5274	7000	NOP	
5275	7110	CLL RAR	/,005 INCH SCALE FACTOR (0.5)
5276	7000	NOP	
5277	7110	CLL RAR	/,01 INCH SCALE FACTOR (0.25)
5300	7110	CLL RAR	

/INIT REPOSE MSG POINTERS

5301	5121	PSM1P,	PSM1
5302	5136	PSM2P,	PSM2
5303	5140	PSM3P,	PSM3
5304	5142	PTM1P,	PTM1
5305	5164	PTM2P,	PTM2
5306	5174	PTM3P,	PTM3
5307	5207	PTM4P,	PTM4
5310	5222	VTM1P,	VTM1
5311	5223	VTM2P,	VTM2
5312	5225	ITM1P,	ITM1
5313	5231	ITM2P,	ITM2
5314	5237	ITM3P,	ITM3
5315	5245	ITM4P,	ITM4
5316	5252	ITM5P,	ITM5

/ACTIVE PATTERN POINTERS. (FILLED IN BY INIT)

5317	5317	PATS,	.	
5320	0000		0	/PAT1
5321	0000		0	/PAT2
5322	0000		0	/PAT3
5323	0000		0	/PAT4
5324	0000		0	/PAT5
5325	0000		0	/PAT6
5326	6721		PAT07	/PAT07

/24 VECTOR PATTERN POINTERS.

5327	5327	PATS24,	.	
5330	6271		PAT124	/PAT1
5331	6341		PAT224	/PAT2
5332	6407		PAT324	/PAT3
5333	6447		PAT424	/PAT4
5334	6556		PAT524	/PAT5
5335	6647		PAT624	/PAT6

/8 VECTOR PATTERN POINTERS.

5336	5336	PAT58,	.	
5337	5711		PAT18	/PAT1
5340	5761		PAT28	/PAT2
5341	6027		PAT38	/PAT3
5342	6067		PAT48	/PAT4
5343	6140		PAT58	/PAT5
5344	6217		PAT68	/PAT6

/NEXT TO LAST LINE IN A GIVEN ACTIVE PATTERN. (FILLED IN BY INIT.)
/DOES NOT APPLY FOR PAT7.

5345	5345	LASLIN,	.	
5346	0000		0	/PAT1
5347	0000		0	/PAT2
5350	0000		0	/PAT3
5351	0000		0	/PAT4
5352	0000		0	/PAT5
5353	0000		0	/PAT6

/NEXT TO LAST LINE NUMBER IN A GIVEN 24 VECTOR PATTERN.
/DOES NOT APPLY TO PAT7.

5354	5354	LAS24,	.	
5355	0032		32	/PAT1
5356	0022		22	/PAT2
5357	0013		13	/PAT3
5360	0071		71	/PAT5
5361	0067		67	/PAT5
5362	0017		17	/PAT6

/NEXT TO LAST LINE NUMBER IN A GIVEN 8 VECTOR PATTERN.

5363	5363	LAS8,	.	
5364	0032		32	/PAT1
5365	0022		22	/PAT2
5366	0013		13	/PAT3
5367	0041		41	/PAT4
5370	0055		55	/PAT5
5371	0017		17	/PAT6

/SERIES 500 PEN SUBROUTINE POINTERS.

5372	5372	P500,	.	
5373	4054	CPU5P,	CPU5	/CONTROL PEN UP.
5374	4060	CPD5P,	CPD5	/CONTROL PEN DOWN.
5375	4064	DPUSP,	DPUS	/DISPLAY PEN UP.
5376	4073	DPD5P,	DPD5	/DISPLAY PEN DOWN.

/SERIES 600-700 PEN SUBROUTINE POINTERS.

5377	5377	P678,	.	
5400	4102	CPU8P,	CPU8	/CONTROL PEN UP.
5401	4110	CPD8P,	CPD8	/CONTROL PEN DOWN

5402 4116 DPU8P, DPU8 /DISPLAY PEN UP
 5403 4127 DPD8P, DPD8 /DISPLAY PEN DOWN

/DIRECTION CONSTANTS. THOSE DEFINED AS ZERO ARE REDEFINED BY
 /PLOTTER SERIES CONNECTED. ADDRESS SENSITIVE RELATIVE TO
 /DKAYS".

5404	5404	DKAYS,	.	/COORDINATE LID CODE	
5405	0000	NK,	0	/+Y	01
5406	0000	NEK,	0	/+Y +X	02
5407	0000	EK,	0	/	03
5410	0000	SEK,	0	/-Y +X	04
5411	0000	SK,	0	/-Y	05
5412	0000	SWK,	0	/-Y -X	06
5413	0000	WK,	0	/	07
5414	0000	NWK,	0	/+Y -X	10
5415	0072	NNEK,	72	/+Y +X/2	11
5416	0070	ENEK,	70	/+Y/2 +X	12
5417	0074	ESEK,	74	/-Y/2 +X	13
5420	0076	SSEK,	76	/-Y +X/2	14
5421	0077	SSWK,	77	/-Y -X/2	15
5422	0075	WSWK,	75	/-Y/2 -X	16
5423	0071	WNWK,	71	/+Y/2 -X	17
5424	0073	NNWK,	73	/+Y -X/2	20
5425	0050	LNK,	50	/+Y/2	21
5426	0051	LNEK,	51	/+Y/2 +X/2	22
5427	0052	LEK,	52	/	23
5430	0053	LSEK,	53	/-Y/2 +X/2	24
5431	0054	LSK,	54	/-Y/2	25
5432	0055	LSWK,	55	/-Y/2 -X/2	26
5433	0056	LWK,	56	/	27
5434	0057	LNWK,	57	/+Y/2 -X/2	30

/8 VECTOR DIRECTION CONSTANTS DEFINITIONS. ADDRESS SENSITIVE
 /RELATIVE TO "DK500" AND "DK678".
 /500 SERIES.

5435	5435	DK500,	.
5436	0020	NK5,	20
5437	0030	NEK5,	30
5440	0010	EK5,	10
5441	0050	SEK5,	50
5442	0040	SK5,	40
5443	0044	SWK5,	44
5444	0004	WK5,	04
5445	0024	NWK5,	24

/600-700 SERIES.

5446	5446	DK678,	.
5447	0010	NK8,	10
5450	0011	NEK8,	11
5451	0012	EK8,	12

5452	0013	SEK8,	13
5453	0014	SK8,	14
5454	0015	SWK8,	15
5455	0016	WK8,	16
5456	0017	NWK8,	17

/LENGTH CONSTANTS: THERE ARE 5 GROUPS OF LENGTH CONSTANTS.

- /1. 0 OR 90 DEGREE FULL VECTORS.
- /2. 45 DEGREE FULL VECTORS.
- /3. MIXED VECTORS.
- /4. 0 OR 90 DEGREE HALF VECTORS.
- /5. 45 DEGREE HALF VECTORS.

	5457	5460	LGTHP,	LGTH	/LID	GRP	INCHES
	5460	0000	LGTH,	0	/00	1	0.1
	5461	0000		0	/01		.25
	5462	0000		0	/02		.50
	5463	0000		0	/03		1.0
	5464	0000		0	/04		1.5
	5465	0000		0	/05		2.5
	5466	0000		0	/06		3.0
	5467	0000		0	/07		4.0
	5470	0000		0	/10	2	0.1
	5471	0000		0	/11		.25
	5472	0000		0	/12		.50
	5473	0000		0	/13		1.0
	5474	0000		0	/14		1.5
	5475	0000		0	/15		2.5
	5476	0000		0	/16		3.0
	5477	0000		0	/17		4.0
	5500	0000		0	/20	3	0.1
	5501	0000		0	/21		.25
	5502	0000		0	/22		.50
	5503	0000		0	/23		1.0
	5504	0000		0	/24		1.5
	5505	0000		0	/25		2.5
	5506	0000		0	/26		3.0
	5507	0000		0	/27		4.0
	5510	0000		0	/30	4	0.1
	5511	0000		0	/31		.25
	5512	0000		0	/32		.50
	5513	0000		0	/33		1.0
	5514	0000		0	/34		1.5
	5515	0000		0	/35		2.5
	5516	0000		0	/36		3.0
	5517	0000		0	/37		4.0
	5520	0000		0	/40	5	0.1
	5521	0000		0	/41		.25
	5522	0000		0	/42		.50
	5523	0000		0	/43		1.0
	5524	0000		0	/44		1.5
	5525	0000		0	/45		2.5
	5526	0000		0	/46		3.0
	5527	0000		0	/47		4.0

/LENGTH CONSTANTS FOR BASIC .0025 INCH INCREMENT SIZE. (IN DECIMAL.)

5530	5530	INCH,	. DECIMAL	/GROUP 1.
5531	0050		40	/00
5532	0144		100	/01
5533	0310		200	/02
5534	0620		400	/03
5535	1130		600	/04
5536	1750		1000	/05
5537	2260		1200	/06
5540	3100		1600	/07
				/GROUP2. (.707 X GROUP1)
5541	0034		28	/10
5542	0107		71	/11
5543	0215		141	/12
5544	0433		283	/13
5545	0650		424	/14
5546	1303		707	/15
5547	1520		848	/16
5550	2153		1131	/17
				/GROUP3. (.447 X GROUP4)
5551	0044		36	/20
5552	0131		89	/21
5553	0263		179	/22
5554	0546		358	/23
5555	1030		536	/24
5556	1576		894	/25
5557	2061		1073	/26
5560	2626		1430	/27
				/GROUP4 (2 X GROUP1)
5561	0120		80	/30
5562	0310		200	/31
5563	0620		400	/32
5564	1440		800	/33
5565	2260		1200	/34
5566	3720		2000	/35
5567	4540		2400	/36
5570	6200		3200	/37
				/GROUP5 (.707 X GROUP4)
5571	0071		57	/40
5572	0215		141	/41
5573	0433		283	/42
5574	1066		566	/43
5575	1520		848	/44
5576	2606		1414	/45
5577	3241		1697	/46
5600	4526		2262	/47

OCTAL

/LENGTH CONSTANTS FOR 0.1 MM INCREMENT SIZE. (IN DECIMAL)

5601	5601	METER,	DECIMAL	
5602	0031		25	/GROUP1
5603	0100		64	/00
5604	0177		127	/01
5605	0376		254	/02
5606	0575		381	/03
5607	1173		635	/04
5610	1372		762	/05
5611	1770		1016	/06
				/07
				/GROUP2 (.707 X GROUP1)
5612	0022		10	/10
5613	0055		45	/11
5614	0132		90	/12
5615	0264		180	/13
5616	0415		269	/14
5617	0701		449	/15
5620	1033		539	/16
5621	1316		718	/17
			OCTAL	

/REPETITION CONSTANTS: (IN POSITIVE FORM.)
/FILLED IN BY INIT.

5622	5622	REPK,	.	/	LID
5623	0000	WIDE,	0	/PLOT AREA WIDTH.	01
5624	0000	WIDE2,	0	/2X WIDE,	02
5625	0000	WIDE4,	0	/4X WIDE	03
5626	0000	WIDE8,	0	/8X WIDE	04
5627	0000	LONG,	0	/PLOT AREA LENGTH	05
5630	0000	LONG2,	0	/2X LONG	06
5631	0000	LONG4,	0	/4X LONG	07
5632	0000	LONG8,	0	/8X LONG	10
5633	0002		2	/2 MULTIPLE	11
5634	0024		24	/20 MULTIPLE	12
5635	0036		36	/30 MULTIPLE	13
5636	0074		74	/60 MULTIPLE	14

/PATTERN-LINE COMMAND HOLDERS.

5637	5637	PLCH,	.
5640	0000	PCH,	0
5641	0000		0
5642	0000		0
5643	0000		0
5644	0000		0
5645	0000		0
5646	0000		0
5647	0000		0
5650	0000		0
5651	0000		0
5652	0000	LCH,	0

5653	0000	0
5654	0000	0
5655	0000	0
5656	0000	0
5657	0000	0
5660	0000	0
5661	0000	0
5662	0000	0
5663	0000	0

/PATTERN-LINE COMMAND TEMPORARIES.

5664	5664	PLCT,	.
5665	0000	PCT,	0
5666	0000		0
5667	0000		0
5670	0000		0
5671	0000		0
5672	0000		0
5673	0000		0
5674	0000		0
5675	0000		0
5676	0000		0
5677	0000	LCT,	0
5700	0000		0
5701	0000		0
5702	0000		0
5703	0000		0
5704	0000		0
5705	0000		0
5706	0000		0
5707	0000		0
5710	0000		0

/PATTERN LISTS.

/1. THERE ARE TWO MAJOR GROUPS OF PATTERNS: 8 VECTOR PATTERNS (PAT 18, 28, 38, 48, 58, AND 68) AND 24 VECTOR PATTERNS (PAT 124, 224, 324, 424, 524, AND 624). PAT 07 /IS A LIST OF DASH AND SHUFFLE VECTORS WHICH ARE CALLED ON A LINE TO LINE BASIS /BY THE USER. THE VECTORS INCLUDED IN PAT07 REPRESENT ALL 16 DIRECTIONS.

/2. THE PATTERN LISTS CONSIST OF LINE IDENTIFIERS (LID) WHICH ARE /IN EFFECT PSEUDO INSTRUCTIONS WHICH DESCRIBE PEN UP OR DOWN, LINE LENGTH, AND /DIRECTION. THERE IS ALSO A SPECIAL GROUP OF LID'S. LID'S ARE INTERPRETED BY THE /"LGEN" ROUTINE. THE FORMATS ARE AS FOLLOWS:

/STANDARD LID BIT ASSIGNMENTS

/0: PEN UP WHEN CLEAR, PEN DOWN WHEN SET.
 /1-6: LENGTH CODE (REFERS TO SPECIFIED LENGTH CONSTANT).
 /7-11: DIRECTION CODE (REFERS TO DIRECTION CONSTANT).

/SPECIAL LID (3XXY) FORMAT.

/3: INDICATES SPECIAL LID.
 /XX: REFERS TO REPETITION CONSTANT.
 /Y: INDICATES HOW MANY STANDARD LID'S FOLLOWING THE SPECIAL

/LID ARE TO BE INCLUDED IN THIS LINE.

/IN ALL CASES +Y=PEN LEFT; -Y=PEN RIGHT; +X=DRUM DOWN; -X=DRUM UP.

/IF VECTOR IS +1/2V, IT INDICATES AUTOMATIC HALF LENGTH VECTOR SWITCHING ON
 /24 VECTOR PLOTTERS ONLY.
 /AT SOME POINTS IN THE PATTERN LISTS NORTH REFERENCED DIRECTIONS ARE INDICATED.
 /THESE ARE BASED ON THE ASSUMPTIONS: NORTH=+Y=PEN LEFT AND EAST=+X=DRUM DOWN.

/8 VECTOR PATTERNS
 /PATTERN 18-PERIMETER OUTLINE.

LINE#	PEN	LGTH	DIR	
5711	3011	PAT18,	3011	/01
5712	0141		0141	/
5713	3011		3011	/02
5714	0145		0145	/
5715	3011		3011	/03
5716	0145		0145	/
5717	0101		0101	/04
5720	3011		3011	/05
5721	0141		0141	/
5722	3051		3051	/06
5723	0143		0143	/
5724	0107		0107	/07
5725	3051		3051	/10
5726	4147		4147	/
5727	3011		3011	/11
5730	4145		4145	/
5731	3051		3051	/12
5732	4143		4143	/
5733	3011		3011	/13
5734	4141		4141	/
5735	3051		3051	/14
5736	0107		0107	/
5737	3011		3011	/15
5740	0105		0105	/
5741	4041		4041	/16
5742	4105		4105	/17
5743	4041		4041	/20
5744	4043		4043	/21
5745	4107		4107	/22
5746	4043		4043	/23
5747	0041		0041	/24
5750	4404		4404	/25
5751	0410		0410	/26
5752	4406		4406	/27
5753	0402		0402	/30
5754	0045		0045	/31
5755	3051	L18,	3051	/32
5756	0107		0107	/
5757	3051		3051	/33
5760	0143		0143	/

ORIENT

PERIMETER

CROSS HAIRS

PAT01 ORIGIN

PAT02 ORIGIN

/PATTERN 28-DASH FILL IN .25 INCHES FROM PERIMETER.

			/LINE	PEN	LGTH	DIR
5761	3011	PAT28, 3011	/01			
5762	0101	0101	/	0	02	01
5763	0047	0047	/02	0	01	07
5764	3022	3022	/03			
5765	4045	4045	/	1	01	05
5766	0045	0045	/	0	01	05
5767	0042	0042	/04	0	01	02
5770	3062	3062	/05			
5771	4047	4047	/	1	01	07
5772	0047	0047	/	0	01	07
5773	0044	0044	/06	0	01	04
5774	3022	3022	/07			
5775	4041	4041	/	1	01	01
5776	0041	0041	/	0	01	01
5777	0046	0046	/10	0	01	06
6000	3062	3062	/11			
6001	4043	4043	/	1	01	03
6002	0043	0043	/	0	01	03
6003	3062	3062	/12			
6004	4047	4047	/	1	01	07
6005	0047	0047	/	0	01	07
6006	0042	0042	/13	0	01	02
6007	3022	3022	/14			
6010	4045	4045	/	1	01	05
6011	0045	0045	/	0	01	05
6012	0050	0050	/15	0	01	10
6013	3062	3062	/16			
6014	4043	4043	/	1	01	03
6015	0043	0043	/	0	01	03
6016	0046	0046	/17	0	01	06
6017	3022	3022	/20			
6020	4041	4041	/	1	01	01
6021	0041	0041	/	0	01	01
6022	0043	0043	/21	0	01	03
6023	3011	L28, 3011	/22			PAT02 ORIGIN
6024	0105	0105	/	0	02	05
6025	3051	3051	/23			PAT03 ORIGIN
6026	0147	0147	/	0	03	07

/PATTERN 38-SHUFFLE .5 INCHES FROM PERIMETER.

			/LINE	PEN	LGTH	DIR
6027	3011	PAT38, 3011	/01			
6030	0101	0101	/	0	02	01
6031	0103	0103	/02	0	02	03
6032	3014	3014	/03			
6033	4105	4105	/	1	02	05
6034	0105	0105	/	0	02	05
6035	4101	4101	/	1	02	01
6036	0105	0105	/	0	02	05
6037	0110	0110	/04	0	02	10
6040	3054	3054	/05			
6041	4103	4103	/	1	02	03

6042	0103	0103	/	0	02	03
6043	4107	4107	/	1	02	07
6044	0103	0103	/	0	02	03
6045	0106	0106	/06	0	02	06
6046	3014	3014	/07			
6047	4101	4101	/	1	02	01
6050	0101	0101	/	0	02	01
6051	4105	4105	/	1	02	05
6052	0101	0101	/	0	02	01
6053	0104	0104	/10	0	02	04
6054	3054	3054	/11			
6055	4107	4107	/	1	02	07
6056	0107	0107	/	0	02	07
6057	4103	4103	/	1	02	03
6060	0107	0107	/	0	02	07
6061	0101	0101	/12	0	02	01
6062	3011	3011	/13			
6063	0105	0105	/	0	02	05
6064	3052	3052	/14			
6065	0103	0103	/	0	02	03
6066	0043	0043	/	0	01	03

L38.

PAT03 ORIGIN

PAT04 ORIGIN

/PATTERN 48-CONCENTRIC AND OBLIQUE SQUARES.

			LINE	PEN	LGTH	DIR.
6067	4341	4341	/01	1	07	01 NORTH
6070	0345	0345	/02	0	07	05
6071	4342	4342	/03	1	07	02 NE
6072	0346	0346	/04	1	07	06
6073	4343	4343	/05	1	07	03 E
6074	0347	0347	/06	0	07	07
6075	4344	4344	/07	1	07	04 SE
6076	0350	0350	/10	0	07	10
6077	4345	4345	/11	1	07	05 S
6100	0341	0341	/12	0	07	01
6101	4346	4346	/13	1	07	06 SW
6102	0342	0342	/14	0	07	02
6103	4347	4347	/15	1	07	07 W
6104	0343	0343	/16	0	07	03
6105	4350	4350	/17	1	07	10 NW
6106	0344	0344	/20	0	07	04
6107	0350	0350	/21	0	07	10
6110	3111	3111	/22			
6111	4343	4343	/	1	07	03 8 INCH SQUARE
6112	3111	3111	/23			
6113	4345	4345	/	1	07	05
6114	3111	3111	/24			
6115	4347	4347	/	1	07	07
6116	3111	3111	/25			
6117	4341	4341	/	1	07	01
6120	3111	3111	/26			
6121	0144	0144	/	0	03	04
6122	4345	4345	/27	1	07	05 4 INCH SQUARE
6123	4343	4343	/30	1	07	03

6124	4341		4341	/31	1	07	01	
6125	4347		4347	/32	1	07	07	
6126	3111		3111	/33				
6127	0144		0144	/	0	03	04	
6130	0341		0341	/34	0	07	01	
6131	4344		4344	/35	1	07	04	4 INCH OBLIQUE SQ.
6132	4346		4346	/36	1	07	06	
6133	4350		4350	/37	1	07	10	
6134	4342		4342	/40	1	07	02	
6135	0345	L48,	0345	/41	0	07	05	PAT04 ORIGIN
6136	3051		3051	/42				PAT05 ORIGIN
6137	0107		0107	/	0	02	07	
				/				
				/				
				/				

/PATTERN 58 - JOINED OCTAGONS

6140	4141	PAT58,	4141	/LINE	PEN	LGTH	DIR	
6141	4041		4041	/01	1	03	01	START TOP OCTAGON
6142	4642		4642	/02	1	01	01	
6143	4243		4243	/03	1	15	02	
6144	4644		4644	/04	1	05	03	
6145	4245		4245	/05	1	15	04	
6146	4646		4646	/06	1	05	05	
6147	4247		4247	/07	1	15	06	
6150	4650		4650	/10	1	05	07	
6151	4241		4241	/11	1	15	10	
6152	4650		4650	/12	1	05	01	BOTTOM OCTAGON
6153	4247		4247	/13	1	15	10	
6154	4646		4646	/14	1	05	07	
6155	4245		4245	/15	1	15	06	
6156	4644		4644	/16	1	05	05	
6157	4243		4243	/17	1	15	04	
6160	4642		4642	/20	1	05	03	
6161	0241		0241	/21	1	15	02	
6162	4344		4344	/22	0	05	01	TOP DIAGS.
6163	4044		4044	/23	1	07	04	
6164	0642		0642	/24	1	01	04	
6165	4350		4350	/25	0	15	02	
6166	4050		4050	/26	1	07	10	
6167	0243		0243	/27	1	01	10	
6170	0544		0544	/30	0	05	03	
6171	0444		0444	/31	0	13	04	
6172	4346		4346	/32	0	11	04	
6173	4046		4046	/33	1	07	06	
6174	0450		0450	/34	1	01	06	
6175	0550		0550	/35	0	11	10	
6176	0241		0241	/36	0	13	10	
6177	4346		4346	/37	0	05	01	
6200	4046		4046	/40	1	07	06	LOW DIAGS.
6201	0650		0650	/41	1	01	06	
6202	4342		4342	/42	0	15	10	
6203	4042		4042	/43	1	07	02	
				/44	1	01	02	

6204	0247		0247	/45	0	05	07	
6205	0546		0546	/46	0	13	06	
6206	0446		0446	/47	0	11	06	
6207	4344		4344	/50	1	07	04	
6210	4044		4044	/51	1	01	04	
6211	0542		0542	/52	0	13	02	
6212	0442		0442	/53	0	11	02	
6213	0141		0141	/54	0	03	01	
6214	0041	L58,	0041	/55	0	01	01	PAT05 ORIGIN
6215	3051		3051	/56				PAT06 ORIGIN
6216	0043		0043	/	0	01	03	

/PATTERN 68 - RAPID ARROWS

				/LINE	PEN	LGTH	DIR
6217	0147	PAT68,	0147	/01	0	03	07
6220	4301		4301	/02	1	06	01
6221	4143		4143	/03	1	03	03
6222	4143		4143	/04	1	03	03
6223	4305		4305	/05	1	06	05
6224	4305		4305	/06	1	06	05
6225	4147		4147	/07	1	03	07
6226	4147		4147	/10	1	03	07
6227	4301		4301	/11	1	06	01
6230	3135		3135	/12			
6231	0001		0001	/	0	00	01
6232	4406		4406	/	1	10	06
6233	0402		0402	/	0	10	02
6234	4404		4404	/	1	10	04
6235	0410		0410	/	0	10	10
6236	3125		3125	/13			
6237	0003		0003	/	0	00	03
6240	4410		4410	/	1	10	10
6241	0404		0404	/	0	10	04
6242	4406		4406	/	1	10	06
6243	0402		0402	/	0	10	02
6244	3145		3145	/14			
6245	0005		0005	/	0	00	05
6246	4402		4402	/	1	10	02
6247	0406		0406	/	0	10	06
6250	4410		4410	/	1	10	10
6251	0404		0404	/	0	10	04
6252	3125		3125	/15			
6253	0007		0007	/	0	00	07
6254	4404		4404	/	1	10	04
6255	0410		0410	/	0	10	10
6256	4402		4402	/	1	10	02
6257	0406		0406	/	0	10	06
6260	3135		3135	/16			
6261	0001		0001	/	0	00	01
6262	4406		4406	/	1	10	06
6263	0402		0402	/	0	10	02
6264	4404		4404	/	1	10	04
6265	0410		0410	/	0	10	10

6266	0143	L68,	0143	/17	0	03	03	PAT06 ORIGIN.
6267	3051		3051	/20				
6270	0107		0107	/	0	02	07	PAT01 ORIGIN.
				/				
				/				

/24 VECTOR PATTERNS
/PATTERN 124-PERIMETER OUTLINE.

6271	3011	PAT124,	3011	/01				ORIENT
6272	0141		0141	/	0	03	01	
6273	3011		3011	/02				
6274	0145		0145	/	0	03	05	
6275	3011		3011	/03				
6276	0145		0145	/	0	03	05	
6277	0101		0101	/04	0	02	01	
6300	3011		3011	/05				
6301	0141		0141	/	0	03	01	
6302	3051		3051	/06				
6303	0143		0143	/	0	03	03	
6304	0107		0107	/07	0	02	07	
6305	3051		3051	/10				PERIMETER
6306	4147		4147	/	1	03	07	
6307	3011		3011	/11				
6310	4145		4145	/	1	03	05	
6311	3051		3051	/12				
6312	4143		4143	/	1	03	03	
6313	3011		3011	/13				
6314	4141		4141	/	1	03	01	
6315	3051		3051	/14				
6316	0107		0107	/	0	02	07	
6317	3011		3011	/15				
6320	0105		0105	/	0	02	05	
6321	4041		4041	/16	1	01	01	CROSS HAIRS
6322	4105		4105	/17	1	02	05	
6323	4041		4041	/20	1	01	01	
6324	4043		4043	/21	1	01	03	
6325	4107		4107	/22	1	02	07	
6326	4043		4043	/23	1	01	03	
6327	0041		0041	/24	0	01	01	
6330	4404		4404	/25	1	10	04	
6331	0410		0410	/26	0	10	10	
6332	4406		4406	/27	1	10	06	
6333	0402		0402	/30	0	10	02	
6334	0045		0045	/31	0	01	05	
6335	3051	L124,	3051	/32				
6336	0107		0107	/	0	02	07	PAT01 ORIGIN
6337	3051		3051	/33				
6340	0143		0143	/	0	03	03	PAT02 ORIGIN
				/				
				/				

/PATTERN 224-DASH FILL IN .25 INCHES FROM PERIMETER.

			/LINE	PEN	LGTH	DIR
6341	3011	PAT224, 3011	/01			
6342	0101	0101	/	0	02	01
6343	0047	0047	/02	0	01	07
6344	3022	3022	/03			
6345	4045	4045	/	1	01	05
6346	0045	0045	/	1	01	05
6347	0042	0042	/04	0	01	02
6350	3062	3062	/05			
6351	4047	4047	/	1	01	07
6352	0047	0047	/	0	01	07
6353	0044	0044	/06	0	01	04
6354	3022	3022	/07			
6355	4041	4041	/	1	01	01
6356	0041	0041	/	0	01	01
6357	0046	0046	/10	0	01	06
6360	3062	3062	/11			
6361	4043	4043	/	1	01	03
6362	0043	0043	/	0	01	03
6363	3062	3062	/12			
6364	4047	4047	/	1	01	07
6365	0047	0047	/	0	01	07
6366	0042	0042	/13	0	01	02
6367	3022	3022	/14			
6370	4045	4045	/	1	01	05
6371	0045	0045	/	0	01	05
6372	0050	0050	/15	0	01	10
6373	3062	3062	/16			
6374	4043	4043	/	1	01	03
6375	0043	0043	/	0	01	03
6376	0046	0046	/17	0	01	06
6377	3022	3022	/20			
6400	4041	4041	/	1	01	01
6401	0041	0041	/	0	01	01
6402	0043	0043	/21	0	01	03
6403	3011	L224, 3011	/22			
6404	0105	0105	/	0	02	05
6405	3051	3051	/23			
6406	0147	0147	/	0	03	07

PAT02 ORIGIN

PAT03 ORIGIN

/PATTERN 324-SHUFFLE .5 INCHES FROM PERIMETER.

			/LINE	PEN	LGTH	DIR
6407	3011	PAT324, 3011	/01			
6410	0101	0101	/	0	02	01
6411	0103	0103	/02	0	02	03
6412	3014	3014	/03			
6413	4105	4105	/	1	02	05
6414	0105	0105	/	0	02	05
6415	4101	4101	/	1	02	01
6416	0105	0105	/	0	02	05
6417	0110	0110	/04	0	02	10
6420	3054	3054	/05			
6421	4103	4103	/	1	02	03

6422	0103	0103	/	0	02	03
6423	4107	4107	/	1	02	07
6424	0103	0103	/	0	02	03
6425	0106	0106	/06	0	02	06
6426	3014	3014	/07			
6427	4101	4101	/	1	02	01
6430	0101	0101	/	0	02	01
6431	4105	4105	/	1	02	05
6432	0101	0101	/	0	02	01
6433	0104	0104	/10	0	02	04
6434	3054	3054	/11			
6435	4107	4107	/	1	02	07
6436	0107	0107	/	0	02	07
6437	4103	4103	/	1	02	03
6440	0107	0107	/	0	02	07
6441	0101	0101	/12	0	02	01
6442	3011	3011	/13			
6443	0105	0105	/	0	02	05
6444	3052	3052	/14			
6445	0103	0103	/	0	02	03
6446	0043	0043	/	0	01	03

L324, PAT03 ORIGIN

PAT04 ORIGIN

/PATTERN 424 - CONCENTRIC AND OBLIQUE SQUARES.

		PAT424,	4341	/LINE	PEN	LGTH	DIR
6447	4341		4341	/01	1	07	01 N
6450	0345		0345	/02	0	07	05
6451	4351		4351	/03	1	07	11 NNE
6452	0355		0355	/04	0	07	15
6453	4342		4342	/05	1	07	02 NE
6454	0346		0346	/06	0	07	06
6455	4352		4352	/07	1	07	12 ENE
6456	0356		0356	/10	0	07	16
6457	4343		4343	/11	1	07	03 E
6460	0347		0347	/12	0	07	07
6461	4353		4353	/13	1	07	13 ESE
6462	0357		0357	/14	0	07	17
6463	4344		4344	/15	1	07	04 SE
6464	0350		0350	/16	0	07	10
6465	4354		4354	/17	1	07	14 SSE
6466	0360		0360	/20	0	07	20
6467	4345		4345	/21	1	07	05 S
6470	0341		0341	/22	0	07	01
6471	4355		4355	/23	1	07	15 SSW
6472	0351		0351	/24	0	07	11
6473	4346		4346	/25	1	07	06 SW
6474	0342		0342	/26	0	07	02
6475	4356		4356	/27	1	07	16 WSW
6476	0352		0352	/30	0	07	12
6477	4347		4347	/31	1	07	07 W
6500	0343		0343	/32	0	07	03
6501	4357		4357	/33	1	07	17 WNW
6502	0353		0353	/34	0	07	13
6503	4350		4350	/35	1	07	10 NW
6504	0344		0344	/36	0	07	04

6505	4360	4360	/37	1	07	20	NNW
6506	0354	0354	/40	0	07	14	
6507	0350	0350	/41	0	07	10	
6510	3111	3111	/42				8 INCH SQUARE
6511	4343	4343	/	1	07	03	
6512	3111	3111	/43				
6513	4345	4345	/	1	07	05	
6514	3111	3111	/44				
6515	4347	4347	/	1	07	07	
6516	3111	3111	/45				
6517	4341	4341	/	1	07	01	
6520	3111	3111	/46				
6521	0144	0144	/	0	03	04	
6522	4345	4345	/47	1	07	05	4 INCH SQUARE
6523	4343	4343	/50	1	07	03	
6524	4341	4341	/51	1	07	01	
6525	4347	4347	/52	1	07	07	
6526	3111	3111	/53				
6527	0142	0142	/	0	03	02	
6530	4344	4344	/54	1	07	04	4 INCH OBLIQUE SQ.
6531	4346	4346	/55	1	07	06	
6532	4350	4350	/56	1	07	10	
6533	4342	4342	/57	1	07	02	
6534	3111	3111	/60				
6535	0143	0143	/	0	03	03	
6536	3111	3111	/61				
6537	4144	4144	/	1	03	04	
6540	0345	0345	/62	0	07	05	
6541	3111	3111	/63				
6542	4146	4146	/	1	03	06	
6543	0347	0347	/64	0	07	07	
6544	3111	3111	/65				
6545	4150	4150	/	1	03	10	
6546	0341	0341	/66	0	07	01	
6547	3111	3111	/67				
6550	4142	4142	/	1	03	02	
6551	3111	3111	/70				
6552	0143	0143	/	0	03	03	
6553	0345	0345	/71	0	07	05	PAT04 ORIGIN
6554	3051	3051	/72				PAT05 ORIGIN
6555	0107	0107	/	0	02	07	

/PATTERN 524 - JOINED 16'ERS (16 SIDED FIGURE WITH ALL MIRRORED SIDES EQUAL).

			/LINE	PEN	LGTH	DIR.	
6556	4101	PAT524, 4101	/01	1	02	01	TOP 16'ER
6557	4151	4151	/02	1	03	11	
6560	4142	4142	/03	1	03	02	
6561	4152	4152	/04	1	03	12	
6562	4143	4143	/05	1	03	03	
6563	4153	4153	/06	1	03	13	
6564	4144	4144	/07	1	03	04	
6565	4154	4154	/10	1	03	14	

6566	4145	4145	/11	1	03	05	
6567	4155	4155	/12	1	03	15	
6570	4146	4146	/13	1	03	06	
6571	4156	4156	/14	1	03	16	
6572	4147	4147	/15	1	03	07	
6573	4157	4157	/16	1	03	17	
6574	4150	4150	/17	1	03	10	
6575	4160	4160	/20	1	03	20	
6576	4141	4141	/21	1	03	01	LOW 16'ER
6577	4160	4160	/22	1	03	20	
6600	4150	4150	/23	1	03	10	
6601	4157	4157	/24	1	03	17	
6602	4147	4147	/25	1	03	07	
6603	4156	4156	/26	1	03	16	
6604	4146	4146	/27	1	03	06	
6605	4155	4155	/30	1	03	15	
6606	4145	4145	/31	1	03	05	
6607	4154	4154	/32	1	03	14	
6610	4144	4144	/33	1	03	04	
6611	4153	4153	/34	1	03	13	
6612	4143	4143	/35	1	03	03	
6613	4152	4152	/36	1	03	12	
6614	4142	4142	/37	1	03	02	
6615	4151	4151	/40	1	03	11	
6616	4244	4244	/41	1	05	04	TOP DIAGS.
6617	0143	0143	/42	0	03	03	
6620	0152	0152	/43	0	03	12	
6621	4350	4350	/44	1	07	10	
6622	0142	0142	/45	0	03	02	
6623	4344	4344	/46	1	07	04	
6624	0151	0151	/47	0	03	11	
6625	0141	0141	/50	0	03	01	
6626	4250	4250	/51	1	05	10	
6627	0246	0246	/52	0	05	06	
6630	0146	0146	/53	0	03	04	
6631	4246	4246	/54	1	05	06	LOW DIAGS
6632	0147	0147	/55	0	03	07	
6633	0157	0157	/56	0	03	17	
6634	4342	4342	/57	1	07	02	
6635	0150	0150	/60	0	03	10	
6636	4346	4346	/61	1	07	06	
6637	0160	0160	/62	0	03	20	
6640	0141	0141	/63	0	03	01	
6641	4242	4242	/64	1	05	02	
6642	0244	0244	/65	0	05	04	
6643	0144	0144	/66	0	03	04	
6644	0101	0101	/67	0	02	01	PAT05 ORIGIN
6645	3051	3051	/70				PAT06 ORIGIN
6646	0043	0043	/	0	01	03	

/PATTERN 624 - RAPID ARROWS

6647 0147 PAT624, 0147

/LINE PEN LGTH DIR
/01 0 03 07

6650	4301	4301	/02	1	06	01
6651	4143	4143	/03	1	03	03
6652	4143	4143	/04	1	03	03
6653	4305	4305	/05	1	06	05
6654	4305	4305	/06	1	06	05
6655	4147	4147	/07	1	03	07
6656	4147	4147	/10	1	03	07
6657	4301	4301	/11	1	06	01
6660	3135	3135	/12			
6661	0001	0001	/	0	00	01
6662	4406	4406	/	1	10	06
6663	0402	0402	/	0	10	02
6664	4404	4404	/	1	10	04
6665	0410	0410	/	0	10	10
6666	3125	3125	/13			
6667	0003	0003	/	0	00	03
6670	4410	4410	/	1	10	10
6671	0404	0404	/	0	10	04
6672	4406	4406	/	1	10	06
6673	0402	0402	/	0	10	02
6674	3145	3145	/14			
6675	0005	0005	/	0	00	05
6676	4402	4402	/	1	10	02
6677	0406	0406	/	0	10	06
6700	4410	4410	/	1	10	10
6701	0404	0404	/	0	10	04
6702	3125	3125	/15			
6703	0007	0007	/	0	00	07
6704	4404	4404	/	1	10	04
6705	0410	0410	/	0	10	10
6706	4402	4402	/	1	10	02
6707	0406	0406	/	0	10	06
6710	3135	3135	/16			
6711	0001	0001	/	0	00	01
6712	4406	4406	/	1	10	06
6713	0402	0402	/	0	10	02
6714	4404	4404	/	1	10	04
6715	0410	0410	/	0	10	10
6716	0143	0143	/17	0	03	03
6717	3051	3051	/20			
6720	0107	0107	/	0	02	07
			/			
			/			

L624,

PAT06 ORIGIN.

PAT01 ORIGIN.

/PATTERN 07 - MAKES EACH VECTOR AVAILABLE IN DASH OR SHUFFLE FOR THE
/WIDTH OR LENGTH OF THE PLOT AREA.

			LINE	PEN	LGTH	DIR	RETURN	LINES
6721	3011	PAT07,	3011					S
6722	4145		4145					
6723	3011		3011					N
6724	4141		4141					
6725	3011		3011					SE

6726	4144	4144	/	1	03	04	
6727	3011	3011	/04				NE
6730	4142	4142	/	1	03	02	DASHES
6731	3012	3012	/05				N
6732	4101	4101	/	1	02	01	
6733	0101	0101	/	0	02	01	
6734	3012	3012	/06				NNE
6735	4111	4111	/	1	02	11	
6736	0111	0111	/	0	02	11	
6737	3012	3012	/07				NE
6740	4102	4102	/	1	02	02	
6741	0102	0102	/	0	02	02	
6742	3012	3012	/10				ENE
6743	4112	4112	/	1	02	12	
6744	0112	0112	/	0	02	12	
6745	3052	3052	/11				E
6746	4103	4103	/	1	02	03	
6747	0103	0103	/	0	02	03	
6750	3012	3012	/12				ESE
6751	4113	4113	/	1	02	13	
6752	0113	0113	/	0	02	13	
6753	3012	3012	/13				SE
6754	4104	4104	/	1	02	04	
6755	0104	0104	/	0	02	04	
6756	3012	3012	/14				SSE
6757	4114	4114	/	1	02	14	
6760	0114	0114	/	0	02	14	
6761	3012	3012	/15				S
6762	4105	4105	/	1	02	05	
6763	0105	0105	/	0	02	05	
6764	3012	3012	/16				SSW
6765	4115	4115	/	1	02	15	
6766	0115	0115	/	0	02	15	
6767	3012	3012	/17				SW
6770	4106	4106	/	1	02	06	
6771	0106	0106	/	0	02	06	
6772	3012	3012	/20				WSW
6773	4116	4116	/	1	02	16	
6774	0116	0116	/	0	02	16	
6775	3052	3052	/21				W
6776	4107	4107	/	1	02	07	
6777	0107	0107	/	0	02	07	
7000	3012	3012	/22				WNW
7001	4117	4117	/	1	02	17	
7002	0117	0117	/	0	02	17	
7003	3012	3012	/23				NW
7004	4110	4110	/	1	02	10	
7005	0110	0110	/	0	02	10	
7006	3012	3012	/24				NNW
7007	4120	4120	/	1	02	20	
7010	0120	0120	/	0	02	20	
7011	3014	3014	/25				SHUFFLES
7012	4101	4101	/	1	02	01	N

7013	0101	0101	/	0	02	01	
7014	4105	4105	/	1	02	05	
7015	0101	0101	/	0	02	01	
7016	3014	3014	/26				NNE
7017	4111	4111	/	1	02	11	
7020	0111	0111	/	0	02	11	
7021	4115	4115	/	1	02	15	
7022	0111	0111	/	0	02	11	
7023	3014	3014	/27				NE
7024	4102	4102	/	1	02	02	
7025	0102	0102	/	0	02	02	
7026	4106	4106	/	1	02	06	
7027	0102	0102	/	0	02	02	
7030	3014	3014	/30				ENE
7031	4112	4112	/	1	02	12	
7032	0112	0112	/	0	02	12	
7033	4116	4116	/	1	02	16	
7034	0112	0112	/	0	02	12	
7035	3054	3054	/31				E
7036	4103	4103	/	1	02	03	
7037	0103	0103	/	0	02	03	
7040	4107	4107	/	1	02	07	
7041	0103	0103	/	0	02	03	
7042	3104	3104	/32				ESE
7043	4113	4113	/	1	02	13	
7044	0113	0113	/	0	02	13	
7045	4117	4117	/	1	02	17	
7046	0113	0113	/	0	02	13	
7047	3014	3014	/33				SE
7050	4104	4104	/	1	02	04	
7051	0104	0104	/	0	02	04	
7052	4110	4110	/	1	02	10	
7053	0104	0104	/	0	02	04	
7054	3014	3014	/34				SSE
7055	4114	4114	/	1	02	14	
7056	0114	0114	/	0	02	14	
7057	4120	4120	/	1	02	20	
7060	0114	0114	/	0	02	14	
7061	3014	3014	/35				S
7062	4105	4105	/	1	02	05	
7063	0105	0105	/	0	02	05	
7064	4101	4101	/	1	02	01	
7065	0105	0105	/	0	02	05	
7066	3014	3014	/36				SSW
7067	4115	4115	/	1	02	15	
7070	0115	0115	/	0	02	15	
7071	4111	4111	/	1	02	11	
7072	0115	0115	/	0	02	15	
7073	3014	3014	/37				SW
7074	4106	4106	/	1	02	06	
7075	0106	0106	/	0	02	06	
7076	4102	4102	/	1	02	02	
7077	0106	0106	/	0	02	06	

7100	3014	3014	/40				WSW
7101	4116	4116	/	1	02	16	
7102	0116	0116	/	0	02	16	
7103	4112	4112	/	1	02	12	
7104	0116	0116	/	0	02	16	
7105	3054	3054	/41				W
7106	4107	4107	/	1	02	07	
7107	0107	0107	/	0	02	07	
7110	4103	4103	/	1	02	03	
7111	0107	0107	/	0	02	07	
7112	3014	3014	/42				WNW
7113	4117	4117	/	1	02	17	
7114	0117	0117	/	0	02	17	
7115	4113	4113	/	1	02	13	
7116	0117	0117	/	0	02	17	
7117	3014	3014	/43				NW
7120	4110	4110	/	1	02	10	
7121	0110	0110	/	0	02	10	
7122	4104	4104	/	1	02	04	
7123	0110	0110	/	0	02	10	
7124	3014	3014	/44				NNW
7125	4120	4120	/	1	02	20	
7126	0120	0120	/	0	02	20	
7127	4114	4114	/	1	02	14	
7130	0120	0120	/	0	02	20	
		S					

